

The Professor's Gauntlet

MDS211 Nervous System — Lectures 1–16 · 300 Questions · Full Distractor Analysis

This is not a past-paper reprint — it's built to think like your professor. Every lecture (1–16) carries at least 15 questions, mixing 150 questions verified against the past-paper archive with 150 original, critical-thinking questions written directly from your lecture slides to test integration, not just recall. Every answer explains WHY the correct choice is right and WHY the most tempting wrong answers are traps — plus recurring PROF'S TRICK call-outs flagging the patterns examiners love to exploit. Questions marked † are original (not verified past-exam items) — everything else is grounded in the compiled archive. This is long, dense, and meant to be worked through properly, not skimmed. Good luck.

Lecture 1 — Organization of the CNS (20 questions)

1. Concerning the parts of the Central Nervous System, which of the following items is NOT included?
 - A. Pineal gland
 - B. Basal ganglia
 - C. Fourth ventricle
 - D. Trigeminal ganglion
 - E. Posterior pituitary gland
2. When you wear a ring for a long time, you stop noticing it. Which receptor's rapid adaptation is responsible?
 - A. Meissner's corpuscle
 - B. Pacinian corpuscle
 - C. Free nerve ending
 - D. Merkel disc
 - E. Ruffini ending
3. Which of the following is NOT a division of the peripheral nervous system?
 - A. Somatic nervous system
 - B. Autonomic nervous system
 - C. Enteric nervous system
 - D. Central nervous system
 - E. Cranial and spinal nerves

† original question, written from lecture content (not a verified past-exam item)
4. A layer-by-layer dissection proceeds from the skull inward. Put the meningeal layers in the correct order.
 - A. Dura → Pia → Arachnoid
 - B. Arachnoid → Dura → Pia
 - C. Dura → Arachnoid → Pia
 - D. Pia → Arachnoid → Dura
 - E. Pia → Dura → Arachnoid

† original question, written from lecture content (not a verified past-exam item)
5. CSF produced in the lateral ventricles must pass through the foramen of Monro to reach the:
 - A. Fourth ventricle
 - B. Third ventricle
 - C. Cerebral aqueduct
 - D. Subarachnoid space directly
 - E. Central canal

† original question, written from lecture content (not a verified past-exam item)

6. Which statement about gray and white matter distribution is correct?

- A. In the spinal cord, gray matter is peripheral and white matter is central
- B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it
- C. The cerebellum has no gray matter
- D. White matter contains neuron cell bodies while gray matter contains only axons
- E. Gray and white matter are distributed identically in brain and spinal cord

† original question, written from lecture content (not a verified past-exam item)

7. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane. This is best classified as:

- A. Spina bifida occulta
- B. Meningocele
- C. Myelomeningocele
- D. Anencephaly
- E. Encephalocele

† original question, written from lecture content (not a verified past-exam item)

8. Which cranial nerve numbers correspond to nerves that are purely sensory?

- A. III, IV, VI
- B. I, II, VIII
- C. V, VII, X
- D. XI, XII
- E. III, VII, X

† original question, written from lecture content (not a verified past-exam item)

9. Dorsal root ganglia contain the cell bodies of which type of neuron?

- A. Lower motor neurons
- B. Preganglionic autonomic neurons
- C. First-order (primary) sensory neurons
- D. Interneurons of the dorsal horn
- E. Postganglionic sympathetic neurons

† original question, written from lecture content (not a verified past-exam item)

10. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit pattern is expected?

- A. Pure sensory loss only
- B. Pure motor loss only
- C. Combined sensory and motor loss at that segmental level
- D. No deficit, since roots regenerate quickly
- E. Only autonomic dysfunction

† original question, written from lecture content (not a verified past-exam item)

11. Which of the following best describes the enteric nervous system?

- A. It requires constant sympathetic and parasympathetic input to function at all
- B. It can regulate gut motility and secretion semi-independently of CNS input
- C. It is anatomically part of the CNS
- D. It contains only efferent (motor) neurons
- E. It is synonymous with the vagus nerve

† original question, written from lecture content (not a verified past-exam item)

12. Which of these structures is a boundary landmark of the fourth ventricle?

- A. Corpus callosum
- B. Cerebral peduncles
- C. Pons and medulla (floor) with cerebellum (roof)
- D. Basal ganglia
- E. Hippocampus

† original question, written from lecture content (not a verified past-exam item)

13. Neural crest cells contribute to all of the following EXCEPT:

- A. Dorsal root ganglia
- B. Schwann cells
- C. Adrenal medulla
- D. Oligodendrocytes
- E. Melanocytes

† original question, written from lecture content (not a verified past-exam item)

14. A 'brainstem' anatomically and functionally includes which three structures, rostral to caudal?

- A. Thalamus, midbrain, pons
- B. Midbrain, pons, medulla oblongata
- C. Cerebellum, pons, medulla
- D. Hypothalamus, midbrain, pons
- E. Pons, medulla, spinal cord

† original question, written from lecture content (not a verified past-exam item)

15. Which statement correctly differentiates somatic and autonomic motor pathways?

- A. Somatic motor pathways use two neurons in series; autonomic use one
- B. Autonomic motor pathways use two neurons in series (pre- and postganglionic); somatic uses one
- C. Both use exactly the same two-neuron chain
- D. Somatic motor neurons synapse in peripheral ganglia
- E. Autonomic motor neurons directly innervate skeletal muscle

† original question, written from lecture content (not a verified past-exam item)

16. How many pairs of spinal nerves are there in total, and which region has one MORE pair than vertebrae?

- A. 31 pairs; cervical
- B. 33 pairs; lumbar
- C. 31 pairs; thoracic
- D. 30 pairs; sacral
- E. 31 pairs; sacral

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17. The scalp, skull, vertebrae, meninges, and CSF together serve what shared function?

- A. Neurotransmitter synthesis
- B. Mechanical and immunological protection of the CNS
- C. Generating the resting membrane potential
- D. Producing myelin
- E. Regulating heart rate

† original question, written from lecture content (not a verified past-exam item)

- 18.** Area postrema, found near the floor of the fourth ventricle, is functionally significant because it:
- A. Has an intact blood-brain barrier, like most brainstem nuclei
 - B. Lacks a complete blood-brain barrier, allowing it to sense blood-borne toxins (chemoreceptor trigger zone)
 - C. Is the primary site of CSF production
 - D. Controls voluntary swallowing exclusively
 - E. Is part of the limbic system

† original question, written from lecture content (not a verified past-exam item)

- 19.** Which best explains why the CNS is described as having a much more limited regenerative capacity than the PNS after axonal injury?

- A. CNS neurons cannot form action potentials
- B. Oligodendrocytes and CNS scar tissue produce inhibitory molecules that block axon regrowth, unlike the permissive environment Schwann cells provide in the PNS
- C. The CNS lacks myelin entirely
- D. PNS axons are thinner and therefore regenerate faster
- E. The CNS has no glial cells to support regeneration

† original question, written from lecture content (not a verified past-exam item)

- 20.** Which meningeal layer is the site of most epidural hematomas, classically from a middle meningeal artery tear?

- A. Between skull and dura (epidural/extradural space)
- B. Between dura and arachnoid (subdural space)
- C. Between arachnoid and pia (subarachnoid space)
- D. Within the brain parenchyma itself (intracerebral)
- E. Within the ventricles (intraventricular)

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Lecture 2 — Neurohistology (Neuron & Neuroglia) (20 questions)

- 21.** Which of the following compounds is transferred from astrocyte to neuron and used for ATP production during starvation?

- A. Fumarate
- B. Citrate
- C. Pyruvate
- D. Succinyl-CoA
- E. Lactate

- 22.** Myelin sheath wrapping nerve axons within the basal ganglia (a CNS structure) is produced by which cell?

- A. Astrocyte
- B. Schwann cell
- C. Satellite cell
- D. Oligodendrocyte
- E. Microglia

- 23.** Which of the following is the glutamate transporter responsible for uptake into astrocytes?

- A. MCT
- B. DCT
- C. IAATS
- D. TCT
- E. EAATs

- 24.** If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?
- A. A- β fiber
 - B. C fiber (unmyelinated)
 - C. Paleospinothalamic tract fibers
 - D. Neo-spinothalamic tract (A- δ) fibers
 - E. A- α fiber
- 25.** Which of the following is NOT a component of the blood-brain barrier?
- A. Oligodendrocyte
 - B. Astrocyte end-feet
 - C. Pericyte of capillary
 - D. Endothelial cell tight junctions
 - E. Basement membrane
- 26.** Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?
- A. Microglia
 - B. Schwann cell
 - C. Oligodendrocyte
 - D. Astrocyte
 - E. Ependymal cell
- 27.** Which part of the axon has the highest density of voltage-gated Na⁺ channels, essential for action potential generation?
- A. Dendrites
 - B. Axon hillock / initial segment
 - C. Distal axon terminal only
 - D. Cell body (soma) uniformly
 - E. Myelin sheath itself
- 28.** Which disease process is defined by the immune system attacking and destroying CNS myelin?
- A. Guillain-Barré syndrome
 - B. Multiple sclerosis
 - C. Charcot-Marie-Tooth disease
 - D. Myasthenia gravis
 - E. Duchenne muscular dystrophy
- 29.** Which statement correctly distinguishes sensory ganglia from autonomic ganglia?
- A. Sensory ganglia contain pseudounipolar neuron cell bodies with no synapses inside the ganglion; autonomic ganglia contain a true synapse between pre- and postganglionic neurons
 - B. Both contain synapses between two neurons
 - C. Sensory ganglia are only found in the CNS
 - D. Autonomic ganglia contain pseudounipolar neurons only
 - E. There is no functional difference

30. Which statement about neurons is INCORRECT?

- A. Neurons are post-mitotic and generally do not divide in adulthood
- B. Neurons can be classified by shape as unipolar, bipolar, or multipolar
- C. All neurons have a myelinated axon
- D. Neurons communicate via chemical or electrical synapses
- E. Dendrites typically receive input while axons conduct output

31. Layers surrounding a peripheral nerve fiber, from innermost to outermost, are correctly ordered as:

- A. Epineurium → Perineurium → Endoneurium
- B. Endoneurium → Perineurium → Epineurium
- C. Perineurium → Endoneurium → Epineurium
- D. Epineurium → Endoneurium → Perineurium
- E. Endoneurium → Epineurium → Perineurium

32. A young woman presents with fatigable muscle weakness that worsens with repetitive activity and improves with rest. This points to a defect at the:

- A. Presynaptic voltage-gated Ca^{2+} channel (Lambert-Eaton syndrome)
- B. Postsynaptic nicotinic ACh receptor (myasthenia gravis)
- C. Node of Ranvier
- D. Dorsal root ganglion
- E. Neuromuscular junction basal lamina only

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33. Which glial cell lines the ventricles and central canal, contributing to CSF circulation?

- A. Ependymal cell
- B. Astrocyte
- C. Microglia
- D. Oligodendrocyte
- E. Satellite cell

† original question, written from lecture content (not a verified past-exam item)

34. A demyelinated CNS axon conducts action potentials more slowly and can fail to propagate altogether. What is the direct biophysical reason?

- A. Loss of myelin decreases membrane capacitance, speeding conduction
- B. Loss of myelin increases current leak across the membrane and removes saltatory jumps between nodes, slowing/failing conduction
- C. Demyelination increases the density of voltage-gated Na^{+} channels along the whole axon
- D. Demyelination has no effect on conduction velocity, only on synaptic transmission
- E. Myelin loss increases the safety factor for conduction

† original question, written from lecture content (not a verified past-exam item)

35. Which best explains why glutamate is normally cleared very quickly from the synaptic cleft?

- A. To prevent excitotoxic Ca^{2+} overload from prolonged NMDA/AMPA receptor activation
- B. Because glutamate is only useful as an inhibitory transmitter
- C. To increase resting membrane potential of the postsynaptic neuron
- D. Glutamate does not need clearance because it is not reused
- E. To prevent action potentials from ever occurring

† original question, written from lecture content (not a verified past-exam item)

36. Satellite cells are glial cells found specifically in which location?

- A. Surrounding neuron cell bodies within peripheral ganglia
- B. Wrapping CNS axons
- C. Lining the ventricles
- D. Within the blood-brain barrier only
- E. Wrapping PNS axons

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37. A axo-axonic synapse, as opposed to an axo-dendritic synapse, is best described as:

- A. One axon terminal synapsing onto another axon (often near its terminal), typically for presynaptic modulation
- B. One axon terminal synapsing onto a dendrite
- C. One dendrite synapsing onto another dendrite
- D. A synapse only found in the retina
- E. A synapse that cannot be inhibitory

† original question, written from lecture content (not a verified past-exam item)

38. Reactive astrogliosis (astrocyte scar formation) after CNS injury has which net effect on axon regeneration?

- A. It always promotes robust axon regrowth across the injury site
- B. It seals off and stabilizes the injury site but the resulting glial scar also physically and chemically inhibits axon regrowth through it
- C. It has no effect on the injury site
- D. It converts astrocytes into oligodendrocytes
- E. It only occurs in the PNS

† original question, written from lecture content (not a verified past-exam item)

39. Multipolar neurons — the most common neuron shape in the CNS — are structurally characterized by:

- A. One axon and one dendrite only
- B. One axon and multiple dendrites
- C. No axon, only dendrites
- D. Two axons and no dendrites
- E. A single process that splits into central and peripheral branches

† original question, written from lecture content (not a verified past-exam item)

40. Which best describes why oligodendrocytes are more vulnerable to certain metabolic/ischemic insults than astrocytes, contributing to selective white matter injury patterns (e.g., in premature infants)?

- A. Oligodendrocytes have particularly high metabolic demand for myelin maintenance and are especially sensitive to oxidative stress and glutamate excitotoxicity during their maturation window
- B. Oligodendrocytes have no metabolic requirements at all
- C. Oligodendrocytes are identical to astrocytes in every functional respect
- D. Oligodendrocytes are immune cells with no relationship to metabolism
- E. Astrocytes are always more vulnerable than oligodendrocytes in every scenario

† original question, written from lecture content (not a verified past-exam item)

Lecture 3 — Neuron Properties: RMP & Action Potential (20 questions)

- 41.** A 40-year-old male needs an urgent hand suture and is given lidocaine as local anesthesia. What is its mechanism of action?
- A. Activation of ligand-gated potassium channels
 - B. Inhibition of nicotinic acetylcholine receptors
 - C. Inhibition of voltage-gated sodium channels
 - D. Inhibition of acetylcholine release
 - E. Activation of muscarinic acetylcholine receptors
- 42.** Hyperpolarization of the postsynaptic membrane in response to glycine or GABA-A receptor activation is produced by the opening of which channel?
- A. Na⁺ channels
 - B. K⁺ channels only
 - C. Cl⁻ channels
 - D. Ca²⁺ channels
 - E. Nonspecific cation channels
- 43.** A 35-year-old male receives a lidocaine injection before incision and drainage of a buttock abscess. Which best describes lidocaine's mechanism?
- A. Blocks voltage-gated K⁺ channels
 - B. Blocks ligand-gated Ca²⁺ channels
 - C. Blocks voltage-gated Na⁺ channels
 - D. Stimulates ligand-gated Na⁺ channels
 - E. Blocks Mg²⁺ channels
- 44.** A patient develops muscle weakness that worsens through the afternoon and evening but is nearly absent in the morning. Which pathology is most consistent?
- A. Guillain-Barré syndrome
 - B. Myasthenia gravis
 - C. Multiple sclerosis
 - D. Duchenne muscular dystrophy
 - E. Amyotrophic lateral sclerosis
- 45.** What is the major consumer of energy (ATP) in the resting brain?
- A. DNA replication in neurons
 - B. Restoring ionic gradients via the Na⁺/K⁺-ATPase after synaptic and action-potential activity
 - C. Myelin synthesis
 - D. Muscle contraction
 - E. Protein degradation
- 46.** Which of the following correctly describes when Na⁺/K⁺-ATPase activity increases the most?
- A. During periods of high neuronal firing, to restore ionic gradients dissipated by Na⁺ influx/K⁺ efflux
 - B. Only during sleep
 - C. Only in glial cells, never in neurons
 - D. It runs at a constant, unchanging rate regardless of activity
 - E. Only during hyperpolarization

47. When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:

- A. Temporal summation
- B. Spatial summation
- C. Facilitation
- D. Long-term potentiation
- E. Accommodation

48. All of the following are true of chemical synapses EXCEPT:

- A. They involve a synaptic delay of roughly 0.5-1 ms
- B. Transmission is unidirectional (pre- to postsynaptic)
- C. They can amplify a small presynaptic signal into a larger postsynaptic response
- D. Transmission is instantaneous with essentially zero delay
- E. Neurotransmitter release is Ca^{2+} -dependent

49. Which statement about graded potentials is INCORRECT?

- A. They decay in amplitude with distance (decremental conduction)
- B. Their amplitude is proportional to stimulus strength
- C. They can summate
- D. They obey the all-or-none law like action potentials
- E. They are typically local, subthreshold events

50. During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?

- A. All voltage-gated K^+ channels are closed
- B. Voltage-gated Na^+ channels are inactivated and cannot reopen until the membrane repolarizes
- C. The membrane is hyperpolarized below threshold, requiring only a slightly stronger stimulus
- D. Cl^- channels are maximally open
- E. The neuron has run out of neurotransmitter

† original question, written from lecture content (not a verified past-exam item)

51. Which factor increases axonal conduction velocity?

- A. Decreasing axon diameter
- B. Demyelination
- C. Increasing axon diameter and/or myelination (saltatory conduction)
- D. Increasing membrane capacitance
- E. Decreasing the density of Na^+ channels at nodes

† original question, written from lecture content (not a verified past-exam item)

52. What determines the equilibrium potential for a given ion, as described by the Nernst equation?

- A. The membrane's permeability to that ion
- B. The ratio of that ion's extracellular to intracellular concentration
- C. The total number of open channels for other ions
- D. The resting membrane potential alone
- E. The speed of the $\text{Na}^+/\text{K}^+-\text{ATPase}$

† original question, written from lecture content (not a verified past-exam item)

53. Presynaptic inhibition, as opposed to postsynaptic inhibition, works primarily by:

- A. Hyperpolarizing the postsynaptic cell body directly
- B. Reducing Ca^{2+} influx into the presynaptic terminal, decreasing neurotransmitter release
- C. Blocking postsynaptic receptors directly
- D. Increasing postsynaptic K^{+} conductance
- E. Destroying the presynaptic terminal

† original question, written from lecture content (not a verified past-exam item)

54. A drug that blocks voltage-gated K^{+} channels in neurons would be expected to:

- A. Speed up repolarization of the action potential
- B. Prolong the action potential by slowing repolarization
- C. Prevent depolarization entirely
- D. Have no effect on the action potential
- E. Hyperpolarize the resting membrane potential

† original question, written from lecture content (not a verified past-exam item)

55. Which best describes the relationship between stimulus intensity and action potential characteristics in a single axon?

- A. A stronger stimulus produces a larger-amplitude action potential
- B. A stronger stimulus produces a higher FREQUENCY of action potentials, not larger amplitude (each AP is all-or-none)
- C. A stronger stimulus produces a longer-duration single action potential
- D. Stimulus intensity has no encoding effect at all
- E. A stronger stimulus lowers the axon's threshold permanently

† original question, written from lecture content (not a verified past-exam item)

56. A toxin selectively blocks voltage-gated Na^{+} channels at the node of Ranvier along an entire myelinated axon. What is the functional consequence?

- A. Conduction velocity increases because resistance drops
- B. Action potential propagation fails completely, since saltatory conduction depends on regenerating the signal at each node
- C. The axon switches to continuous conduction instead
- D. Only the resting potential is affected, conduction is unaffected
- E. Myelin thickness increases to compensate

† original question, written from lecture content (not a verified past-exam item)

57. Long-term potentiation (LTP), a cellular model of learning and memory, is classically initiated by which receptor-level event?

- A. GABA-A receptor mediated Cl^{-} influx
- B. Glutamate binding AMPA receptors causes partial depolarization, relieving the Mg^{2+} block on NMDA receptors and allowing Ca^{2+} entry
- C. Serotonin binding 5-HT₃ receptors
- D. Glycine receptor activation
- E. Muscarinic receptor activation only

† original question, written from lecture content (not a verified past-exam item)

58. Which best explains why increasing extracellular K⁺ concentration (hyperkalemia) can cause cardiac/neuronal membrane dysfunction, based on the Nernst equation?

- A. It has no effect on membrane potential
- B. It depolarizes the resting membrane potential (brings it closer to 0), since the K⁺ equilibrium potential becomes less negative, potentially causing sustained depolarization and eventual inactivation of voltage-gated Na⁺ channels
- C. It always hyperpolarizes the membrane
- D. It only affects skeletal muscle, never neurons or cardiac cells
- E. It has no relationship to the Nernst equation

† original question, written from lecture content (not a verified past-exam item)

59. Which best describes the ionic basis of the falling (repolarizing) phase of the action potential?

- A. Continued Na⁺ influx
- B. Na⁺ channel inactivation combined with delayed voltage-gated K⁺ channel opening (K⁺ efflux)
- C. Cl⁻ influx alone
- D. Ca²⁺ efflux alone
- E. Complete cessation of all ion movement

† original question, written from lecture content (not a verified past-exam item)

60. A drug that selectively blocks only T-type (low-threshold) Ca²⁺ channels would be expected to most directly affect which physiological process?

- A. Skeletal muscle contraction exclusively
- B. Thalamic rhythmic burst firing, relevant to absence seizure pathophysiology (ethosuximide's mechanism)
- C. Only cardiac contraction
- D. Only presynaptic neurotransmitter release
- E. No known physiological role

† original question, written from lecture content (not a verified past-exam item)

Lecture 4 — Metabolic Pathways of Neurons (20 questions)

61. Which of the following is INCORRECT regarding neuronal metabolism?

- A. Neurons preferentially utilize aerobic oxidative phosphorylation for ATP production
- B. Astrocytes preferentially perform anaerobic glycolysis, producing lactate
- C. Neurons store large amounts of glycogen for backup fuel
- D. Neurons contain LDH-1, which favors pyruvate formation for entry into the TCA cycle
- E. Astrocytes contain LDH-5, which favors lactate formation from pyruvate

62. What enzyme hydrolyzes glutamine to glutamate inside the presynaptic neuron, regenerating the neurotransmitter pool?

- A. Glutamine synthetase
- B. Glutaminase
- C. Glutamate decarboxylase
- D. GABA transaminase
- E. Monoamine oxidase

63. Excess extracellular glutamate causes excitotoxic neuronal injury primarily through excessive intracellular accumulation of which ion?

- A. Na^+
- B. K^+
- C. Ca^{2+}
- D. Cl^-
- E. Mg^{2+}

64. A 58-year-old man with cirrhosis develops confusion and drowsiness with markedly elevated serum ammonia (hepatic encephalopathy). Which cell is primarily responsible for detoxifying the ammonia in the brain, and how?

- A. Neurons, via glutaminase
- B. Astrocytes, by converting glutamate + ammonia into glutamine via glutamine synthetase
- C. Microglia, via phagocytosis of ammonia
- D. Oligodendrocytes, via myelin synthesis
- E. Ependymal cells, via CSF filtration

† original question, written from lecture content (not a verified past-exam item)

65. In the same hepatic encephalopathy scenario, why does excessive glutamine accumulation in astrocytes become harmful rather than simply protective?

- A. Glutamine is directly neurotoxic to Na^+ channels
- B. Glutamine acts as an osmolyte, drawing water into astrocytes and causing cellular swelling/cerebral edema
- C. Glutamine competitively inhibits the Na^+/K^+ -ATPase directly
- D. Glutamine converts spontaneously back to ammonia, worsening toxicity
- E. Glutamine has no downstream effect once formed

† original question, written from lecture content (not a verified past-exam item)

66. Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?

- A. Because astrocyte dysfunction/swelling disrupts the glutamate-glutamine cycle and lactate shuttle that neurons depend on for neurotransmitter recycling and fuel
- B. Because ammonia only affects neurons and never astrocytes
- C. Because neurons contain glutamine synthetase too, so they are equally affected
- D. Ammonia directly destroys myelin
- E. There is no real connection between astrocyte and neuron function

† original question, written from lecture content (not a verified past-exam item)

67. Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?

- A. GLUT3
- B. EAAT-2
- C. Monocarboxylate transporters (MCT1/4)
- D. Na^+/K^+ -ATPase
- E. SNAT

† original question, written from lecture content (not a verified past-exam item)

68. Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?

- A. Astrocyte: GLUT1; Neuron: GLUT3
- B. Astrocyte: GLUT3; Neuron: GLUT1
- C. Both use GLUT4 exclusively
- D. Astrocyte: GLUT2; Neuron: GLUT5
- E. Neither cell type expresses glucose transporters

† original question, written from lecture content (not a verified past-exam item)

69. The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product important for antioxidant defense?

- A. ATP directly
- B. NADPH, which regenerates reduced glutathione (GSH) to neutralize reactive oxygen species
- C. Lactate
- D. Glycogen
- E. Acetyl-CoA

† original question, written from lecture content (not a verified past-exam item)

70. Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?

- A. Vitamin B1 (thiamine)
- B. Vitamin C
- C. Vitamin B12 only
- D. Folate only
- E. Vitamin D

† original question, written from lecture content (not a verified past-exam item)

71. Astrocytic glycogen stores are mobilized during which physiological circumstance?

- A. Never, since astrocytes cannot break down glycogen
- B. Periods of high neuronal activity or low glucose availability, to supply additional lactate to neurons
- C. Only during REM sleep
- D. Only in the peripheral nervous system
- E. Exclusively during fetal development

† original question, written from lecture content (not a verified past-exam item)

72. Which statement about ascorbic acid (vitamin C) recycling in the CNS is correct?

- A. Astrocytes help regenerate reduced ascorbate using NADPH from the pentose phosphate pathway
- B. Ascorbate has no antioxidant role in the CNS
- C. Ascorbate is synthesized de novo by neurons
- D. Ascorbate recycling requires no reducing cofactors
- E. Ascorbate is toxic to astrocytes

† original question, written from lecture content (not a verified past-exam item)

73. Which of the following best summarizes the overall direction of the glutamate-glutamine cycle across the synapse?

- A. Glutamate released by the neuron → taken up by astrocyte via EAAT → converted to glutamine by glutamine synthetase → glutamine shuttled back to neuron → converted back to glutamate by glutaminase
- B. Glutamine is released directly by the neuron as the neurotransmitter itself
- C. Astrocytes release glutamate directly onto the postsynaptic neuron, bypassing the presynaptic terminal entirely
- D. The cycle runs in the opposite direction, from neuron to neuron with no astrocyte involvement
- E. Glutamate is destroyed permanently after each synaptic event, with no recycling

† original question, written from lecture content (not a verified past-exam item)

74. Why do astrocytes convert glutamate to glutamine before returning it to neurons, rather than simply shuttling glutamate straight back?

- A. Glutamine is inert/non-excitotoxic and safe to transport through the extracellular space, unlike glutamate which would over-stimulate receptors along the way
- B. Glutamine is a better energy source than glutamate
- C. There is no functional reason; it is metabolically wasteful with no benefit
- D. Glutamate cannot cross any cell membrane at all
- E. Neurons cannot process glutamate under any circumstances

† original question, written from lecture content (not a verified past-exam item)

75. A patient with a genetic deficiency of astrocytic glutamine synthetase would be expected to show which combination of findings?

- A. Elevated extracellular glutamate (impaired clearance/recycling) with risk of excitotoxicity, plus low glutamine levels
- B. Normal glutamate handling with no clinical consequence
- C. Elevated glutamine with normal glutamate
- D. Complete absence of any neurotransmission
- E. Enhanced GABA synthesis with no other changes

† original question, written from lecture content (not a verified past-exam item)

76. Compared to resting conditions, during intense neuronal activity astrocytic ATP levels tend to:

- A. Increase, because astrocytes are not metabolically engaged during activity
- B. Drop, because both glutamate uptake (via Na^+/K^+ -ATPase-dependent EAAT-linked transport) and its conversion to glutamine by glutamine synthetase are energy-intensive processes
- C. Stay completely unchanged
- D. Increase because glycolysis is turned off
- E. Have no relationship to neuronal firing whatsoever

† original question, written from lecture content (not a verified past-exam item)

77. Which best explains why neurons are said to 'prefer' lactate over glucose as a fuel source when both are available in sufficient quantity?

- A. Lactate yields more ATP per molecule than glucose
- B. Lactate can be converted directly to pyruvate by neuronal LDH-1 and fed straight into the TCA cycle, and this pathway is kinetically favored in neurons under these conditions
- C. Neurons cannot metabolize glucose at all
- D. Lactate bypasses the need for oxygen entirely
- E. There is no evidence neurons prefer either substrate

† original question, written from lecture content (not a verified past-exam item)

78. Why does the brain, despite being only about 2% of body weight, consume roughly 20% of the body's resting oxygen and glucose supply?

- A. Because neurons have unusually low metabolic demands
- B. Because of the continuous, high energetic cost of restoring ionic gradients (Na^+/K^+ -ATPase) after action potentials and synaptic activity across billions of neurons, plus baseline maintenance of resting potentials
- C. Because the brain does not use oxidative phosphorylation
- D. Because the brain stores enormous glycogen reserves, requiring no external glucose
- E. There is no real metabolic demand from brain tissue

† original question, written from lecture content (not a verified past-exam item)

79. Which vitamin, besides thiamine, is a critical cofactor for the glutamate-glutamine cycle and broader amino acid metabolism, with deficiency linked to neurological symptoms including peripheral neuropathy?

- A. Vitamin B6 (pyridoxine), a cofactor for transaminase/decarboxylase reactions including GABA synthesis
- B. Vitamin A
- C. Vitamin K
- D. Vitamin E only
- E. Vitamin D only

† original question, written from lecture content (not a verified past-exam item)

80. During intense, sustained neuronal activity beyond what the lactate shuttle can immediately supply, which backup fuel source can neurons utilize, especially during prolonged fasting/ketosis?

- A. Ketone bodies (e.g., beta-hydroxybutyrate), which cross the BBB and can be oxidized for ATP production
- B. Free fatty acids directly, which cross the BBB easily
- C. Only ingested protein
- D. Neurons cannot use any alternative fuel ever
- E. Only dietary cholesterol

† original question, written from lecture content (not a verified past-exam item)

Lecture 5 — Embryonic Development of the Nervous System (18 questions)

81. Which embryonic structure gives rise to the substantia nigra?

- A. Metencephalon
- B. Myelencephalon
- C. Telencephalon
- D. Diencephalon
- E. Mesencephalon

82. The optic cup is an evagination (outgrowth) of which structure?

- A. Surface ectoderm
- B. Neuroectoderm of the diencephalon (optic vesicle)
- C. Mesoderm
- D. Neural crest
- E. Endoderm

83. A meningocele is characterized by protrusion of which structure(s) through a vertebral/cranial defect?

- A. Meninges and CSF only, no neural tissue
- B. Neural tissue with no meninges
- C. Bone only
- D. Meninges plus spinal cord tissue
- E. Skin only

84. Which embryonic structure gives rise to the pineal gland?

- A. Roof of the diencephalon (epithalamus)
- B. Floor of the mesencephalon
- C. Telencephalon
- D. Roof of the myelencephalon
- E. Neural crest

- 85.** Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?
- A. GSA (general somatic afferent)
 - B. SVA (special visceral afferent)
 - C. GVA (general visceral afferent)
 - D. SVE (special visceral efferent)
 - E. GSE (general somatic efferent)
- 86.** Which structure does the optic cup differentiate into?
- A. Lens
 - B. Cornea epithelium
 - C. Retina (both neural and pigmented layers) and epithelium of the iris/ciliary body
 - D. Vitreous humor only
 - E. Extraocular muscles
- 87.** Which birth defect of the spinal cord/vertebral column is classically associated with hydrocephalus (via Arnold-Chiari II malformation)?
- A. Spina bifida occulta
 - B. Myelomeningocele
 - C. Simple meningocele
 - D. Anencephaly
 - E. Craniosynostosis
- 88.** Failure of the neural folds to fuse (neural tube defect) most directly results in which condition, if it occurs at the cranial end?
- A. Spina bifida occulta
 - B. Anencephaly
 - C. Hydrocephalus alone
 - D. Syringomyelia
 - E. Meningocele
- 89.** What is the developmental origin of the pituitary gland (hypophysis)?
- A. Entirely from neuroectoderm (diencephalon)
 - B. Dual origin: Rathke's pouch (surface ectoderm, oral roof) forms the anterior lobe; a diencephalic downgrowth forms the posterior lobe
 - C. Entirely from surface ectoderm
 - D. Entirely from neural crest
 - E. Entirely from mesoderm
- 90.** The corneal endothelium develops from which embryonic source?
- A. Surface ectoderm
 - B. Neural crest cell migration
 - C. Optic cup neuroectoderm
 - D. Mesoderm exclusively
 - E. Lens vesicle

91. Failure of closure of the anterior neuropore results in which condition?

- A. Anencephaly
- B. Spina bifida occulta
- C. Meningocele
- D. Syringomyelia
- E. Diastematomyelia

92. The lens of the eye develops from which structure?

- A. Optic cup neuroectoderm
- B. Surface ectoderm (via the lens placode/lens vesicle)
- C. Neural crest
- D. Mesoderm
- E. Retinal pigment epithelium

93. Which structure forms the walls of the lateral ventricles, and what is its embryonic layer of origin?

- A. Ependyma, derived from the neuroepithelium lining the neural tube lumen
- B. Dura mater, derived from mesoderm
- C. Pia mater, derived from neural crest
- D. Skull bones, derived from surface ectoderm
- E. Choroid plexus stroma, derived from endoderm

† original question, written from lecture content (not a verified past-exam item)

94. Neural crest cells that fail to migrate normally from the developing neural tube can cause which category of clinical disorders?

- A. Neurocristopathies (e.g., Hirschsprung disease, some forms of neurofibromatosis)
- B. Disorders exclusively of the retina
- C. Disorders exclusively of skeletal muscle
- D. No known clinical disorders
- E. Only cardiac septal defects

† original question, written from lecture content (not a verified past-exam item)

95. At which week of embryonic development does neural tube closure (both neuropores) normally complete, making this the critical window for folic acid supplementation to prevent neural tube defects?

- A. By approximately the end of the 4th week
- B. By the end of the first trimester (week 12)
- C. By week 20
- D. Immediately at fertilization
- E. Not until after birth

† original question, written from lecture content (not a verified past-exam item)

96. A newborn has a tuft of hair and a dimple overlying the lower lumbar spine, with an otherwise normal neurological exam and no palpable defect. This is most consistent with:

- A. Myelomeningocele
- B. Spina bifida occulta
- C. Anencephaly
- D. Meningocele with active CSF leak
- E. Arnold-Chiari II malformation

† original question, written from lecture content (not a verified past-exam item)

97. Which best describes the relationship between the neural tube lumen and the adult ventricular system?

- A. The neural tube lumen is completely obliterated in the adult brain
- B. The neural tube lumen persists and enlarges/reshapes into the ventricular system and central canal
- C. The neural tube lumen becomes the subarachnoid space
- D. The neural tube lumen has no relationship to the ventricles
- E. The neural tube lumen becomes the extradural space

† original question, written from lecture content (not a verified past-exam item)

98. Which structure connects the third ventricle to the fourth ventricle, and what would obstruction here (e.g., from a tectal plate tumor) cause?

- A. Foramen of Monro; obstruction causes unilateral lateral ventricle enlargement
- B. Cerebral aqueduct (of Sylvius); obstruction causes non-communicating (obstructive) hydrocephalus affecting the lateral and third ventricles
- C. Foramen of Luschka; obstruction has no effect on ventricular size
- D. Central canal; obstruction causes syringomyelia only
- E. Foramen magnum; obstruction always causes brain death immediately

† original question, written from lecture content (not a verified past-exam item)

Lecture 6 — Development of the Eye (18 questions)

99. Sustained high intracranial pressure classically produces which triad of findings (Cushing's triad)?

- A. Hypertension, bradycardia, irregular respirations
- B. Hypotension, tachycardia, tachypnea
- C. Fever, rash, photophobia
- D. Hypertension, tachycardia, hyperventilation
- E. Hypotension, bradycardia, apnea

100. What does the ventral (inferior) part of the optic vesicle, at the point where it connects to the forebrain, give rise to?

- A. The lens
- B. The retinal fissure (choroid fissure), through which the hyaloid vessels pass into the developing eye
- C. The extraocular muscles
- D. The eyelids
- E. The lacrimal gland

101. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?

- A. Coloboma (a gap/cleft in the iris, retina, or optic disc)
- B. Cataract
- C. Anophthalmia
- D. Strabismus
- E. Glaucoma

102. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?

- A. Telencephalon
- B. Alar plate of the developing brainstem (sensory column)
- C. Basal plate of the developing brainstem (motor column)
- D. Neural crest exclusively
- E. Notochord

103. What is the embryonic origin of the vitreous body of the eye?

- A. Mesenchyme (partly neural crest, partly mesoderm) within the optic cup, plus contributions from the hyaloid vascular system
- B. Surface ectoderm exclusively
- C. The lens vesicle
- D. The retinal pigment epithelium alone
- E. Endoderm

† original question, written from lecture content (not a verified past-exam item)

104. The extraocular muscles that move the eyeball are derived embryologically from which tissue?

- A. Neural crest
- B. Prechordal and paraxial head mesoderm (specialized somitomeres)
- C. Surface ectoderm
- D. Neuroectoderm of the optic cup
- E. Endoderm

† original question, written from lecture content (not a verified past-exam item)

105. Persistence of the hyaloid artery after birth (a common incidental finding) reflects failure of which normal developmental process?

- A. Failure of the hyaloid vessels to regress after supplying the developing lens/vitreous
- B. Failure of the optic stalk to form
- C. Failure of the retina to differentiate
- D. Overgrowth of the choroid fissure
- E. Failure of lens placode induction

† original question, written from lecture content (not a verified past-exam item)

106. A newborn is found to have microphthalmia (an abnormally small eye) with an associated inferior coloboma. What is the most likely embryological explanation?

- A. Normal, unrelated variation with no embryological link
- B. A disruption during optic vesicle/cup formation and/or choroid fissure closure occurring early in the 5th-7th weeks of development
- C. A purely postnatal, acquired process
- D. Failure of neural crest migration to the limbs
- E. A defect confined entirely to the lens placode with no other structures involved

† original question, written from lecture content (not a verified past-exam item)

107. Which layer of the optic cup becomes the retinal pigment epithelium (RPE), and which becomes the neurosensory retina?

- A. Outer layer → RPE; inner layer → neurosensory retina
- B. Outer layer → neurosensory retina; inner layer → RPE
- C. Both layers become RPE
- D. Both layers become neurosensory retina
- E. Neither layer is retinal tissue

† original question, written from lecture content (not a verified past-exam item)

108. Congenital cataract in a newborn (lens opacity present at birth) most directly implies a disruption of which developmental process?

- A. Lens placode induction/lens vesicle formation and differentiation of lens fibers
- B. Cornea vascularization
- C. Retinal pigment epithelium proliferation
- D. Extraocular muscle attachment
- E. Choroid fissure closure exclusively

† original question, written from lecture content (not a verified past-exam item)

109. Which statement correctly relates the timing of eye development to overall CNS/neural tube development?

- A. Eye development is entirely independent of neural tube development, occurring in a completely separate organ system
- B. The eye develops as an outgrowth of the diencephalon, so major eye anomalies often co-occur with, or hint at, other forebrain developmental problems
- C. The eye develops only after birth
- D. The eye develops from the hindbrain exclusively
- E. The retina has no connection to the brain whatsoever

† original question, written from lecture content (not a verified past-exam item)

110. Which of the following eye structures is NOT derived from neuroectoderm?

- A. Neural retina
- B. Retinal pigment epithelium
- C. Optic nerve axons
- D. Lens
- E. Epithelium of the iris and ciliary body

† original question, written from lecture content (not a verified past-exam item)

111. A patient with anophthalmia (complete absence of one eye) most likely had a disruption at which developmental stage?

- A. Failure of optic vesicle formation/outgrowth from the diencephalon very early in development
- B. A late third-trimester event only
- C. Normal development with a postnatal traumatic cause
- D. A defect isolated entirely to the lens
- E. A defect isolated entirely to the extraocular muscles

† original question, written from lecture content (not a verified past-exam item)

112. Why are the extraocular muscles innervated by cranial nerves III, IV, and VI classified as GSE (general somatic efferent) rather than SVE (special visceral efferent)?

- A. Because they derive from paraxial head mesoderm/somitomeres (like body wall/limb somatic muscle), not from pharyngeal arch (branchiomeric) mesoderm
- B. Because they are located inside the skull
- C. Because they move the eyeball
- D. There is no real embryological reason, it is an arbitrary naming convention
- E. Because they are innervated by more than one nerve

† original question, written from lecture content (not a verified past-exam item)

113. Which statement about the development of the cornea is correct?

- A. The corneal epithelium is from surface ectoderm; the stroma and endothelium are largely from neural crest-derived mesenchyme
- B. The entire cornea is from neuroectoderm
- C. The entire cornea is from mesoderm only
- D. The corneal endothelium is from surface ectoderm
- E. The cornea has no distinct embryonic layers

† original question, written from lecture content (not a verified past-exam item)

114. Which structure gives rise to the ciliary body and its ciliary muscle, important for lens accommodation?

- A. Surface ectoderm exclusively
- B. The rim of the optic cup (neuroectoderm) for the epithelium, with associated mesenchyme for the ciliary muscle
- C. Endoderm
- D. The lens vesicle directly
- E. Notochord

† original question, written from lecture content (not a verified past-exam item)

115. Aniridia (congenital absence of the iris) is most directly linked to mutations affecting which category of developmental genes?

- A. PAX6, a master regulatory transcription factor for eye development
- B. Genes with no known role in eye development
- C. Only genes affecting the lens
- D. Only genes affecting the cornea
- E. Only genes affecting extraocular muscles

† original question, written from lecture content (not a verified past-exam item)

116. Congenital rubella syndrome can cause cataracts, sensorineural deafness, and cardiac defects together. What does this triad best illustrate about teratogen exposure timing?

- A. A single teratogen exposure during a critical shared developmental window can simultaneously disrupt multiple organ systems that are undergoing critical morphogenesis at that same time
- B. These three findings are always caused by three completely unrelated, coincidental processes
- C. Rubella only ever affects the eye
- D. Teratogen timing is irrelevant to which organs are affected
- E. This triad has no known common explanation

† original question, written from lecture content (not a verified past-exam item)

Lecture 7 — Spinal Cord: Blood Supply & Clinical Correlation (18 questions)

117. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which finding is expected on the SAME (right) side at and below L2?

- A. Loss of pain and temperature
- B. Loss of conscious proprioception and vibration, plus spastic paralysis
- C. Complete anesthesia
- D. Isolated loss of pain only
- E. No deficit on the ipsilateral side

118. A 65-year-old man has lost vibration, position sense, and joint discrimination in the legs, with preserved pain and temperature sensation. Which tract/pathway is most likely damaged?

- A. Lateral spinothalamic tract
- B. Dorsal column-medial lemniscus pathway
- C. Anterior spinothalamic tract
- D. Corticospinal tract only
- E. Spinocerebellar tract

119. Nuchal rigidity (stiff neck) in a patient is a compensatory sign classically seen with which condition?

- A. Simple tension headache
- B. Meningeal irritation (e.g., meningitis or subarachnoid hemorrhage)
- C. Peripheral neuropathy
- D. Migraine without aura
- E. Bell's palsy

120. An injury damages the LATERAL HORN of the spinal cord gray matter (present at T1-L2/L3). Which function is most directly lost?

- A. Somatic motor function to skeletal muscle
- B. Preganglionic sympathetic output
- C. Dorsal column sensation
- D. Spinothalamic sensation
- E. Cerebellar coordination

121. All of the following statements concerning the anterior spinal artery are correct EXCEPT:

- A. It supplies the anterior two-thirds of the spinal cord cross-section
- B. It arises from paired branches of the vertebral arteries that fuse into a single midline vessel
- C. Its territory includes the corticospinal and spinothalamic tracts
- D. Occlusion spares the dorsal columns
- E. It is reinforced only at the cervical level, with no other reinforcing radicular arteries lower down

122. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is characteristically SPARED?

- A. Pain
- B. Temperature
- C. Vibration and proprioception (dorsal column)
- D. Crude touch
- E. Motor function

123. Which of the following is NOT part of the central nervous system?

- A. Spinal cord
- B. Cerebellum
- C. Dorsal root ganglion
- D. Brainstem
- E. Cerebral cortex

124. If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?

- A. Corticospinal tract function
- B. Pain and temperature sensation
- C. Dorsal column (vibration/proprioception) function, supplied by the posterior spinal arteries instead
- D. Anterior horn motor neurons
- E. Lateral spinothalamic tract

125. Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?

- A. Loss of pain sensation
- B. Spastic paralysis (initially flaccid in spinal shock, then spastic) below the lesion
- C. Loss of vibration sense
- D. Autonomic dysreflexia exclusively
- E. Loss of smell

126. Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating dorsal (posterior) from ventral (anterior) spinal nerve roots?

- A. Denticulate ligament
- B. Filum terminale
- C. Ligamentum flavum
- D. Arachnoid trabeculae
- E. Falx cerebri

127. A patient with posterior spinal artery territory infarction would be expected to lose which function, with relative sparing of motor and pain/temperature?

- A. Vibration and proprioception (dorsal columns)
- B. Motor strength
- C. Pain sensation
- D. Temperature sensation
- E. Bladder control exclusively

128. A patient with a complete spinal cord lesion at C4 would be expected to have which respiratory consequence?

- A. No respiratory effect at all, since breathing is entirely brainstem-controlled
- B. Potential diaphragmatic paralysis / respiratory failure, since the phrenic nerve (C3-C5) supplying the diaphragm would be disconnected from higher control
- C. Only mild shortness of breath with normal diaphragm function
- D. Hyperventilation with no risk of failure
- E. Isolated loss of smell affecting breathing pattern

† original question, written from lecture content (not a verified past-exam item)

129. Which spinal cord region is most vulnerable to ischemia because it lies in a watershed zone between the anterior spinal artery and the posterior spinal arteries/segmental radicular supply?

- A. The cervical enlargement, richly supplied by multiple vertebral artery branches
- B. The mid-thoracic cord (roughly T4-T8), which has relatively sparse radicular artery reinforcement
- C. The lumbar enlargement, always well perfused
- D. The conus medullaris exclusively, never at risk
- E. There is no watershed region in the spinal cord

† original question, written from lecture content (not a verified past-exam item)

130. Which combination of findings would distinguish a central cord syndrome from a complete cord transection?

- A. Central cord syndrome causes disproportionately greater weakness in the arms than the legs, with variable sensory loss, sparing some function; complete transection causes total loss of everything below the level
- B. They are clinically identical and cannot be distinguished
- C. Central cord syndrome always spares all motor function
- D. Complete transection always spares bladder function
- E. Central cord syndrome only affects the legs

† original question, written from lecture content (not a verified past-exam item)

131. A 25-year-old man sustains a traumatic complete transection of the spinal cord at T6. In the IMMEDIATE hours after injury, which finding is expected below the lesion?

- A. Spastic paralysis with hyperreflexia immediately
- B. Flaccid paralysis with absent reflexes (spinal shock)
- C. Normal strength with only sensory loss
- D. Isolated loss of bladder control with normal motor and sensory function
- E. Hyperreflexia with clonus immediately

† original question, written from lecture content (not a verified past-exam item)

132. Which spinal cord segment level is most classically associated with the anal wink reflex (afferent and efferent both via the pudendal nerve)?

- A. L1-L2
- B. S2-S4
- C. T10-T12
- D. C5-C6
- E. S1 only, never S2-S4

† original question, written from lecture content (not a verified past-exam item)

133. Which vascular territory watershed makes the conus medullaris and cauda equina especially vulnerable during aortic surgery involving cross-clamping?

- A. Anterior spinal artery, especially where reinforcement from the artery of Adamkiewicz is compromised
- B. Middle cerebral artery
- C. Posterior cerebral artery
- D. Basilar artery
- E. Ophthalmic artery

† original question, written from lecture content (not a verified past-exam item)

134. Which spinal level marks the typical inferior extent of the adult spinal cord (conus medullaris), below which lumbar puncture is safely performed?

- A. L1-L2 in adults
- B. S1-S2
- C. T12 only
- D. C7-T1
- E. The cord extends to the sacral hiatus in adults

† original question, written from lecture content (not a verified past-exam item)

Lecture 8 — PNS: Cranial & Spinal Nerve Organization (18 questions)

135. Which functional component is found in the hypoglossal nucleus?

- A. GSA
- B. SVA
- C. GVA
- D. SVE
- E. GSE

136. According to the organization of the PNS, which statement below is INCORRECT?

- A. Spinal nerves are formed by the union of anterior and posterior nerve roots
- B. The trigeminal ganglion contains pseudounipolar sensory neurons for face/oral cavity sensation
- C. There is only one functional component in a peripheral nerve
- D. Secretomotor functions from the brainstem are transmitted via CN III, VII, IX, X
- E. There are 12 pairs of cranial nerves, and only CN III-XII emerge from the brainstem

137. Collections of nerve cell bodies located inside the peripheral nervous system are called:

- A. Tracts
- B. Nerve fibers
- C. Ganglia
- D. Nerves
- E. Nuclei

138. Which functional component is NOT found in the facial nerve (CN VII)?

- A. SVE (motor to muscles of facial expression)
- B. GVE (parasympathetic to lacrimal/salivary glands)
- C. SVA (taste, anterior 2/3 tongue)
- D. GSA (general sensation from a small area of the external ear)
- E. GVA (general visceral afferent from thoracic/abdominal viscera)

139. Which spinal nerves are NOT included in a somatic nerve plexus?

- A. C1-C4 (cervical plexus)
- B. T2-T12 (thoracic nerves, which mostly run as individual intercostal nerves)
- C. C5-T1 (brachial plexus)
- D. L1-S4 (lumbosacral plexus)
- E. All spinal nerves form a plexus

140. A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which nerve is most likely involved?

- A. Facial nerve (Bell's palsy)
- B. Trigeminal nerve (trigeminal neuralgia)
- C. Glossopharyngeal nerve
- D. Vagus nerve
- E. Accessory nerve

141. Which functional component of nerve fibers is NOT found in a typical spinal nerve?

- A. GSA
- B. GSE
- C. GVA
- D. GVE
- E. SVE

- 142.** Which nucleus contains general visceral efferent (GVE) preganglionic parasympathetic neurons supplying the salivary and lacrimal glands?
- A. Nucleus ambiguus
 - B. Superior and inferior salivatory nuclei
 - C. Hypoglossal nucleus
 - D. Edinger-Westphal nucleus only
 - E. Dorsal motor nucleus of vagus only
- 143.** Which statement about a transverse section of the caudal pons is correct?
- A. It contains the abducens (VI) and facial (VII) nerve nuclei/fascicles, plus the facial colliculus on the 4th ventricle floor
 - B. It contains the hypoglossal nucleus
 - C. It contains the substantia nigra
 - D. It contains the red nucleus
 - E. It has no cranial nerve nuclei at all
- 144.** A patient cannot detect light with the optic disc (physiologic blind spot). What is the anatomical reason?
- A. The optic disc is where retinal blood vessels and the optic nerve exit, and it contains no photoreceptors
 - B. The optic disc has the highest photoreceptor density in the retina
 - C. The optic disc is part of the sclera
 - D. The optic disc is where the fovea is located
 - E. The optic disc is covered by the iris
- 145.** Which dermatome landmark is listed INCORRECTLY?
- A. T4 - nipple line
 - B. T10 - umbilicus
 - C. L1 - inguinal ligament / groin
 - D. C6 - thumb
 - E. S1 - forehead
- 146.** Which nerve fiber type is responsible for increased salivation and constriction of the urinary sphincter... wait, which fiber type mediates PARASYMPATHETIC responses like increased salivation, and which mediates SYMPATHETIC responses like urinary sphincter contraction?
- A. Both are mediated by the same GVE fiber type with no distinction
 - B. Increased salivation = parasympathetic GVE (cranial outflow); urinary sphincter contraction = sympathetic GVE (thoracolumbar outflow)
 - C. Both are GSE fibers
 - D. Both are GSA fibers
 - E. Increased salivation is sympathetic; sphincter contraction is parasympathetic
- 147.** A lesion of the vagus nerve (CN X) on one side would be expected to cause which finding on indirect laryngoscopy/palate exam?
- A. Uvula deviates toward the side of the lesion
 - B. Uvula deviates AWAY from the side of the lesion (toward the intact side), because the weak side cannot elevate the palate
 - C. No change in uvula position ever occurs with vagus lesions
 - D. Tongue deviates toward the lesion
 - E. Only hearing loss occurs
- † original question, written from lecture content (not a verified past-exam item)*

148. Which structure, if compressed by a mass at the cerebellopontine angle, would classically cause combined sensorineural hearing loss AND facial weakness on the same side?

- A. Isolated CN VIII compression only
- B. A mass affecting both CN VII and CN VIII together as they run through the internal acoustic meatus (e.g., vestibular schwannoma / acoustic neuroma)
- C. CN V compression only
- D. CN XII compression
- E. Optic chiasm compression

† original question, written from lecture content (not a verified past-exam item)

149. A patient shows loss of the corneal reflex and facial sensory loss, but normal facial MOVEMENT (motor function intact). Which nerve is most likely affected?

- A. Facial nerve (CN VII) motor branch
- B. Trigeminal nerve (CN V), sensory (ophthalmic) division
- C. Glossopharyngeal nerve (CN IX)
- D. Vagus nerve (CN X)
- E. Accessory nerve (CN XI)

† original question, written from lecture content (not a verified past-exam item)

150. Which of the following is the correct composition of a typical mixed spinal nerve regarding its roots?

- A. Dorsal root (sensory, GSA+GVA) + Ventral root (motor, GSE+GVE) unite to form the mixed spinal nerve
- B. Only a single unified root with no dorsal/ventral distinction
- C. Dorsal root is motor, ventral root is sensory (reversed)
- D. Spinal nerves have three roots, not two
- E. Spinal nerves have no roots at all

† original question, written from lecture content (not a verified past-exam item)

151. Which cranial nerve provides both taste to the anterior 2/3 of the tongue AND general sensation to a small area of the external ear?

- A. Glossopharyngeal nerve (CN IX)
- B. Vagus nerve (CN X)
- C. Facial nerve (CN VII)
- D. Trigeminal nerve (CN V)
- E. Hypoglossal nerve (CN XII)

† original question, written from lecture content (not a verified past-exam item)

152. Damage to the recurrent laryngeal nerve (a branch of the vagus) would be expected to cause which finding?

- A. Hoarseness/vocal cord paralysis, due to loss of motor innervation to most intrinsic laryngeal muscles
- B. Complete loss of taste
- C. Facial paralysis
- D. Loss of pupillary light reflex
- E. Loss of hearing

† original question, written from lecture content (not a verified past-exam item)

Lecture 9 — Brainstem I: Internal Structures & Localization (17 questions)

153. An MRI shows a tumor compressing the right tuberculum cinereum at the closed (caudal) medulla. Which nerve's nucleus is most directly compressed?

- A. Trigeminal nerve (spinal trigeminal nucleus/tract)
- B. Facial nerve
- C. Glossopharyngeal nerve
- D. Vagus nerve
- E. Accessory nerve

154. Which reflex test best represents intact function of the medulla oblongata?

- A. Gag reflex
- B. Knee-jerk reflex
- C. Babinski reflex
- D. Pupillary light reflex
- E. Corneal reflex

155. A 20-year-old man has loss of muscle power on the LEFT side of his body plus strabismus and ptosis of the RIGHT eye. Where is the lesion?

- A. Left cerebral cortex
- B. Right pons
- C. Rostral midbrain, right side (crossed hemiplegia — Weber syndrome)
- D. Left cerebellum
- E. Right medulla

156. Which neuronal pathway performs voluntary control of body and limb movements?

- A. Reticulospinal tract
- B. Corticopontine tract
- C. Posterior column-medial lemniscus pathway
- D. Hypothalamospinal tract
- E. Lateral corticospinal (pyramidal) tract

157. Which CNS structure lies at approximately the same horizontal (axial) level as the tentorial notch/incisura, useful for localizing herniation syndromes?

- A. Medulla oblongata
- B. Midbrain
- C. Pons
- D. Cervical spinal cord
- E. Cerebellar tonsils

158. A 32-year-old man has loss of muscle power and a positive Babinski sign on the LEFT side, with right-sided tongue deviation on protrusion. Where is the lesion?

- A. Left cerebral cortex
- B. Right medial medulla (medial medullary syndrome — affecting CN XII and the pyramid/medial lemniscus)
- C. Left cerebellum
- D. Right pons
- E. Left midbrain

159. A patient has a left pupil larger than the right, plus ptosis and an eye deviated 'down and out.' Which structure is most likely compressed?

- A. Left optic nerve
- B. Left oculomotor nerve (CN III), e.g., from uncal herniation
- C. Right abducens nerve
- D. Left trochlear nerve only
- E. Left trigeminal nerve

160. A patient has loss of pain and temperature sensation on the CONTRALATERAL body plus an IPSILATERAL facial pain/temperature deficit, plus ipsilateral ataxia, Horner's syndrome, and hoarseness. This constellation is:

- A. Weber syndrome
- B. Lateral medullary (Wallenberg) syndrome — PICA territory
- C. Medial medullary syndrome
- D. Locked-in syndrome
- E. Benedikt syndrome

161. Which tract's second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the spinal cord)?

- A. Lateral spinothalamic tract (crosses within 1-2 spinal segments)
- B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)
- C. Anterior spinothalamic tract
- D. Corticospinal tract (crosses at the medulla too, but as a first-order/UMN axon, not a sensory 2nd-order neuron)
- E. None of the sensory pathways cross in the medulla

162. Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?

- A. Cerebellum
- B. Spinal cord below C4
- C. Peripheral nerves
- D. Pons and midbrain (rostral brainstem reticular formation)
- E. Basal ganglia exclusively

163. What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?

- A. Cerebral cortex alone, unilaterally
- B. Pons and midbrain reticular formation
- C. Cerebellar vermis
- D. Spinal cord
- E. Peripheral autonomic ganglia

164. A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical picture most directly suggests dysfunction at which level?

- A. Peripheral nerve only
- B. Brainstem (reticular formation) dysfunction, possibly from herniation or direct injury
- C. Isolated muscle disease
- D. Isolated dorsal root ganglion pathology
- E. Isolated retinal disease

165. Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?

- A. CN III, IV
- B. CN V, VI, VII, VIII
- C. CN IX, X, XI, XII
- D. CN I, II
- E. None; the pons has no cranial nerve nuclei

† original question, written from lecture content (not a verified past-exam item)

166. Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye movement, but quadriplegia and inability to speak)?

- A. Bilateral ventral pons (corticospinal and corticobulbar tracts), sparing the tegmentum/reticular formation
- B. Bilateral cerebral cortex
- C. Bilateral cerebellum
- D. Bilateral thalamus
- E. Bilateral occipital lobe

† original question, written from lecture content (not a verified past-exam item)

167. Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?

- A. The MLF connects the abducens nucleus on one side to the contralateral oculomotor (medial rectus) subnucleus, coordinating conjugate horizontal gaze
- B. The MLF is a purely sensory tract with no motor function
- C. The MLF controls only vertical gaze
- D. The MLF has no connection to cranial nerve nuclei
- E. The MLF is located in the spinal cord, unrelated to eye movement

† original question, written from lecture content (not a verified past-exam item)

168. Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather than some other eye movement?

- A. Because the MLF specifically carries the signal from the contralateral abducens nucleus to the ipsilateral oculomotor nucleus's medial rectus subnucleus, which is what drives eye adduction during conjugate horizontal gaze
- B. Because the MLF has no relationship to any specific eye muscle
- C. Because the MLF controls only vertical eye movements
- D. Because the MLF is part of the visual (not motor) pathway
- E. Because adduction is controlled by the trochlear nerve, unrelated to the MLF

† original question, written from lecture content (not a verified past-exam item)

169. Which best explains why lesions affecting the pyramids (corticospinal tract) at the level of the medulla produce contralateral limb weakness, while a lesion of the same tract just above the pyramidal decussation would also eventually cause contralateral weakness?

- A. Because the pyramidal decussation is where corticospinal fibers cross from one side to the other; a lesion at ANY point along this pathway (above or at the decussation, but before it has crossed) will still ultimately produce weakness on the side OPPOSITE the lesion, once you account for where the crossing occurs relative to the lesion
- B. Corticospinal fibers never cross at all
- C. The decussation has no bearing on lateralization
- D. Lesions above and below the decussation always produce identical, ipsilateral effects
- E. Only spinal cord lesions can be lateralized, brainstem lesions cannot

† original question, written from lecture content (not a verified past-exam item)

Lecture 10 — Reticular Formation & ARAS (17 questions)

170. Which of the following correctly pairs a reticular formation column with its function?

- A. Median column (raphe nuclei) — ascending projections for arousal and attention
- B. Lateral column — controls eye movements only
- C. Medial column — has no function
- D. All columns have identical functions
- E. Median column — controls only descending spinal reflexes

171. Fibers from which reticular formation structure synapse with the motor nuclei of CN V and CN VII to help mediate brainstem reflexes like the jaw jerk and blink?

- A. Median column (raphe nuclei)
- B. Medial (central) column, via short intrinsic interneuronal connections
- C. Lateral column exclusively
- D. Locus coeruleus only
- E. Substantia nigra

172. Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?

- A. Thalamus (intralaminar nuclei)
- B. Basal forebrain
- C. Hypothalamus
- D. Cerebral cortex, diffusely (via thalamic relay)
- E. The cerebellar deep nuclei as its primary/defining target

173. What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla to the midbrain?

- A. Reticular formation
- B. Corticospinal tract
- C. Medial lemniscus
- D. Cerebellar peduncle
- E. Corpus callosum

† original question, written from lecture content (not a verified past-exam item)

174. The reticular formation contributes to autonomic regulation of which vital functions via its lateral (descending) column?

- A. Respiration and blood pressure (cardiovascular) control
- B. Vision only
- C. Hearing only
- D. Taste only
- E. Smell only

† original question, written from lecture content (not a verified past-exam item)

175. Damage to serotonergic raphe nuclei neurons has been implicated in which category of clinical conditions?

- A. Mood disorders (e.g., depression) and sleep-wake regulation abnormalities
- B. Isolated hearing loss
- C. Isolated visual field defects
- D. Isolated loss of smell
- E. Isolated motor weakness only

† original question, written from lecture content (not a verified past-exam item)

176. Which reticular formation nucleus group is the principal source of noradrenergic (norepinephrine) projections implicated in arousal and the fight-or-flight-associated attentional response?

- A. Raphe nuclei
- B. Locus coeruleus
- C. Substantia nigra
- D. Red nucleus
- E. Nucleus ambiguus

† original question, written from lecture content (not a verified past-exam item)

177. Why is the reticular formation considered anatomically and functionally distinct from discrete, well-bounded nuclei like the facial motor nucleus?

- A. It consists of a diffuse, interconnected network of neurons with widespread dendritic/axonal arborization rather than a tight, well-circumscribed nuclear boundary
- B. It has no neurons at all, only glial cells
- C. It is located entirely outside the brainstem
- D. It has only a single, discrete cell body cluster
- E. It has no known function

† original question, written from lecture content (not a verified past-exam item)

178. Which best describes the circadian rhythm-related function attributed in part to reticular formation and related brainstem/hypothalamic circuits?

- A. Regulating the sleep-wake cycle in coordination with the suprachiasmatic nucleus
- B. Regulating heart valve function directly
- C. Regulating bone growth
- D. Regulating red blood cell production
- E. No connection to circadian rhythm exists

† original question, written from lecture content (not a verified past-exam item)

179. A patient with a lesion selectively damaging the lateral reticular formation column (sparing the median and medial columns) would most likely show which combination of deficits?

- A. Impaired arousal only, with normal autonomic function
- B. Disrupted descending control of muscle tone, respiration, and blood pressure, with relatively preserved arousal and intrinsic brainstem reflexes
- C. Isolated loss of taste
- D. Isolated loss of smell
- E. Isolated loss of hearing

† original question, written from lecture content (not a verified past-exam item)

180. Which of the following would be the MOST specific bedside test to assess pontomesencephalic (upper brainstem) ARAS integrity in a comatose patient?

- A. Deep tendon reflexes in the legs
- B. Pupillary light reflex and oculocephalic (dolls eyes) / oculovestibular (caloric) reflexes
- C. Babinski sign alone
- D. Blood pressure measurement alone
- E. Grip strength testing

† original question, written from lecture content (not a verified past-exam item)

181. Which neurotransmitter system, originating largely from the reticular formation, is most associated with the transition FROM NREM sleep TO wakefulness?

- A. Increased norepinephrine (locus coeruleus) and serotonin (raphe), decreased acetylcholine
- B. Decreased norepinephrine and serotonin, increased acetylcholine (this describes the shift INTO REM, not wakefulness)
- C. No neurotransmitter changes occur across sleep-wake states
- D. Only dopamine changes matter
- E. Only histamine changes matter

† original question, written from lecture content (not a verified past-exam item)

182. The reticular formation's role in modulating pain (e.g., via descending projections to the dorsal horn) is mediated in part through which descending pathway?

- A. Descending raphespinal (serotonergic) and reticulospinal (noradrenergic) pathways that can inhibit dorsal horn pain transmission
- B. The corticospinal tract exclusively
- C. The dorsal column pathway (an ascending, not descending, pathway)
- D. The optic nerve
- E. The vestibulocochlear nerve

† original question, written from lecture content (not a verified past-exam item)

183. Damage to the ARAS from a large bilateral thalamic stroke would most likely produce which clinical picture?

- A. Isolated unilateral limb weakness with full alertness
- B. Coma or severely depressed consciousness, since the thalamus is a critical relay for ARAS projections to the cortex
- C. Isolated loss of smell
- D. Isolated loss of taste
- E. No effect on consciousness whatsoever

† original question, written from lecture content (not a verified past-exam item)

184. Why is a UNILATERAL cortical stroke, however large, generally insufficient to cause coma on its own (unless secondary herniation occurs)?

- A. Because the ARAS provides bilateral, diffuse projections to both hemispheres, so the intact hemisphere and brainstem can still maintain arousal
- B. Because the cortex has no role in consciousness at all
- C. Because unilateral strokes never affect brain function
- D. Because the brainstem is always destroyed in any stroke
- E. Because coma requires spinal cord involvement

† original question, written from lecture content (not a verified past-exam item)

185. Which best distinguishes the concepts of 'arousal' (wakefulness) from 'awareness' (content of consciousness), both relevant to reticular formation function?

- A. Arousal and awareness are the same thing with no distinction
- B. Arousal (eyes-open wakefulness, ARAS-dependent) can be present without full awareness (as in a vegetative state), showing these are dissociable aspects of consciousness
- C. Awareness always requires eyes to be open
- D. Arousal is controlled only by the cerebellum
- E. Awareness is controlled only by the spinal cord

† original question, written from lecture content (not a verified past-exam item)

186. Which best explains why damage to the reticular formation can simultaneously threaten both consciousness AND vital signs (breathing, blood pressure), unlike damage to most other brain regions?

- A. Because the reticular formation's different functional columns (arousal, autonomic, motor-reflex) are anatomically intermingled within the same compact brainstem core, so injury severe enough to affect one system often affects the others too
- B. Because consciousness and vital signs have no anatomical relationship anywhere in the brain
- C. Because the reticular formation controls only vision
- D. Because vital sign control is located exclusively in the cerebral cortex
- E. Because arousal and autonomic control are handled by completely separate, distant brain regions with no overlap

† original question, written from lecture content (not a verified past-exam item)

Lecture 11 — Special Sense Organs: Histology & Development (17 questions)

187. The inner ear (membranous labyrinth) develops embryologically from which structure?

- A. First pharyngeal cleft
- B. Otic placode
- C. First pharyngeal pouch
- D. Neural crest cells
- E. Surface ectoderm overlying the eye

† original question, written from lecture content (not a verified past-exam item)

188. The external auditory meatus develops from which embryonic structure?

- A. First pharyngeal pouch
- B. First pharyngeal cleft
- C. Otic vesicle
- D. Second pharyngeal arch
- E. First pharyngeal arch mesenchyme

† original question, written from lecture content (not a verified past-exam item)

189. The malleus and incus (middle ear ossicles) are derived embryologically from which cartilage?

- A. Reichert's cartilage (2nd arch)
- B. Meckel's cartilage (1st arch)
- C. Third arch cartilage
- D. Otic capsule cartilage
- E. Laryngeal cartilage

† original question, written from lecture content (not a verified past-exam item)

190. Which cell type in the olfactory epithelium serves as the stem cell population for continuous regeneration of olfactory receptor neurons?

- A. Sustentacular cells
- B. Bowman's gland cells
- C. Basal cells
- D. Olfactory receptor neurons themselves
- E. Schwann cells

† original question, written from lecture content (not a verified past-exam item)

191. The nasal placodes are induced by signals from which adjacent embryonic structure?

- A. The developing forebrain (telencephalon)
- B. The developing heart
- C. The otic vesicle
- D. The pharyngeal pouches
- E. The notochord

† original question, written from lecture content (not a verified past-exam item)

192. Which taste bud cell type is directly responsible for chemoreceptive taste transduction?

- A. Basal (Type IV) cells
- B. Dark cells
- C. Gustatory (taste receptor) cells
- D. Supporting cells only
- E. Merkel cells

† original question, written from lecture content (not a verified past-exam item)

193. Which special sense epithelium is uniquely capable of continuous neurogenesis (producing new sensory NEURONS, not just supporting cells) throughout adult life?

- A. Cochlea (organ of Corti)
- B. Retina
- C. Olfactory epithelium
- D. Vestibular maculae
- E. Taste buds

† original question, written from lecture content (not a verified past-exam item)

194. The cochlear duct (scala media) contains which fluid, and how does its ionic composition differ from the adjacent scalae?

- A. Perilymph, similar in composition to CSF
- B. Endolymph, which is unusually high in K⁺ (more like intracellular fluid)
- C. Blood plasma directly
- D. Pure water with no ions
- E. The same fluid as the scala vestibuli and scala tympani

† original question, written from lecture content (not a verified past-exam item)

195. Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?

- A. Geniculate ganglion
- B. Spiral ganglion (within the modiolus of the cochlea)
- C. Vestibular ganglion
- D. Petrosal ganglion
- E. Dorsal root ganglion

† original question, written from lecture content (not a verified past-exam item)

196. Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair cells?

- A. About 5%
- B. About 50%
- C. About 95%
- D. 0%, they have no afferent innervation
- E. Exactly equal to outer hair cells

† original question, written from lecture content (not a verified past-exam item)

197. Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?

- A. They are embedded in an overlying membrane (e.g., tectorial membrane in the cochlea) and their deflection relative to the kinocilium determines depolarization vs. hyperpolarization
- B. They have no relationship to any overlying structure
- C. They are located inside the cell body, not on the apical surface
- D. They always cause hyperpolarization regardless of direction
- E. Kinocilium is absent in all mature mammalian hair cells

† original question, written from lecture content (not a verified past-exam item)

198. The three semicircular canals detect which type of head movement?

- A. Linear acceleration only (e.g., gravity, forward/backward motion)
- B. Angular (rotational) acceleration, each canal oriented in a different plane
- C. Sound frequency only
- D. Static head position only, with no dynamic component
- E. Taste changes

† original question, written from lecture content (not a verified past-exam item)

199. Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?

- A. The maculae of the utricle/saccule have an otolithic membrane embedded with calcium carbonate crystals (otoconia), adding inertial mass; the ampullary crista instead has a gelatinous cupula with no crystals
- B. They have identical histology with no differences
- C. Only the ampullae contain hair cells
- D. Only the maculae contain hair cells
- E. The utricle has no sensory epithelium at all

† original question, written from lecture content (not a verified past-exam item)

200. Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to different frequencies (tonotopic organization)?

- A. Tectorial membrane
- B. Basilar membrane
- C. Reissner's membrane
- D. Stria vascularis
- E. Round window membrane

† original question, written from lecture content (not a verified past-exam item)

201. Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?

- A. Low frequencies first, affecting the cochlear apex
- B. High frequencies first, affecting outer hair cells at the cochlear base
- C. All frequencies equally, with no particular pattern
- D. Only affects middle ear ossicles, not the cochlea
- E. Only affects the vestibular system, not hearing

† original question, written from lecture content (not a verified past-exam item)

202. Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching between air and cochlear fluid?

- A. The three ossicles (malleus, incus, stapes) acting as a lever system
- B. The Eustachian tube
- C. The round window
- D. The semicircular canals
- E. The vestibule alone

† original question, written from lecture content (not a verified past-exam item)

203. The Eustachian (auditory) tube connects the middle ear to which structure, and what is its main physiological function?

- A. The nasopharynx; equalizing middle ear pressure with atmospheric pressure
- B. The oral cavity directly; food digestion
- C. The inner ear; endolymph production
- D. The brain directly; CSF drainage
- E. The eye; tear drainage

† original question, written from lecture content (not a verified past-exam item)

Lecture 12 — Somatosensory Pathways (17 questions)

204. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which neuronal tract/fasciculus passes through the anterior white commissure at the level of decussation?

- A. Corticospinal tract
- B. Dorsal columns
- C. Lateral spinothalamic tract, as its second-order fibers cross to the opposite side
- D. Spinocerebellar tract
- E. Rubrospinal tract

205. A patient has loss of proprioception on the LEFT side and loss of pain/temperature on the RIGHT side of the body below a spinal lesion. This pattern is most consistent with:

- A. Complete cord transection
- B. Left-sided Brown-Séquard hemisection (ipsilateral dorsal column loss, contralateral spinothalamic loss)
- C. Anterior spinal artery syndrome
- D. Posterior spinal artery syndrome
- E. Peripheral neuropathy

206. Which tract lesion classically gives a positive Romberg test (patient sways/falls with eyes closed but is stable with eyes open)?

- A. Corticospinal tract
- B. Dorsal column-medial lemniscus pathway (proprioception loss compensated visually)
- C. Lateral spinothalamic tract
- D. Cerebellar vermis (this instead causes a Romberg-NEGATIVE, i.e., always-unsteady, truncal ataxia)
- E. Optic nerve

207. Which of the following conditions primarily affects the PNS rather than the CNS?

- A. Multiple sclerosis
- B. Guillain-Barré syndrome
- C. Stroke (ischemic)
- D. Parkinson's disease
- E. Amyotrophic lateral sclerosis (upper motor neuron component)

208. A patient has right-sided pain and temperature loss below T12, and left-sided pain and temperature loss below T10 (staggered levels). What does this pattern suggest about the lesion?

- A. A single, precise transverse hemisection at exactly one level
- B. An intramedullary lesion (e.g., syrinx) expanding asymmetrically, or two separate/expanding lesions affecting decussating spinothalamic fibers at slightly different levels on each side
- C. A pure peripheral nerve lesion
- D. Normal variant anatomy with no pathology
- E. A pure dorsal column lesion

209. Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar tract) toward the cerebellum, synapsing first at Clarke's nucleus (C8-L2/3)?

- A. Dorsal root ganglion neurons directly to the cerebellum, with no relay
- B. First-order neuron → Clarke's nucleus (second-order neuron) → dorsal spinocerebellar tract → cerebellum
- C. Thalamus → cerebellum
- D. Directly via the corticospinal tract
- E. Via the spinothalamic tract

210. What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's nucleus does not extend this far caudally?

- A. Direct dorsal column ascent with no relay
- B. Sacral proprioceptive afferents ascend and effectively synapse in the region of Clarke's nucleus equivalent higher up, or use the ventral spinocerebellar pathway
- C. They have no route to the cerebellum at all
- D. They travel exclusively via the spinothalamic tract
- E. They bypass the spinal cord entirely

211. Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive spinocerebellar pathways?

- A. Lateral spinothalamic tract
- B. Dorsal column-medial lemniscus (DCML) pathway
- C. Dorsal spinocerebellar tract
- D. Ventral spinocerebellar tract
- E. Anterior spinothalamic tract

† original question, written from lecture content (not a verified past-exam item)

212. A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to cause which finding?

- A. Loss of pain sensation with normal proprioception
- B. Sensory ataxia with a positive Romberg sign, loss of vibration and proprioception, normal pain/temperature
- C. Isolated motor weakness with no sensory loss
- D. Isolated loss of hearing
- E. Isolated Horner syndrome

† original question, written from lecture content (not a verified past-exam item)

213. Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the most medial part of the dorsal columns?

- A. Fasciculus cuneatus (upper limb)
- B. Fasciculus gracilis (lower limb, medial dorsal column)
- C. Lateral spinothalamic tract
- D. Anterior spinothalamic tract
- E. Rubrospinal tract

† original question, written from lecture content (not a verified past-exam item)

214. A patient has a central spinal cord lesion (e.g., early syringomyelia) that selectively damages decussating spinothalamic fibers at the cervical level, sparing the dorsal columns and corticospinal tracts. What is the expected sensory pattern?

- A. Bilateral, symmetric cape-like loss of pain and temperature sensation over the shoulders/arms, with preserved proprioception/vibration and motor strength
- B. Complete loss of all sensory and motor function below the lesion
- C. Isolated unilateral loss of proprioception only
- D. Normal sensory exam throughout
- E. Isolated loss of hearing

† original question, written from lecture content (not a verified past-exam item)

215. Which fiber type conducts the FASTEST among peripheral sensory afferents, corresponding to proprioceptive information from muscle spindles (Ia afferents)?

- A. C fibers (unmyelinated, slowest)
- B. A-delta fibers
- C. A-beta fibers
- D. A-alpha fibers (largest diameter, most heavily myelinated, fastest conducting)
- E. B fibers

† original question, written from lecture content (not a verified past-exam item)

216. Which best explains why a lesion of the dorsal columns produces a POSITIVE Romberg sign, while a cerebellar lesion produces a NEGATIVE Romberg sign (i.e., patient is unsteady with eyes both open and closed)?

- A. Dorsal column loss removes proprioceptive input that vision can otherwise substitute for; cerebellar ataxia is a movement-coordination problem that visual input cannot compensate for at all
- B. There is no real mechanistic difference between the two
- C. Cerebellar disease always improves with eyes closed
- D. Dorsal column disease has nothing to do with balance
- E. Vision has no role in either condition

† original question, written from lecture content (not a verified past-exam item)

217. Two-point discrimination (the ability to distinguish two separate touch points as distinct) is best in which body region, and why?

- A. The back, because of thick skin
- B. The fingertips/lips, because of high mechanoreceptor density and correspondingly larger cortical representation (small receptive fields)
- C. It is equal everywhere on the body
- D. The thigh, because of large surface area
- E. The sole of the foot exclusively

† original question, written from lecture content (not a verified past-exam item)

218. Which best describes the concept of a 'dermatome' and why overlapping dermatomes make it hard to detect a single-root sensory loss on exam?

- A. A dermatome is the skin area innervated by a single spinal nerve root; because adjacent dermatomes overlap significantly, a single nerve root lesion may produce only SUBTLE or no detectable sensory loss on exam
- B. Dermatomes never overlap, so any single root lesion is always obvious
- C. A dermatome refers to a muscle group, not a skin area
- D. Dermatomes are identical to peripheral nerve territories, which never overlap
- E. Dermatomes have no clinical relevance

† original question, written from lecture content (not a verified past-exam item)

219. Which best describes the difference between 'fast pain' and 'slow pain' in terms of BOTH fiber type AND the resulting behavioral/emotional response?

- A. Fast pain (A-delta) is sharp, well-localized, and prompts immediate withdrawal; slow pain (C fiber) is dull, poorly-localized, and drives a more prolonged, emotionally unpleasant, protective response
- B. There is no difference between fast and slow pain
- C. Fast pain is always worse than slow pain in every clinical scenario
- D. Slow pain is carried by A-alpha fibers
- E. Fast pain has no protective function

† original question, written from lecture content (not a verified past-exam item)

220. Which best summarizes why understanding the ANATOMICAL LEVEL at which a sensory tract crosses the midline is the single most useful tool for localizing spinal cord lesions?

- A. Because it lets you predict, for any given tract, whether a lesion at a specific spinal level will produce a deficit on the SAME side or OPPOSITE side of the body relative to the lesion, and matching that predicted pattern against the patients actual exam findings is how you pinpoint both the level AND the tract(s) involved
- B. Crossing level has no clinical usefulness whatsoever
- C. All spinal tracts cross at exactly the same level, so this information is redundant
- D. Only motor tracts have a crossing level worth knowing
- E. This information is only useful for the brainstem, never the spinal cord

† original question, written from lecture content (not a verified past-exam item)

Lecture 13 — Taste, Sensory Receptors & Neurotransmitters (17 questions)

221. The ganglia housing first-order taste neurons are located in which structures?

- A. Geniculate, Petrosal, Nodose ganglia (for CN VII, IX, X respectively)
- B. Superior, Ciliary, Otic ganglia
- C. Stellate, Geniculate, Nodose ganglia
- D. Trigeminal, Geniculate, Nodose ganglia
- E. Geniculate, Spiral, Pterygopalatine ganglia

222. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?

- A. Satellite cells
- B. Schwann cells
- C. Basal cells
- D. Stem keratinocytes
- E. Merkel cells

223. The gustatory (taste) pathway includes all of the following EXCEPT:

- A. Solitary nucleus (rostral/gustatory part)
- B. VPM nucleus of the thalamus
- C. Geniculate/petrosal/nodose ganglia
- D. Insula and frontal operculum (primary gustatory cortex)
- E. Trapezoid body

224. An unpleasant or distorted taste sensation is medically termed:

- A. Dysphagia
- B. Dysphonia
- C. Dysgeusia
- D. Dysarthria
- E. Dysmetria

225. When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?

- A. Hyperpolarization occurs
- B. K⁺ channels close permanently
- C. Depolarization occurs
- D. Ca²⁺ efflux increases
- E. Action potential frequency decreases

226. In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:

- A. Blocking Na⁺ channels presynaptically
- B. Opening presynaptic Ca²⁺ channels
- C. Inhibiting presynaptic Ca²⁺ influx and opening postsynaptic K⁺ channels, reducing substance P release and dampening pain transmission
- D. Stimulating NMDA receptors
- E. Increasing glutamate release

227. Which of the following correctly matches a neurotransmitter to its vesicular/reuptake transporter pair?

- A. Glutamate = VGLUT (vesicular) / EAAT (reuptake)
- B. GABA = VGLUT / EAAT
- C. Serotonin = VGAT / GAT
- D. Glutamate = VMAT / SERT
- E. GABA = VMAT / SERT

228. During long-term potentiation (LTP), after Ca²⁺ enters through NMDA receptors, which retrograde messenger diffuses back to the PRESynaptic terminal to help further increase glutamate release?

- A. Nitric oxide (NO)
- B. Acetylcholine
- C. Serotonin
- D. GABA
- E. Histamine

229. What is the mechanism of action of diazepam (a benzodiazepine) at the GABA-A receptor?

- A. It directly opens the Cl⁻ channel independent of GABA
- B. It increases the FREQUENCY of Cl⁻ channel opening, but only in the presence of GABA (allosteric potentiation)
- C. It blocks GABA from binding entirely
- D. It converts GABA-A receptors into GABA-B receptors
- E. It has no effect on chloride conductance

230. Which sensory receptor type is a rapidly-adapting mechanoreceptor responsible for detecting vibration and deep pressure, with a characteristic 'onion-layered' capsule?

- A. Meissner's corpuscle
- B. Pacinian corpuscle
- C. Merkel disc
- D. Ruffini ending
- E. Free nerve ending

† original question, written from lecture content (not a verified past-exam item)

231. Which type of thermoreceptor channel (TRP family) is activated by noxious heat and capsaicin (the 'hot' component of chili peppers)?

- A. TRPM8
- B. TRPV1
- C. TRPA1 exclusively
- D. TRPC1
- E. None of the TRP channels respond to temperature

† original question, written from lecture content (not a verified past-exam item)

232. Lateral inhibition in a sensory system serves what functional purpose?

- A. It sharpens spatial resolution/contrast by suppressing the response of less-stimulated neighboring afferents, enhancing the perceived edge/peak of a stimulus
- B. It amplifies all afferents equally with no sharpening effect
- C. It only occurs in the auditory system
- D. It eliminates all sensory input
- E. It has no role in perception

† original question, written from lecture content (not a verified past-exam item)

233. A 71-year-old man reports his face 'feels funny' on one side. Examination shows loss of light touch, temperature, and taste sensation in a pattern consistent with a lesion affecting multiple cranial nerve territories near the internal acoustic meatus/cerebellopontine angle. Which combination of nerves is most likely involved?

- A. CN II and III only
- B. CN V, VII, and VIII (given their close anatomical relationship at the cerebellopontine angle)
- C. CN IX and XII only
- D. CN I and II only
- E. CN X and XI only

234. Which taste modality is most closely associated with glutamate detection (via a specific metabotropic receptor), important in the perception of savory/meaty flavors?

- A. Sweet
- B. Umami (via T1R1/T1R3 receptor heterodimer, and related metabotropic glutamate receptors)
- C. Sour
- D. Bitter
- E. Salty

† original question, written from lecture content (not a verified past-exam item)

235. Which taste modality is transduced by direct H⁺ ion entry/detection, rather than a G-protein-coupled receptor mechanism?

- A. Sweet
- B. Umami
- C. Sour
- D. Bitter
- E. None; all tastes use GPCRs

† original question, written from lecture content (not a verified past-exam item)

236. A patient reports that food 'tastes like metal' and has reduced overall taste sensitivity after starting a new medication. Which mechanism is most likely responsible, given that zinc is a required cofactor for normal taste bud turnover?

- A. The medication is chelating/depleting zinc, impairing normal taste receptor cell regeneration/turnover (basal cell-dependent)
- B. The medication has directly destroyed the gustatory cortex
- C. The medication has cut CN X entirely
- D. This has no relationship to zinc or taste bud biology
- E. The medication has caused permanent CN I damage, unrelated to taste

† original question, written from lecture content (not a verified past-exam item)

237. Why can a strong, unpleasant taste (e.g., bitter) trigger a protective gag or spitting reflex even before conscious taste identification is complete?

- A. Because gustatory afferents also connect to brainstem reflex circuitry (via the solitary nucleus and its connections to motor nuclei) that can trigger protective reflexes rapidly, in parallel with the slower pathway to conscious cortical taste perception
- B. Because bitter taste has no connection to any reflex pathway
- C. Because taste and the gag reflex are entirely unrelated systems
- D. Because the gag reflex requires full cortical processing first, ruling out any fast pathway
- E. Because only visual input can trigger a gag reflex

† original question, written from lecture content (not a verified past-exam item)

Lecture 14 — Sensory Receptors, Neurotransmitters & Pharmacology (22 questions)

238. A woman presents with depressive symptoms and suicidal thoughts. Which neurotransmitter is classically implicated as abnormal in this case?

- A. Glutamate
- B. Acetylcholine
- C. Serotonin
- D. GABA
- E. Dopamine

239. What is the mechanism of action of botulinum toxin?

- A. Increased sodium channel activity at the presynaptic neuron
- B. Blocks acetylcholine receptor at the postsynaptic neuron
- C. Blocks presynaptic fusion of neurotransmitter vesicles (cleaves SNARE proteins)
- D. Blocks calcium channels at the presynaptic neuron
- E. Increased reabsorption of acetylcholine

240. An 80-year-old man has progressive short-term memory loss and word-finding difficulty, diagnosed with Alzheimer disease. Which statement about its pathogenesis is INCORRECT?

- A. Atrophy of gyri in the cerebral cortex
- B. Formation of amyloid plaques
- C. Formation of neurofibrillary tangles
- D. Increased activity of beta-secretase enzyme
- E. Blocking of dopamine reuptake transporters

241. A patient arrives with continuous generalized tonic-clonic seizures for 15 minutes (status epilepticus). What is the most appropriate FIRST-line treatment?

- A. Diazepam (benzodiazepine)
- B. Valproate
- C. Phenytoin
- D. Fluoxetine
- E. Memantine

242. The summation of EPSPs arriving from numerous different presynaptic fibers converging onto the same postsynaptic neuron at roughly the same moment is called:

- A. Spatial summation
- B. Temporal summation
- C. Presynaptic inhibition
- D. Long-term potentiation
- E. Presynaptic facilitation

243. A 20-year-old college student has depressive symptoms including sleep disturbance and anorexia for 1 month, with no prior mood disorder history. What is the mechanism of the drug most appropriate for this patient (an SSRI)?

- A. Stimulation of GABA receptors
- B. Inhibition of dopamine receptors
- C. Inhibition of the serotonin transporter (SERT), increasing synaptic serotonin availability
- D. Inhibition of acetylcholinesterase
- E. Stimulation of norepinephrine release

244. A woman presents with progressive muscle weakness, dysphagia, and ptosis that worsens with repeated blinking. Which finding is most likely present?

- A. Ryanodine receptor antibody
- B. Anti-presynaptic Ca^{2+} channel antibody (Lambert-Eaton)
- C. Anti-nicotinic acetylcholine receptor antibody (myasthenia gravis)
- D. Inhibition of dopamine receptors
- E. Stimulation of GABA receptors

245. A 45-year-old man has a deep gardening wound, then develops jaw stiffness (lockjaw) and muscle spasms. Which mechanism best explains the responsible toxin's action?

- A. Directly inhibits neurotransmitter release at the neuromuscular junction
- B. Autoantibody formation against nicotinic acetylcholine receptors
- C. Inhibits calcium channels at the postsynaptic membrane
- D. Blocks release of inhibitory neurotransmitters (glycine/GABA) from spinal inhibitory interneurons, causing disinhibited, excessive motor neuron firing (tetanus)

246. A chemotherapy patient has nausea and vomiting. Which antiemetic receptor target is most appropriate?

- A. 5-HT₃ receptor (e.g., ondansetron)
- B. Dopamine receptor
- C. Muscarinic receptor
- D. Nicotinic receptor
- E. Adrenergic receptor

247. Which of the following agents does NOT stimulate nociceptors (i.e., is not itself an inflammatory/pain-sensitizing mediator)?

- A. Prostaglandin
- B. K⁺
- C. Substance P
- D. Histamine
- E. Vibration

248. What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic membrane?

- A. Synaptic delay
- B. Synaptic fatigue

249. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?

- A. AMPA receptor
- B. mGluR6
- C. GABA-B receptor
- D. 5-HT₁ receptor

250. What is the mechanism of fluoxetine?

- A. Selective serotonin reuptake inhibitor (blocks SERT)
- B. Blocks D₂ dopamine receptor
- C. Blocks peripheral and central 5-HT₃ receptors

251. What is the mechanism of the antiemetic drug ondansetron?

- A. Selective serotonin reuptake inhibitor
- B. Blocks D₂ receptor
- C. Blocks peripheral and central 5-HT₃ receptors

252. What is the mechanism of haloperidol?

- A. Selective serotonin reuptake inhibitor
- B. Blocks D₂ dopamine receptor
- C. Blocks peripheral and central 5-HT₃ receptors

253. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?

- A. TRPV2 (very high heat threshold, extreme noxious heat)
- B. TRPV4
- C. TRPM8 (cold receptor — wrong category entirely)

254. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?

- A. Dopamine
- B. Glutamate
- C. Serotonin
- D. Acetylcholine
- E. Epinephrine

255. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry receptor for SARS-CoV-2?

- A. Dopamine D2 receptor
- B. Muscarinic acetylcholine receptor
- C. ACE2 (angiotensin-converting enzyme 2)
- D. 5-HT3 receptor
- E. NMDA receptor

256. Which neurotransmitter-transporter pairing is correct?

- A. VGAT - GABA (correct vesicular pairing)
- B. SERT - Dopamine
- C. DAT - Acetylcholine
- D. EAATs - Norepinephrine
- E. Glutaminase - Glutamine

257. A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound, potentially fatal respiratory depression. Which drug class does this describe, and why is it more dangerous than benzodiazepines in overdose?

- A. Benzodiazepines, because they always require GABA to be present
- B. Barbiturates, because at high doses they can directly gate the Cl⁻ channel independent of GABA, producing unopposed, potentially fatal CNS/respiratory depression
- C. SSRIs, which have no GABA-A activity at all
- D. Opioids, which act on a completely unrelated receptor system with no ceiling effect described here
- E. Antipsychotics, which block dopamine receptors, not GABA-A

† original question, written from lecture content (not a verified past-exam item)

258. Why does chronic SSRI use often take 2-4 weeks to produce clinical antidepressant benefit, even though synaptic serotonin levels rise within hours of the first dose?

- A. The delay reflects downstream neuroadaptive changes — including desensitization of presynaptic serotonin autoreceptors (allowing sustained increased release) and slower changes in postsynaptic receptor density/signaling and neurotrophic factors (e.g., BDNF) — not simply the acute rise in synaptic serotonin itself
- B. SSRIs actually have no effect on serotonin levels at all
- C. The drug takes 2-4 weeks just to reach the brain
- D. Serotonin itself directly causes mood improvement within minutes, contradicting the delay
- E. There is no known reason for this delay

† original question, written from lecture content (not a verified past-exam item)

259. Which best explains the clinical observation that opioid analgesics are highly effective for nociceptive (tissue-injury) pain but often less effective for neuropathic pain?

- A. Opioids primarily act on mu-opioid receptors in the pain-modulating circuits (periaqueductal gray, dorsal horn) engaged by normal nociceptive signaling, whereas neuropathic pain often involves aberrant, opioid-less-responsive mechanisms (ectopic firing, central sensitization, altered ion channel expression) in damaged nerves themselves
- B. Opioids have no mechanism of action at all
- C. Neuropathic pain does not exist as a real phenomenon
- D. Opioids are equally effective for all pain types with no exceptions
- E. Nociceptive pain does not involve any receptors

† original question, written from lecture content (not a verified past-exam item)

Lecture 15 — Visual System (19 questions)

260. A patient exhibits a near-vision reflex defect. Which of the following is NOT a normal characteristic of the near vision reflex?

- A. Medial deviation (convergence) of both eyes
- B. Levator palpebrae superioris contraction (eyelid elevation)
- C. Smaller pupil diameter (constriction)
- D. Increased convexity of the lens (accommodation)

261. A patient has a tumor compressing the optic chiasm. Which visual field defect results?

- A. Left homonymous hemianopia
- B. Right homonymous hemianopia
- C. Bitemporal heteronymous hemianopia
- D. Binasal heteronymous hemianopia

262. Which part of the retina has NO photoreceptor cells at all?

- A. Macula lutea
- B. Fovea centralis
- C. Retina near the ora serrata
- D. Optic disc (the physiologic blind spot)
- E. Paramacular field

263. What structure in the eye is primarily responsible for absorbing stray light, reducing internal reflection/glare?

- A. Choroid (its pigmented layer)
- B. Retina
- C. Cornea
- D. Sclera

264. Which condition is indicated by an intraocular pressure (IOP) greater than 25 mmHg?

- A. Cataract
- B. Retinal detachment
- C. Macular degeneration
- D. Glaucoma

265. Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?

- A. Rods/cones → Inner nuclear → Outer nuclear → Ganglion cell → Optic nerve fiber
- B. Rods/cones → Outer nuclear → Inner plexiform → Inner nuclear → Optic nerve fiber
- C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber
- D. Rods/cones → Ganglion cell → Inner nuclear → Optic nerve fiber

266. A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye deviation. Where is the lesion?

- A. Optic tract
- B. Optic nerve
- C. Optic chiasm
- D. CN III (oculomotor nerve)
- E. Optic radiation

267. Surface ectoderm gives rise to which eye structure?

- A. Lens
- B. Iris
- C. Corneal endothelium
- D. Fibrous tunic (sclera) connective tissue
- E. Retina

268. Which specialization of the fovea centralis increases visual acuity there?

- A. Abundant blood vessels overlying it
- B. A high ratio of photoreceptor cells to ganglion cells
- C. Lateral displacement of overlying ganglion/inner retinal cell layers, minimizing light scatter before reaching photoreceptors
- D. Increased density of rod cells

269. What structure is damaged to produce a RIGHT homonymous hemianopia?

- A. Left optic tract (or fibers further downstream on the left visual pathway)
- B. Optic chiasm
- C. Superior left lateral geniculate body region alone
- D. Inferior left lateral geniculate body region alone

270. Which abnormality of retinal cells causes the majority of color blindness cases?

- A. Abnormal rod cells
- B. Abnormal ganglion cells
- C. Abnormal cone cells (cone photopigment gene defects)
- D. Abnormal bipolar cells
- E. Abnormal pigment epithelium cells

271. What is the effect of sympathetic fiber activation on the eye?

- A. Pupillary constriction (miosis)
- B. Increased lens convexity (accommodation)
- C. Pupillary dilation (mydriasis)
- D. Enhanced visual acuity specifically in bright light
- E. Reduced rod cell sensitivity directly

272. What is the physiological effect of retinal detachment?

- A. Loss of blood supply to the photoreceptor layer (from the choroid), causing photoreceptor ischemia/degeneration
- B. Loss of blood supply to bipolar cells specifically
- C. Loss of blood supply to ganglion cells specifically

273. A lesion of the RIGHT Meyer's loop (temporal lobe optic radiation) causes which visual field defect?

- A. Left homonymous SUPERIOR quadrantanopia
- B. Right homonymous inferior quadrantanopia
- C. Right homonymous superior quadrantanopia
- D. Left homonymous inferior quadrantanopia
- E. Left homonymous hemianopia with macular sparing

274. A patient with a right temporal lobe lesion (affecting Meyer's loop) most likely shows which visual field defect?

- A. Right homonymous hemianopia
- B. Bitemporal heteronymous hemianopia
- C. Left homonymous superior quadrantanopia
- D. Left homonymous inferior quadrantanopia
- E. Left homonymous hemianopia

275. Touching the RIGHT cornea produces no blink in either eye; touching the LEFT cornea produces bilateral blinking. Where is the lesion?

- A. Left facial nerve
- B. Right optic nerve
- C. Right trigeminal nerve (afferent limb of the corneal reflex)
- D. Left trigeminal nerve
- E. Right oculomotor nerve

276. A patient has anisocoria noted incidentally — pupils differ in size — but reacts normally to light and accommodation, worse in dim light. This mild, benign finding without other neurological signs most likely represents:

- A. A dangerous CN III compression requiring emergent imaging
- B. Physiologic (simple) anisocoria, a common benign variant present in a meaningful percentage of the healthy population
- C. Horner syndrome, always
- D. Argyll Robertson pupil, always
- E. Complete bilateral optic nerve failure

† original question, written from lecture content (not a verified past-exam item)

277. Which layer of the eye is most directly responsible for providing nutrients to the outer retina via diffusion (choriocapillaris), making it critical in conditions like age-related macular degeneration?

- A. Sclera
- B. Choroid
- C. Cornea
- D. Conjunctiva
- E. Iris

† original question, written from lecture content (not a verified past-exam item)

278. Why does damage to the optic nerve (before the chiasm) cause a MONOCULAR visual deficit, while damage to the optic tract (after the chiasm) causes a BINOCULAR (but homonymous) deficit?

- A. Because the chiasm is the point where fibers from both eyes' nasal retinas cross and combine with the intact temporal retina fibers of the other eye, so everything AFTER the chiasm carries information from BOTH eyes; everything BEFORE the chiasm carries information from only ONE eye
- B. There is no anatomical reason for this difference
- C. The optic nerve carries information from both eyes already
- D. The optic tract carries information from only one eye
- E. Chiasm crossing is irrelevant to this distinction

† original question, written from lecture content (not a verified past-exam item)

Lecture 16 — Auditory & Vestibular Systems (22 questions)

279. Which structure is found at the level of the rostral pons?

- A. Cerebral aqueduct
- B. Abducens nucleus
- C. Area postrema
- D. Motor and principal (chief) sensory nuclei of CN V
- E. Inferior vestibular nucleus

280. A patient has facial nerve damage resulting in hyperacusis (abnormally loud perception of sound). Which structure's function is most likely interrupted?

- A. Trapezoid body
- B. Superior olivary complex
- C. Tensor tympani reflex
- D. Stapedius reflex (stapedius muscle, innervated by CN VII)
- E. Lateral lemniscus

281. Which vestibular nucleus sends axons directly to the oculomotor nucleus (contributing to the vestibulo-ocular reflex)?

- A. Medial vestibular nucleus
- B. Lateral vestibular nucleus
- C. Superior vestibular nucleus
- D. Inferior vestibular nucleus

282. Which structure releases (dissipates) the pressure wave of sound energy out of the inner ear, allowing continued fluid movement within the cochlea?

- A. Eardrum (tympanic membrane)
- B. Oval window
- C. Round window
- D. Helicotrema
- E. Scala vascularis (not a real structure)

283. Which statement regarding sound stimulation and cochlear transduction is TRUE?

- A. Vibration within the endolymph of the scala vestibuli is transmitted through the vestibular (Reissner's) membrane
- B. Outer hair cells vibrate and touch the tectorial membrane, generating the main afferent signal carried to the cochlear nerve
- C. Inner hair cells are stimulated by relative movement against the tectorial membrane and generate action potentials carried via the peripheral processes of the SPIRAL ganglion
- D. High-frequency sound waves stimulate the organ of Corti at the APEX of the cochlear duct

- 284.** What structure detects vertical linear acceleration, as when an elevator starts moving up or down?
- A. Superior semicircular canal
 - B. Lateral (horizontal) semicircular canal
 - C. Posterior semicircular canal
 - D. Saccule (oriented to detect vertical linear acceleration)
 - E. Utricle
- 285.** What structure primarily detects tilting the head UP (vertical linear head position change)?
- A. Utricle
 - B. Saccule
 - C. Horizontal semicircular canal
 - D. Superior semicircular canal
 - E. Cochlea
- 286.** What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?
- A. Utricle
 - B. Saccule
 - C. Horizontal semicircular canal
 - D. Superior (anterior) semicircular canal
 - E. Cochlea
- 287.** When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?
- A. Superior semicircular canal
 - B. Lateral semicircular canal
 - C. Posterior semicircular canal
 - D. Saccule
 - E. Utricle
- 288.** Where is the vomiting center located?
- A. Vestibular nuclei
 - B. Nucleus ambiguus
 - C. Area postrema
 - D. Superior and inferior salivatory nuclei
 - E. Nucleus solitarius
- 289.** What structure reduces sensitivity to background/surrounding sound (by inhibiting outer hair cells) to help focus on speech in a noisy environment?
- A. Stapedius muscle (and the broader olivocochlear efferent system inhibiting outer hair cells)
 - B. Tectorial membrane
 - C. Cochlear nerve
 - D. Superior olivary complex acting alone with no efferent output
 - E. Inferior colliculus
- 290.** Trouble hearing LOW-frequency sounds suggests a possible lesion/occlusion where along the basilar membrane?
- A. Apex of the vestibular (Reissner's) membrane
 - B. Apex of the basilar membrane (tuned to low frequencies)
 - C. Base of the cochlear duct
 - D. Base of the vestibular membrane
 - E. Central portion of the basilar membrane

291. Which of the following correctly describes the central auditory pathway?

- A. Cochlear nerve → Cochlear nucleus → Superior olivary complex → Medial geniculate body (of the thalamus)
- B. Cochlear nucleus → Cochlear nerve → Superior olivary → Medial geniculate body
- C. Cochlear nerve → Cochlear nucleus → Superior olivary → Lateral geniculate body
- D. Cochlear nucleus → Cochlear nerve → Superior olivary → Inferior colliculus → Lateral geniculate body

292. An 80-year-old patient presents with presbycusis (age-related hearing loss, 50-8000 Hz range affected). Which structure is most likely damaged?

- A. Endolymph
- B. Outer hair cells (cumulative damage, especially at the cochlear base)
- C. Tectorial membrane
- D. Stria vascularis
- E. Inner hair cells

293. A man slips and is about to fall; he reflexively extends his upper and lower limbs to catch himself. Which pathway mediates this reflex?

- A. Lateral spinoventricular tract (not a standard named tract)
- B. Lateral vestibulospinal tract (extensor/antigravity tone control)
- C. Medial vestibulospinal tract
- D. Medial spinoventricular tract (not standard)
- E. Cerebellospinal tract (not standard)

294. What is the underlying pathological cause of presbycusis?

- A. Damage to the tympanic membrane
- B. Stiffening of the ear ossicles
- C. Progressive damage to cochlear hair cells (especially outer hair cells at the base)
- D. Blockage of the Eustachian tube
- E. Excess fluid in the semicircular canals

295. What is INCORRECT about stimulation of the crista ampullaris (within the semicircular canal ampulla)?

- A. It detects angular acceleration of the head
- B. Its hair cells are stimulated by movement of endolymph
- C. It contains sensory hair cells whose stereocilia are embedded in a gelatinous cupula
- D. It detects LINEAR acceleration of the head (this is the false statement — that is the otolith organs' job)

296. Which tract can inhibit/modulate the vestibulospinal reflex's effect on limb muscles, allowing voluntary override of purely reflexive postural responses?

- A. Medial vestibular tract (not the modulating pathway being asked about)
- B. Corticospinal tract (descending voluntary motor control that can override/modulate reflexive vestibulospinal output)
- C. Spinocerebellar tract (ascending, sensory, not a descending inhibitory pathway)

297. A presbycusis test is performed on patients aged 45, 60, and 90. At which decibel (dB) level does hearing loss typically begin to be noticeable in this progressive, age-related pattern?

- A. >90 dB
- B. 25-50 dB
- C. >50 dB

298. Why does damage to the outer hair cells (rather than inner hair cells) cause a disproportionately large loss of hearing SENSITIVITY, even though outer hair cells carry only ~5% of the afferent auditory signal?

- A. Outer hair cells function as the cochlear amplifier — using electromotility to actively amplify basilar membrane vibration at specific frequencies BEFORE inner hair cells transduce it, so losing them means inner hair cells receive a much weaker mechanical signal to begin with
- B. Outer hair cells carry the primary afferent signal, so this premise is false
- C. Outer hair cells have no functional role and their loss has no effect
- D. Outer hair cells are purely structural with no electromotile properties
- E. Inner hair cells compensate completely for any outer hair cell loss, so sensitivity is unaffected

† original question, written from lecture content (not a verified past-exam item)

299. Why does conductive hearing loss (e.g., from otosclerosis or a perforated eardrum) typically preserve the ability to hear via bone conduction, while sensorineural hearing loss impairs BOTH air and bone conduction?

- A. Conductive loss is a problem in the outer/middle ear mechanical transmission pathway (which bone conduction bypasses entirely by vibrating the skull/cochlea directly); sensorineural loss is a problem in the cochlea/nerve itself, which bone conduction cannot bypass
- B. There is no real difference between the two types of hearing loss
- C. Bone conduction only works in sensorineural loss
- D. Air conduction always exceeds bone conduction in every scenario
- E. Conductive loss always also damages the cochlea equally

† original question, written from lecture content (not a verified past-exam item)

300. Why does spinning in place and then suddenly stopping cause a strong sensation of continued rotation (post-rotatory vertigo/nystagmus)?

- A. Endolymph in the semicircular canals has inertia and continues moving briefly after the head stops, still deflecting the cupula and signaling rotation even though the head is now still
- B. The otolith organs are entirely responsible with no semicircular canal involvement
- C. This phenomenon has no physiological explanation
- D. The cochlea is responsible for this sensation
- E. This only occurs in pathological states, never in healthy people

† original question, written from lecture content (not a verified past-exam item)

Answer Key & Full Distractor Analysis

Lectures 1–16 · 300 Questions

Lecture 1 — Organization of the CNS (20 questions)

Q1. Concerning the parts of the Central Nervous System, which of the following items is NOT included?

✓ **Correct: D. Trigeminal ganglion**

The trigeminal ganglion sits outside the brain proper (in Meckel's cave) and houses first-order sensory neuron cell bodies — that makes it a PNS structure, not CNS.

✗ **A is wrong:** The pineal gland is neuroectoderm-derived and physically part of the diencephalon roof — it counts as CNS even though it's an endocrine gland.

✗ **E is wrong:** The posterior pituitary (neurohypophysis) is a direct downgrowth of the hypothalamus/diencephalon, so it's CNS tissue despite sitting outside the cranial vault proper.

PROF'S TRICK: Classic trick: examiners mix in endocrine-sounding structures (pineal, posterior pituitary) that are secretly CNS because of their embryonic origin from the neural tube.

Q2. When you wear a ring for a long time, you stop noticing it. Which receptor's rapid adaptation is responsible?

✓ **Correct: B. Pacinian corpuscle**

Pacinian corpuscles are the fastest-adapting (phasic) mechanoreceptors, firing only at the onset/offset of a stimulus — perfect for 'you stop feeling constant pressure.'

✗ **A is wrong:** Meissner's corpuscles are also rapidly adapting, but they encode light touch/flutter at the fingertips, not sustained deep pressure like a ring.

✗ **D is wrong:** Merkel discs are slowly adapting (tonic) — they'd keep firing the whole time you wore the ring, which is the opposite of what the question describes.

PROF'S TRICK: Sort receptors into two adaptation buckets first: SLOW (Merkel, Ruffini) = keep firing, sense duration/pressure; FAST (Meissner, Pacinian) = fire once, sense change/vibration.

Q3. Which of the following is NOT a division of the peripheral nervous system?

✓ **Correct: D. Central nervous system**

By definition the CNS (brain + spinal cord) is the other half of the nervous system, not a subdivision of the PNS.

✗ **C is wrong:** The enteric nervous system is often forgotten, but it's a genuine PNS division — it can function semi-autonomously even when vagal/sympathetic input is cut.

† original question, not a verified past-exam item

Q4. A layer-by-layer dissection proceeds from the skull inward. Put the meningeal layers in the correct order.

✓ **Correct: C. Dura → Arachnoid → Pia**

From outside in: Dura (tough, periosteal+meningeal layers) → Arachnoid (web-like, subarachnoid space beneath it holds CSF) → Pia (hugs the brain surface, follows every gyrus/sulcus).

✗ **D is wrong:** This reverses the order completely — pia is the innermost layer, directly adherent to neural tissue, not the outermost.

† original question, not a verified past-exam item

PROF'S TRICK: Mnemonic: 'Dr. AP' — Dura, Arachnoid, Pia, outside to in. The subdural space is a potential space (between dura & arachnoid); the subarachnoid space is real and full of CSF.

Q5. CSF produced in the lateral ventricles must pass through the foramen of Monro to reach the:

✓ **Correct: B. Third ventricle**

The interventricular foramen (of Monro) connects each lateral ventricle to the third ventricle — the next stop on the CSF circuit.

✗ **A is wrong:** The fourth ventricle is two stops away — CSF must still cross the cerebral aqueduct (of Sylvius) from the third ventricle before reaching the fourth.

† original question, not a verified past-exam item

PROF'S TRICK: Know the full CSF circuit cold: Lateral ventricles → (foramen of Monro) → 3rd ventricle → (cerebral aqueduct) → 4th ventricle → (foramina of Luschka/Magendie) → subarachnoid space → arachnoid granulations → venous sinuses.

Q6. Which statement about gray and white matter distribution is correct?

✓ **Correct: B. In the cerebrum, gray matter forms the cortex and white matter lies deep to it**

Cerebral/cerebellar cortex = gray matter (cell bodies) on the outside; the underlying white matter is myelinated axon tracts connecting cortical areas.

✗ **A is wrong:** This is backwards for the spinal cord — the spinal cord actually has gray matter centrally (the classic 'butterfly') surrounded by white matter tracts peripherally, the opposite arrangement to the cerebrum.

† original question, not a verified past-exam item

PROF'S TRICK: Spinal cord and cerebrum have OPPOSITE gray/white arrangements — a favorite flip-the-diagram trick question.

Q7. A newborn is found to have a portion of spinal cord and meninges protruding through a vertebral defect, covered only by a thin membrane....

✓ **Correct: C. Myelomeningocele**

Myelomeningocele = neural tissue (spinal cord/roots) PLUS meninges herniate through the vertebral defect — the most severe open form of spina bifida.

✗ **B is wrong:** A meningocele herniates meninges and CSF only — no neural tissue — so it usually has milder or no neurological deficit.

✗ **A is wrong:** Spina bifida occulta is the mildest form: just a bony vertebral arch defect with skin intact over it, often an incidental finding.

† original question, not a verified past-exam item

PROF'S TRICK: Rank spina bifida severity: occulta (bone only, hidden) < meningocele (meninges herniate) < myelomeningocele (meninges + cord herniate, worst).

Q8. Which cranial nerve numbers correspond to nerves that are purely sensory?

✓ **Correct: B. I, II, VIII**

CN I (olfactory), CN II (optic), and CN VIII (vestibulocochlear) carry only special sensory information — no motor fibers.

✗ **A is wrong:** III, IV, and VI are the purely MOTOR nerves (oculomotor, trochlear, abducens) — the opposite category from what's being asked.

† original question, not a verified past-exam item

PROF'S TRICK: Group the 12 CNs into 3s: Sensory only (I, II, VIII), Motor only (III, IV, VI, XI, XII), Mixed (V, VII, IX, X).

Q9. Dorsal root ganglia contain the cell bodies of which type of neuron?

✓ **Correct: C. First-order (primary) sensory neurons**

DRG house the pseudounipolar cell bodies of primary afferent (first-order sensory) neurons carrying input from the periphery into the spinal cord.

✗ **A is wrong:** Lower motor neuron cell bodies live in the ventral horn of the spinal cord gray matter, not in a peripheral ganglion.

† original question, not a verified past-exam item

Q10. A patient has a lesion that damages both anterior and posterior spinal nerve roots equally at one level. Which functional deficit...

✓ **Correct: C. Combined sensory and motor loss at that segmental level**

The posterior (dorsal) root is sensory and the anterior (ventral) root is motor; damaging both together at a spinal nerve level knocks out both modalities in that dermatome/myotome.

✗ **A is wrong:** This only accounts for the dorsal root — ignoring the ventral (motor) root damage described in the stem.

† original question, not a verified past-exam item

PROF'S TRICK: Dorsal = sensory (Dorsal = Down = sensory comes down from the periphery... or just remember 'Bell-Magendie law': dorsal root sensory, ventral root motor).

Q11. Which of the following best describes the enteric nervous system?

✓ **Correct: B. It can regulate gut motility and secretion semi-independently of CNS input**

The enteric nervous system has its own local reflex circuits (sensory neurons, interneurons, motor neurons) in the gut wall and can function even after vagal/sympathetic denervation — hence 'the second brain.'

✗ **A is wrong:** This is the opposite of what makes the ENS special — it's celebrated precisely because it does NOT require constant extrinsic autonomic input to run basic reflexes.

† original question, not a verified past-exam item

Q12. Which of these structures is a boundary landmark of the fourth ventricle?

✓ **Correct: C. Pons and medulla (floor) with cerebellum (roof)**

The 4th ventricle's floor is the rhomboid fossa formed by the dorsal pons/medulla, and its roof is formed by the cerebellum — classic exam-image landmark.

✗ **B is wrong:** Cerebral peduncles are midbrain structures related to the cerebral aqueduct, one level rostral to the 4th ventricle.

† original question, not a verified past-exam item

Q13. Neural crest cells contribute to all of the following EXCEPT:

✓ **Correct: D. Oligodendrocytes**

Oligodendrocytes are CNS glia derived from the neuroectoderm of the neural tube itself, not from migrating neural crest cells.

✗ **B is wrong:** Schwann cells are the PNS myelinating glia and ARE neural-crest derived — a commonly confused pair with oligodendrocytes (which are NOT).

† original question, not a verified past-exam item

PROF'S TRICK: Neural TUBE → CNS neurons + astrocytes + oligodendrocytes. Neural CREST → PNS neurons/ganglia + Schwann cells + adrenal medulla + melanocytes + much of the facial skeleton.

Q14. A 'brainstem' anatomically and functionally includes which three structures, rostral to caudal?

✓ **Correct: B. Midbrain, pons, medulla oblongata**

The brainstem = midbrain (mesencephalon) + pons + medulla oblongata, connecting the cerebrum to the spinal cord.

✗ **A is wrong:** The thalamus is part of the diencephalon, not the brainstem, even though it sits just rostral to the midbrain.

† original question, not a verified past-exam item

Q15. Which statement correctly differentiates somatic and autonomic motor pathways?

✓ **Correct: B. Autonomic motor pathways use two neurons in series (pre- and postganglionic); somatic uses one**

Autonomic efferents are a two-neuron relay (preganglionic → ganglion → postganglionic → target); somatic motor is a single lower motor neuron running straight from the spinal cord/brainstem to skeletal muscle, no ganglion synapse.

✗ **A is wrong:** This reverses the pattern — somatic is actually the single-neuron pathway, autonomic is the two-neuron relay.

† original question, not a verified past-exam item

PROF'S TRICK: If a pathway synapses in a ganglion outside the CNS before reaching the target organ, it's autonomic. Somatic motor neurons never stop at a peripheral ganglion.

Q16. How many pairs of spinal nerves are there in total, and which region has one MORE pair than vertebrae?

✓ **Correct: A. 31 pairs; cervical**

31 pairs total (8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal); there are only 7 cervical vertebrae but 8 cervical nerve pairs because C1 exits above C1 vertebra and C8 exits below C7.

✗ **C is wrong:** Thoracic nerves match their vertebral count 1:1 (12 thoracic nerves, 12 thoracic vertebrae) — no extra pair there.

† original question, not a verified past-exam item

PROF'S TRICK: C1-C7 vertebrae but C1-C8 NERVES — the 'extra' cervical nerve is the classic trick. From C8 down, spinal nerve numbering matches the vertebra below which it exits.

Q17. The scalp, skull, vertebrae, meninges, and CSF together serve what shared function?

✓ **Correct: B. Mechanical and immunological protection of the CNS**

These are the layered physical defenses of the CNS — bone for rigid protection, meninges + CSF for cushioning (buoyancy reduces effective brain weight) and immune surveillance.

† original question, not a verified past-exam item

Q18. Area postrema, found near the floor of the fourth ventricle, is functionally significant because it:

✓ **Correct: B. Lacks a complete blood-brain barrier, allowing it to sense blood-borne toxins (chemoreceptor trigger zone)**

Area postrema is one of the circumventricular organs with a leaky BBB, letting it directly sample blood for toxins/emetic substances and trigger vomiting.

✗ **A is wrong:** This is the opposite — its defining feature is precisely that it does NOT have the typical tight BBB other brainstem regions have.

† original question, not a verified past-exam item

PROF'S TRICK: Circumventricular organs (area postrema, median eminence, OVLT, pineal, etc.) are the exam's favorite 'blood-brain barrier exceptions' — know that area postrema = vomiting center that samples blood directly.

Q19. Which best explains why the CNS is described as having a much more limited regenerative capacity than the PNS after axonal injury?

✓ **Correct: B. Oligodendrocytes and CNS scar tissue produce inhibitory molecules that block axon regrowth, unlike the permissive environment Schwann cells provide in the PNS**

Schwann cells in the PNS proliferate, clear debris, and form a growth-permissive tube (bands of Büngner) guiding regrowth; CNS oligodendrocytes and reactive astrocytic/glia scarring instead release growth-inhibitory molecules (e.g., Nogo, chondroitin sulfate proteoglycans).

✗ **C is wrong:** The CNS has plenty of myelin (from oligodendrocytes) — it's the myelin-associated inhibitory proteins that are the problem, not an absence of myelin.

† original question, not a verified past-exam item

Q20. Which meningeal layer is the site of most epidural hematomas, classically from a middle meningeal artery tear?

✓ **Correct: A. Between skull and dura (epidural/extradural space)**

Epidural hematomas classically form between the skull and dura, often from a tear in the middle meningeal artery after temporal bone (pterion) trauma — arterial bleeding into this potential space produces the classic rapidly-expanding, lens (biconvex)-shaped hematoma on CT.

✗ **B is wrong:** Subdural hematomas instead form between dura and arachnoid, typically from torn bridging VEINS (not the artery), producing a slower-onset, crescent-shaped hematoma — a classic paired distinction (epidural=artery=fast=biconvex vs subdural=vein=slow=crescent) tested constantly.

† original question, not a verified past-exam item

PROF'S TRICK: Epidural = artery = biconvex/lens-shaped = fast, dramatic. Subdural = bridging veins = crescent-shaped = slower, especially in the elderly/atrophied brain where veins are more stretched and vulnerable. This lens-vs-crescent shape distinction is one of the highest-yield facts in all of neuroanatomy/radiology.

Lecture 2 — Neurohistology (Neuron & Neuroglia) (20 questions)

Q21. Which of the following compounds is transferred from astrocyte to neuron and used for ATP production during starvation?

✓ **Correct: E. Lactate**

The astrocyte-neuron lactate shuttle exports lactate (via MCT) from glycolytic astrocytes to fuel neuronal oxidative metabolism, especially when demand is high or glucose delivery is limited.

✗ **C is wrong:** Pyruvate is the end-product of glycolysis inside a cell, but it isn't the molecule classically shuttled between astrocyte and neuron — lactate (which astrocytes preferentially generate via LDH5) is.

PROF'S TRICK: 'Astrocyte-neuron LACTATE shuttle' — if you remember only one word from this topic, remember 'lactate,' not 'pyruvate' or 'glucose.'

Q22. Myelin sheath wrapping nerve axons within the basal ganglia (a CNS structure) is produced by which cell?

✓ **Correct: D. Oligodendrocyte**

Oligodendrocytes myelinate CNS axons, and each oligodendrocyte can myelinate multiple axon segments simultaneously.

✗ **B is wrong:** Schwann cells myelinate PNS axons only (one Schwann cell = one internode on one axon) — a very common mix-up since both cells 'do the same job' in different compartments.

PROF'S TRICK: CNS myelin = Oligodendrocyte (multiple axons at once). PNS myelin = Schwann cell (one axon, one cell). If the question mentions basal ganglia, cortex, spinal cord tracts = CNS = oligodendrocyte.

Q23. Which of the following is the glutamate transporter responsible for uptake into astrocytes?

✓ **Correct: E. EAATs**

Excitatory Amino-Acid Transporters (EAATs, especially EAAT-2/GLT-1) on astrocytic membranes clear synaptic glutamate to prevent excitotoxicity.

✗ **A is wrong:** MCT (monocarboxylate transporter) moves lactate/pyruvate/ketone bodies, not glutamate — easy to confuse since both transporters live on astrocytes and both are exam favorites.

Q24. If the tract carrying FAST, sharp, well-localized pain is damaged, which fiber type is responsible for that lost sensation?

✓ **Correct: D. Neo-spinothalamic tract (A-δ) fibers**

Fast, sharp, well-localized ('first') pain is carried by thinly myelinated A-δ fibers via the neospinothalamic tract, which projects rather directly to VPL and then somatosensory cortex — hence good localization.

✗ **B is wrong:** C fibers carry SLOW, dull, poorly-localized ('second') pain via the paleospinothalamic tract — the opposite pain quality from what's described.

PROF'S TRICK: Fast/sharp/localized = A-δ = neospinothalamic. Slow/dull/burning/poorly localized = C fiber = paleospinothalamic (more diffuse projections including to reticular formation/limbic system, which is why it's harder to pinpoint and more 'emotionally' unpleasant).

Q25. Which of the following is NOT a component of the blood-brain barrier?

✓ **Correct: A. Oligodendrocyte**

The BBB's neurovascular unit is built from capillary endothelial cells (tight junctions), the basement membrane, pericytes, and astrocytic end-feet — oligodendrocytes play no structural role in it.

✗ **C is wrong:** Pericytes are a genuine, often-forgotten component that helps regulate BBB permeability and capillary blood flow — students frequently (wrongly) exclude this one instead.

PROF'S TRICK: BBB team: Endothelium (tight junctions, the main seal) + Basement membrane + Pericytes + Astrocyte end-feet. Oligodendrocytes are not invited to this party — they're busy myelinating.

Q26. Which cell type is the resident immune cell of the CNS, analogous to macrophages elsewhere in the body?

✓ **Correct: A. Microglia**

Microglia are mesoderm/yolk-sac-derived resident CNS immune cells that phagocytose debris and pathogens and mediate neuroinflammation.

✗ **D is wrong:** Astrocytes support the BBB, metabolism, and repair (glial scarring) but are not the primary phagocytic immune surveillance cell — that's specifically microglia's job.

Q27. Which part of the axon has the highest density of voltage-gated Na⁺ channels, essential for action potential generation?

✓ **Correct: B. Axon hillock / initial segment**

The axon initial segment (axon hillock) has the lowest threshold and highest density of voltage-gated Na⁺ channels, making it the site where action potentials are actually triggered.

✗ **E is wrong:** Myelin itself is insulating and essentially channel-free — voltage-gated channels cluster at the UNMYELINATED nodes of Ranvier, not within the myelin sheath.

Q28. Which disease process is defined by the immune system attacking and destroying CNS myelin?

✓ **Correct: B. Multiple sclerosis**

Multiple sclerosis is an autoimmune demyelinating disease targeting CNS oligodendrocytes/myelin, producing periventricular white matter plaques.

✗ *A is wrong: Guillain-Barré is also autoimmune demyelination, but it targets PNS myelin (Schwann cells) — a classic paired-distractor testing CNS vs PNS demyelination.*

PROF'S TRICK: MS = CNS demyelination (oligodendrocyte target). GBS = PNS demyelination (Schwann cell target). Both are autoimmune — the exam differentiates them by WHICH compartment is hit.

Q29. Which statement correctly distinguishes sensory ganglia from autonomic ganglia?

✓ **Correct: A. Sensory ganglia contain pseudounipolar neuron cell bodies with no synapses inside the ganglion; autonomic ganglia contain a true synapse between pre- and postganglionic neurons**

Sensory (dorsal root/cranial) ganglia are just way-stations for pseudounipolar cell bodies — the impulse passes through without synapsing. Autonomic ganglia are actual synaptic relay stations between pre- and postganglionic neurons.

✗ *D is wrong: Autonomic ganglion neurons are typically multipolar, not pseudounipolar — pseudounipolar morphology is the signature of sensory ganglion cells.*

Q30. Which statement about neurons is INCORRECT?

✓ **Correct: C. All neurons have a myelinated axon**

Not all axons are myelinated — many are unmyelinated (particularly small-diameter, slow-conducting C fibers) and still function perfectly well, just more slowly.

Q31. Layers surrounding a peripheral nerve fiber, from innermost to outermost, are correctly ordered as:

✓ **Correct: B. Endoneurium → Perineurium → Epineurium**

Endoneurium wraps each individual nerve fiber; perineurium bundles fibers into fascicles; epineurium wraps the whole nerve trunk — inside to outside.

✗ *A is wrong: This is the exact reverse order — a favorite examiner trick, so always double-check direction (inner vs outer) before matching.*

PROF'S TRICK: Nerve = En-Peri-Epi, small to big, just like muscle's Endo-Peri-Epimysium. Same naming logic, different tissue.

Q32. A young woman presents with fatigable muscle weakness that worsens with repetitive activity and improves with rest. This points to a...

✓ **Correct: B. Postsynaptic nicotinic ACh receptor (myasthenia gravis)**

Fatigable weakness that WORSENS with use and improves with rest is the classic presentation of myasthenia gravis, caused by autoantibodies against postsynaptic nicotinic ACh receptors, progressively depleting available receptors with repeated stimulation.

✗ *A is wrong: Lambert-Eaton syndrome (presynaptic Ca²⁺ channel antibodies) classically does the OPPOSITE — strength IMPROVES with repeated use as Ca²⁺ accumulates presynaptically, a key discriminator the exam loves to test.*

† original question, not a verified past-exam item

PROF'S TRICK: Worse with use = postsynaptic (myasthenia gravis). Better with use = presynaptic (Lambert-Eaton). This pairing shows up constantly.

Q33. Which glial cell lines the ventricles and central canal, contributing to CSF circulation?

✓ **Correct: A. Ependymal cell**

Ependymal cells form a ciliated epithelial lining of the ventricular system and central canal, helping circulate CSF; specialized ependymal cells (choroid plexus epithelium) also help produce it.

† original question, not a verified past-exam item

Q34. A demyelinated CNS axon conducts action potentials more slowly and can fail to propagate altogether. What is the direct biophysical reason?

✓ **Correct: B. Loss of myelin increases current leak across the membrane and removes saltatory jumps between nodes, slowing/failing conduction**

Myelin normally insulates the axon (increases resistance, decreases capacitance) so current jumps rapidly node-to-node (saltatory conduction). Losing it increases capacitance/leak and removes the insulated internodes, so conduction slows or can block entirely if the safety factor drops too low.

✗ *A is wrong: This has the capacitance effect backwards — myelin loss INCREASES membrane capacitance (more charge needed to depolarize each patch), which slows, not speeds, conduction.*

† original question, not a verified past-exam item

Q35. Which best explains why glutamate is normally cleared very quickly from the synaptic cleft?

✓ **Correct: A. To prevent excitotoxic Ca²⁺ overload from prolonged NMDA/AMPA receptor activation**

Rapid EAAT-mediated reuptake (mainly by astrocytes) limits how long glutamate can activate ionotropic receptors, protecting neurons from Ca²⁺-mediated excitotoxic injury (relevant in stroke/hypoglycemia).

† original question, not a verified past-exam item

Q36. Satellite cells are glial cells found specifically in which location?

✓ **Correct: A. Surrounding neuron cell bodies within peripheral ganglia**

Satellite cells are the PNS ganglion equivalent of astrocytes — they surround and support neuronal cell bodies inside sensory and autonomic ganglia.

✗ *E is wrong: Wrapping PNS AXONS is the Schwann cell's job — satellite cells surround cell BODIES within ganglia, a distinction that's easy to blur.*

† original question, not a verified past-exam item

PROF'S TRICK: PNS glia, sorted by job: Schwann cell = wraps AXONS. Satellite cell = surrounds cell BODIES in ganglia.

Q37. A axo-axonic synapse, as opposed to an axo-dendritic synapse, is best described as:

✓ **Correct: A. One axon terminal synapsing onto another axon (often near its terminal), typically for presynaptic modulation**

Axo-axonic synapses let one neuron modulate (facilitate or, more classically, presynaptically inhibit) neurotransmitter release from another axon terminal, without directly firing an action potential in the postsynaptic cell.

† original question, not a verified past-exam item

Q38. Reactive astrogliosis (astrocyte scar formation) after CNS injury has which net effect on axon regeneration?

✓ **Correct: B. It seals off and stabilizes the injury site but the resulting glial scar also physically and chemically inhibits axon regrowth through it**

Reactive astrocytes wall off damaged tissue (limiting spread of injury/inflammation) but the resulting dense glial scar, rich in chondroitin sulfate proteoglycans, is a major physical/chemical barrier to axon regeneration — a double-edged sword.

✗ *A is wrong: This ignores the well-documented inhibitory side of the glial scar — it's protective in the short term but a major obstacle to long-term regeneration.*

† original question, not a verified past-exam item

Q39. Multipolar neurons — the most common neuron shape in the CNS — are structurally characterized by:

✓ **Correct: B. One axon and multiple dendrites**

Multipolar neurons (e.g., motor neurons, most CNS interneurons) have a single axon plus multiple dendrites branching from the soma — the dominant neuron morphology in the brain and spinal cord.

✗ *E is wrong: That description is a pseudounipolar neuron (like DRG sensory neurons), which has one process that bifurcates into peripheral and central branches — a different, less common shape.*

† original question, not a verified past-exam item

Q40. Which best describes why oligodendrocytes are more vulnerable to certain metabolic/ischemic insults than astrocytes, contributing to...

✓ **Correct: A. Oligodendrocytes have particularly high metabolic demand for myelin maintenance and are especially sensitive to oxidative stress and glutamate excitotoxicity during their maturation window**

Immature/maturing oligodendrocytes have high metabolic demands (myelin synthesis) and limited antioxidant defenses, making them especially vulnerable to hypoxic-ischemic and excitotoxic injury during a specific developmental window — this vulnerability underlies periventricular leukomalacia, a classic white matter injury pattern in premature infants.

† original question, not a verified past-exam item

Lecture 3 — Neuron Properties: RMP & Action Potential (20 questions)

Q41. A 40-year-old male needs an urgent hand suture and is given lidocaine as local anesthesia. What is its mechanism of action?

✓ **Correct: C. Inhibition of voltage-gated sodium channels**

Lidocaine (an amide local anesthetic) blocks voltage-gated Na⁺ channels from the intracellular side, preventing the rising phase of the action potential and thus blocking nociceptive transmission.

✗ *D is wrong: That describes botulinum toxin's mechanism (blocking ACh vesicle fusion), not a local anesthetic — a classic mechanism mix-up between two totally different drug classes.*

PROF'S TRICK: Amide local anesthetics all end in 'i-caine' with two i's (lidocaine, bupivacaine) vs ester types with one i (procaine, tetracaine) — useful for allergy history questions too, since esters cross-react via PABA metabolites and amides generally don't.

Q42. Hyperpolarization of the postsynaptic membrane in response to glycine or GABA-A receptor activation is produced by the opening of which...

✓ **Correct: C. Cl⁻ channels**

GABA-A and glycine receptors are both ionotropic ligand-gated Cl⁻ channels; Cl⁻ influx (down its electrochemical gradient in most mature neurons) hyperpolarizes/stabilizes the membrane — classic fast inhibition.

✗ *B is wrong: K⁺ channels do contribute to some inhibitory postsynaptic potentials (e.g., via GABA-B, a metabotropic receptor), but the FAST, ionotropic GABA-A/glycine response specifically is a Cl⁻ channel, not K⁺.*

Q43. A 35-year-old male receives a lidocaine injection before incision and drainage of a buttock abscess. Which best describes lidocaine's...

✓ **Correct: C. Blocks voltage-gated Na⁺ channels**

Same core fact as any lidocaine question: it's a voltage-gated Na⁺ channel blocker, preventing depolarization and action potential propagation in nociceptive fibers.

PROF'S TRICK: If you see 'lidocaine,' 'bupivacaine,' or 'procaine' anywhere in a stem, your first instinct should always be 'voltage-gated Na⁺ channel blockade' unless the question is specifically asking about something else (e.g., allergy class).

Q44. A patient develops muscle weakness that worsens through the afternoon and evening but is nearly absent in the morning. Which pathology...

✓ **Correct: B. Myasthenia gravis**

Myasthenia gravis produces fatigable weakness that accumulates over the course of the day/with use as postsynaptic ACh receptors become progressively less available, then partially recovers with rest overnight.

Q45. What is the major consumer of energy (ATP) in the resting brain?

✓ **Correct: B. Restoring ionic gradients via the Na⁺/K⁺-ATPase after synaptic and action-potential activity**

The largest share of brain ATP consumption goes to running the Na⁺/K⁺-ATPase, restoring the Na⁺/K⁺ gradients dissipated by action potentials and synaptic currents — the brain is metabolically expensive largely because of this pump.

Q46. Which of the following correctly describes when Na⁺/K⁺-ATPase activity increases the most?

✓ **Correct: A. During periods of high neuronal firing, to restore ionic gradients dissipated by Na⁺ influx/K⁺ efflux**

Every action potential lets Na⁺ in and K⁺ out; the pump has to work harder (using more ATP) proportional to how much firing has occurred, to restore the resting gradients.

Q47. When depolarizations from multiple simultaneous synaptic inputs on different parts of a neuron add together, this is called:

✓ **Correct: B. Spatial summation**

Spatial summation = combining inputs arriving at different locations on the neuron at roughly the same time; temporal summation is combining repeated inputs at the SAME synapse arriving in rapid succession.

✗ *A is wrong: Temporal summation is the time-based version (same synapse, rapid-fire inputs) — the question specifically describes different synapses/locations, which is spatial.*

PROF'S TRICK: SPATIAL = different SPACE (different synapses) at the same time. TEMPORAL = same synapse, different TIME (rapid repeats).

Q48. All of the following are true of chemical synapses EXCEPT:

✓ **Correct: D. Transmission is instantaneous with essentially zero delay**

Chemical synapses always have a measurable synaptic delay (vesicle fusion, diffusion, receptor binding take real time) — instantaneous, zero-delay transmission describes ELECTRICAL synapses (gap junctions), not chemical ones.

PROF'S TRICK: Electrical synapses (gap junctions) = fast, bidirectional, no delay, no amplification, but also no plasticity. Chemical synapses = slower, unidirectional, delayed, but capable of amplification and plasticity (learning/memory). Know both columns of this comparison table.

Q49. Which statement about graded potentials is INCORRECT?

✓ **Correct: D. They obey the all-or-none law like action potentials**

Graded potentials are explicitly NOT all-or-none — their amplitude scales with stimulus intensity. The all-or-none law is the signature property of the action potential, the opposite category.

Q50. During the absolute refractory period, why can no new action potential be generated no matter how strong the stimulus?

✓ **Correct: B. Voltage-gated Na⁺ channels are inactivated and cannot reopen until the membrane repolarizes**

Na⁺ channels have an inactivation gate that closes shortly after opening and won't reset (become available again) until the membrane has repolarized back toward resting potential — no stimulus, however strong, can force them open during this state.

✗ *C is wrong: That describes the RELATIVE refractory period (a stronger-than-normal stimulus CAN trigger an AP), not the absolute refractory period where triggering a new AP is flatly impossible.*

† original question, not a verified past-exam item

PROF'S TRICK: Absolute refractory = impossible, period (Na⁺ channels inactivated). Relative refractory = possible but needs a bigger stimulus (partially hyperpolarized + some channels still inactivated).

Q51. Which factor increases axonal conduction velocity?

✓ **Correct: C. Increasing axon diameter and/or myelination (saltatory conduction)**

Larger diameter lowers internal resistance (faster passive current spread) and myelination enables saltatory conduction (current jumps node-to-node) — both independently speed up propagation, and combine multiplicatively in large myelinated axons.

† original question, not a verified past-exam item

PROF'S TRICK: Two independent 'speed knobs': (1) bigger diameter = faster (lower resistance), (2) myelination = faster (saltatory). The fastest fibers (A- α , e.g. proprioception/alpha motor) max out both knobs.

Q52. What determines the equilibrium potential for a given ion, as described by the Nernst equation?

✓ **Correct: B. The ratio of that ion's extracellular to intracellular concentration**

The Nernst equation calculates the voltage at which the electrical gradient exactly balances the chemical (concentration) gradient for one specific ion, and it depends only on that ion's concentration ratio (and temperature/valence) — not on permeability.

✗ *A is wrong: Permeability is what the GOLDMAN (GHK) equation weighs when combining multiple ions to predict the actual resting membrane potential — Nernst calculates a single ion's own equilibrium point regardless of how permeable the membrane is to it.*

† original question, not a verified past-exam item

PROF'S TRICK: Nernst = one ion's own equilibrium potential (concentration ratio only). Goldman-Hodgkin-Katz = the ACTUAL resting membrane potential, a permeability-weighted average across several ions (mainly Na⁺, K⁺, Cl⁻).

Q53. Presynaptic inhibition, as opposed to postsynaptic inhibition, works primarily by:

✓ **Correct: B. Reducing Ca²⁺ influx into the presynaptic terminal, decreasing neurotransmitter release**

Presynaptic inhibition (e.g., via axo-axonic synapses releasing GABA onto a presynaptic terminal) reduces voltage-gated Ca²⁺ channel opening in that terminal, so less neurotransmitter is released — a selective way to dampen one specific input without silencing the whole postsynaptic cell.

✗ *A is wrong: Direct postsynaptic hyperpolarization is exactly what POSTsynaptic inhibition does — this question is specifically asking about the presynaptic mechanism, which acts one step earlier, at the release machinery itself.*

† original question, not a verified past-exam item

Q54. A drug that blocks voltage-gated K⁺ channels in neurons would be expected to:

✓ **Correct: B. Prolong the action potential by slowing repolarization**

K⁺ efflux through voltage-gated K⁺ channels drives repolarization; blocking these channels delays that efflux, prolonging the depolarized phase of the action potential (this is, for example, the basis of some antiarrhythmic and toxin mechanisms).

† original question, not a verified past-exam item

Q55. Which best describes the relationship between stimulus intensity and action potential characteristics in a single axon?

✓ **Correct: B. A stronger stimulus produces a higher FREQUENCY of action potentials, not larger amplitude (each AP is all-or-none)**

Because action potentials are all-or-none (fixed amplitude/shape once threshold is crossed), the nervous system encodes stimulus intensity through FIRING RATE (frequency coding), not by making individual spikes bigger.

✗ *A is wrong: This violates the all-or-none principle — no matter how strong the stimulus (above threshold), each individual action potential looks the same amplitude.*

† original question, not a verified past-exam item

PROF'S TRICK: All-or-none means AMPLITUDE can't encode intensity — so the nervous system had to invent frequency (rate) coding instead. This logical link is a favorite short-answer/critical-thinking pivot.

Q56. A toxin selectively blocks voltage-gated Na⁺ channels at the node of Ranvier along an entire myelinated axon. What is the functional...

✓ **Correct: B. Action potential propagation fails completely, since saltatory conduction depends on regenerating the signal at each node**

Saltatory conduction relies on the action potential being actively regenerated at each successive node (where Na⁺ channels are concentrated); block those channels and the depolarizing current decays passively under the myelin and dies out before reaching the next node — conduction block.

† original question, not a verified past-exam item

PROF'S TRICK: This is the mechanism behind conditions like tetrodotoxin poisoning, and conceptually related to why local anesthetics (Na⁺ channel blockers) achieve regional nerve block even though they're only applied at one point along the nerve.

Q57. Long-term potentiation (LTP), a cellular model of learning and memory, is classically initiated by which receptor-level event?

✓ **Correct: B. Glutamate binding AMPA receptors causes partial depolarization, relieving the Mg²⁺ block on NMDA receptors and allowing Ca²⁺ entry**

LTP requires coincident presynaptic glutamate release AND sufficient postsynaptic depolarization (from AMPA receptor activation) to expel the voltage-dependent Mg²⁺ block from NMDA receptors, letting Ca²⁺ flood in — this Ca²⁺ signal triggers the downstream cascade (more AMPA receptor insertion, etc.) that strengthens the synapse.

† original question, not a verified past-exam item

PROF'S TRICK: NMDA receptors are BOTH ligand-gated (need glutamate) AND voltage-gated (need Mg²⁺ block relieved by depolarization) — that dual requirement is exactly why LTP needs simultaneous pre- and postsynaptic activity (Hebbian: 'cells that fire together, wire together').

Q58. Which best explains why increasing extracellular K⁺ concentration (hyperkalemia) can cause cardiac/neuronal membrane dysfunction, based...

✓ **Correct: B. It depolarizes the resting membrane potential (brings it closer to 0), since the K⁺ equilibrium potential becomes less negative, potentially causing sustained depolarization and eventual inactivation of voltage-gated Na⁺ channels**

Since resting potential is heavily K⁺-dominated, raising extracellular K⁺ makes the K⁺ equilibrium potential (per Nernst) less negative, depolarizing the resting membrane potential; if depolarization is sustained enough, it inactivates voltage-gated Na⁺ channels, ultimately impairing excitability — this is exactly the mechanism behind hyperkalemia's dangerous cardiac conduction effects.

† original question, not a verified past-exam item

PROF'S TRICK: This is a great 'basic physiology explains a life-threatening clinical scenario' question — hyperkalemia's ECG changes and cardiac risk trace directly back to the same Nernst equation logic used for any other resting-potential question, just applied to a clinical extreme.

Q59. Which best describes the ionic basis of the falling (repolarizing) phase of the action potential?

✓ **Correct: B. Na⁺ channel inactivation combined with delayed voltage-gated K⁺ channel opening (K⁺ efflux)**

Repolarization results from the combination of Na⁺ channels inactivating (shutting off further depolarizing current) and voltage-gated K⁺ channels opening with a delay, driving K⁺ efflux that brings the membrane back toward (and often transiently past) resting potential.

† original question, not a verified past-exam item

Q60. A drug that selectively blocks only T-type (low-threshold) Ca²⁺ channels would be expected to most directly affect which physiological...

✓ **Correct: B. Thalamic rhythmic burst firing, relevant to absence seizure pathophysiology (ethosuximide's mechanism)**

T-type Ca²⁺ channels underlie the rhythmic burst-firing behavior of thalamocortical neurons implicated in absence seizure generation — this is exactly the target of ethosuximide, a first-line absence seizure medication.

† original question, not a verified past-exam item

Lecture 4 — Metabolic Pathways of Neurons (20 questions)

Q61. Which of the following is INCORRECT regarding neuronal metabolism?

✓ **Correct: C. Neurons store large amounts of glycogen for backup fuel**

Neurons store essentially NO glycogen — astrocytes are the brain's glycogen reservoir, not neurons; this is a key reason neurons depend on a steady glucose/lactate supply from surrounding glia.

PROF'S TRICK: Glycogen storage in the CNS = astrocyte job, not neuron. This single fact underlies why the astrocyte-neuron lactate shuttle exists in the first place — neurons need a buffer they don't keep themselves.

Q62. What enzyme hydrolyzes glutamine to glutamate inside the presynaptic neuron, regenerating the neurotransmitter pool?

✓ **Correct: B. Glutaminase**

Glutaminase (phosphate-activated glutaminase, PAG) converts glutamine back into glutamate inside neurons, completing the glutamate-glutamine cycle after astrocytes have converted the neuron's released glutamate into (non-excitotoxic) glutamine.

✗ *A is wrong: Glutamine synthetase runs the OPPOSITE reaction (glutamate → glutamine) and is located in ASTROCYTES, not neurons — a very commonly reversed pair.*

✗ *C is wrong: Glutamate decarboxylase converts glutamate into GABA (in GABAergic neurons), a completely different fate for glutamate, not glutamine regeneration.*

PROF'S TRICK: Glutamine SYNTHETASE (astrocyte, makes glutamine) vs GlutaminASE (neuron, breaks down glutamine back to glutamate). One letter of difference in the name, opposite direction, opposite cell — the exam's favorite pair to swap.

Q63. Excess extracellular glutamate causes excitotoxic neuronal injury primarily through excessive intracellular accumulation of which ion?

✓ **Correct: C. Ca²⁺**

Overactivation of NMDA (and AMPA) receptors lets excessive Ca²⁺ flood into the postsynaptic neuron, triggering destructive enzyme cascades (proteases, lipases, free radical formation) — the hallmark of excitotoxicity seen in stroke/ischemia and hyperammonemia.

Q64. A 58-year-old man with cirrhosis develops confusion and drowsiness with markedly elevated serum ammonia (hepatic encephalopathy). Which...

✓ **Correct: B. Astrocytes, by converting glutamate + ammonia into glutamine via glutamine synthetase**

Astrocytes are the brain's main ammonia detoxifiers: glutamine synthetase combines ammonia with glutamate to form glutamine, trapping the nitrogen in a non-toxic form — but this consumes large amounts of glutamate and ATP, and excess glutamine itself is what drives astrocyte swelling in hepatic encephalopathy.

✗ *A is wrong: Neurons run glutaminase, the opposite reaction (breaking glutamine down) — they don't have significant ammonia-detoxifying capacity; that job belongs to astrocytic glutamine synthetase.*

† original question, not a verified past-exam item

PROF'S TRICK: This is a real exam-style integrative case: liver failure → ↑ammonia → astrocytes overwork glutamine synthetase → glutamine accumulates intracellularly → osmotic astrocyte swelling → cerebral edema/confusion. Multi-step 'why' chains like this are exactly what a good prof will test.

Q65. In the same hepatic encephalopathy scenario, why does excessive glutamine accumulation in astrocytes become harmful rather than simply...

✓ **Correct: B. Glutamine acts as an osmolyte, drawing water into astrocytes and causing cellular swelling/cerebral edema**

Glutamine is osmotically active; when astrocytes are forced to make large amounts of it to buffer ammonia, intracellular osmolarity rises and water follows, causing astrocyte (and eventually brain) swelling — a major contributor to the cerebral edema seen in acute hepatic encephalopathy.

† original question, not a verified past-exam item

Q66. Why can hyperammonemia impair neuronal function even though ammonia detoxification happens mainly in astrocytes rather than neurons?

✓ **Correct: A. Because astrocyte dysfunction/swelling disrupts the glutamate-glutamine cycle and lactate shuttle that neurons depend on for neurotransmitter recycling and fuel**

Neurons and astrocytes are metabolically interdependent (lactate shuttle, glutamate-glutamine cycle); when astrocytes are overwhelmed and swollen from ammonia detoxification, this whole support system for neurons breaks down, indirectly impairing neuronal neurotransmission and energy supply even though neurons themselves aren't doing the detoxifying.

† original question, not a verified past-exam item

PROF'S TRICK: Big-picture takeaway questions like this ('why does X affect Y indirectly') are how a prof tests whether you understand the SYSTEM, not just isolated facts. Always ask yourself 'what depends on what?' when reviewing a pathway.

Q67. Which transporter class is primarily responsible for astrocytic uptake of the lactate that astrocytes then export to neurons?

✓ **Correct: C. Monocarboxylate transporters (MCT1/4)**

Monocarboxylate transporters (MCT1/4 on astrocytes, MCT2 on neurons) move lactate (and pyruvate/ketone bodies) across cell membranes — the physical conduit of the astrocyte-neuron lactate shuttle.

✗ *A is wrong: GLUT3 is a neuronal GLUCOSE transporter, not a lactate transporter — glucose and lactate use entirely different transporter families.*

† *original question, not a verified past-exam item*

Q68. Which glucose transporter is characteristic of astrocytes, versus the one preferentially expressed in neurons?

✓ **Correct: A. Astrocyte: GLUT1; Neuron: GLUT3**

Astrocytes preferentially express GLUT1 (lower affinity, matches their role as the glucose-sensing/glycolytic 'first responder'); neurons preferentially express GLUT3 (higher affinity, ensures neurons can still pull in glucose even at low extracellular concentrations).

† *original question, not a verified past-exam item*

PROF'S TRICK: Higher-affinity transporter goes to the cell type that absolutely cannot afford to run out of substrate — neurons (GLUT3) win the 'first dibs on glucose' contest over astrocytes (GLUT1).

Q69. The pentose phosphate pathway (PPP), activated in neurons especially under high glucose availability, primarily generates which product...

✓ **Correct: B. NADPH, which regenerates reduced glutathione (GSH) to neutralize reactive oxygen species**

The PPP's main output relevant here is NADPH, the essential reducing cofactor for regenerating glutathione (GSSG → GSH) and recycling ascorbate — both central to neutralizing oxidative stress, to which neurons are especially vulnerable given their high metabolic rate.

† *original question, not a verified past-exam item*

Q70. Which micronutrient deficiency is classically associated with Wernicke-Korsakoff syndrome, disrupting neuronal energy metabolism?

✓ **Correct: A. Vitamin B1 (thiamine)**

Thiamine is a required cofactor for pyruvate dehydrogenase and α -ketoglutarate dehydrogenase (both key TCA cycle enzymes); its deficiency (classically in chronic alcoholism) starves neurons of ATP production and causes the classic triad of confusion, ataxia, and ophthalmoplegia.

† *original question, not a verified past-exam item*

PROF'S TRICK: Thiamine deficiency = TCA cycle bottleneck (can't run pyruvate/ α -KG dehydrogenase efficiently) = ATP crisis in the most metabolically-demanding brain regions first (mammillary bodies, thalamus) — that's why Wernicke's classically involves those structures.

Q71. Astrocytic glycogen stores are mobilized during which physiological circumstance?

✓ **Correct: B. Periods of high neuronal activity or low glucose availability, to supply additional lactate to neurons**

Astrocytic glycogenolysis provides an additional local lactate source during periods when neuronal demand outpaces glucose delivery (e.g., intense synaptic activity, transient hypoglycemia) — a buffering role neurons themselves cannot perform since they store essentially no glycogen.

† *original question, not a verified past-exam item*

Q72. Which statement about ascorbic acid (vitamin C) recycling in the CNS is correct?

✓ **Correct: A. Astrocytes help regenerate reduced ascorbate using NADPH from the pentose phosphate pathway**

Oxidized ascorbate (from neutralizing free radicals) is regenerated back to its reduced, active form using reducing power ultimately derived from the PPP/NADPH system in astrocytes — part of the broader antioxidant network alongside glutathione.

† *original question, not a verified past-exam item*

Q73. Which of the following best summarizes the overall direction of the glutamate-glutamine cycle across the synapse?

✓ **Correct: A. Glutamate released by the neuron → taken up by astrocyte via EAAT → converted to glutamine by glutamine synthetase → glutamine shuttled back to neuron → converted back to glutamate by glutaminase**

This is the full loop: presynaptic neuron releases glutamate → astrocyte EAAT uptake protects against excitotoxicity → astrocytic glutamine synthetase converts it to glutamine (non-excitotoxic, energy-requiring) → glutamine transported back to the neuron → neuronal glutaminase regenerates glutamate for repackaging into vesicles.

† original question, not a verified past-exam item

PROF'S TRICK: Draw this cycle out as a loop diagram when you study it — 'released → uptaken → converted → shuttled back → reconverted' — because exam questions love to test you on ANY single arrow in this loop out of context, and you need to know which cell and which enzyme belongs to each step.

Q74. Why do astrocytes convert glutamate to glutamine before returning it to neurons, rather than simply shuttling glutamate straight back?

✓ **Correct: A. Glutamine is inert/non-excitotoxic and safe to transport through the extracellular space, unlike glutamate which would over-stimulate receptors along the way**

Converting glutamate to glutamine 'neutralizes' it — glutamine doesn't activate glutamate receptors, so it can safely diffuse through the extracellular space back to neurons without inappropriately exciting other nearby synapses, and neurons then reconvert it locally right where it's needed.

† original question, not a verified past-exam item

Q75. A patient with a genetic deficiency of astrocytic glutamine synthetase would be expected to show which combination of findings?

✓ **Correct: A. Elevated extracellular glutamate (impaired clearance/recycling) with risk of excitotoxicity, plus low glutamine levels**

Without glutamine synthetase, astrocytes cannot efficiently convert taken-up glutamate into glutamine — glutamate handling backs up (risking excitotoxic buildup) and downstream glutamine supply to neurons (needed to regenerate the neurotransmitter pool) drops, a genuinely described rare metabolic disease (glutamine synthetase deficiency) causing severe neonatal encephalopathy.

† original question, not a verified past-exam item

PROF'S TRICK: This is a 'predict the phenotype from the enzyme defect' question — a favorite critical-thinking format. Always trace: which reaction is blocked → what backs up (increases) → what runs short (decreases) → what's the clinical consequence?

Q76. Compared to resting conditions, during intense neuronal activity astrocytic ATP levels tend to:

✓ **Correct: B. Drop, because both glutamate uptake (via Na⁺/K⁺-ATPase-dependent EAAT-linked transport) and its conversion to glutamine by glutamine synthetase are energy-intensive processes**

Astrocytes pay a real ATP cost to support active neurons — both the Na⁺-dependent glutamate uptake and the ATP-requiring glutamine synthetase reaction draw down astrocytic ATP during periods of high synaptic activity, part of why they ramp up glycolysis/lactate production to compensate.

† original question, not a verified past-exam item

Q77. Which best explains why neurons are said to 'prefer' lactate over glucose as a fuel source when both are available in sufficient quantity?

✓ **Correct: B. Lactate can be converted directly to pyruvate by neuronal LDH-1 and fed straight into the TCA cycle, and this pathway is kinetically favored in neurons under these conditions**

Experimental and kinetic evidence shows that when both substrates are present, neuronal LDH-1 (which favors converting lactate to pyruvate) and the overall enzymatic/kinetic profile of neurons favors oxidizing lactate first via the TCA cycle — this is the basis for the 'lactate preference' described in the astrocyte-neuron lactate shuttle hypothesis.

✗ **A is wrong:** Lactate does not yield more ATP per molecule than full glucose oxidation — the preference is about kinetics/pathway efficiency under the shuttle model, not raw energy yield per molecule.

† original question, not a verified past-exam item

Q78. Why does the brain, despite being only about 2% of body weight, consume roughly 20% of the body's resting oxygen and glucose supply?

✓ **Correct: B. Because of the continuous, high energetic cost of restoring ionic gradients (Na⁺/K⁺-ATPase) after action potentials and synaptic activity across billions of neurons, plus baseline maintenance of resting potentials**

The brain's disproportionate energy consumption reflects the sheer scale of continuous electrochemical signaling occurring across its ~86 billion neurons — restoring ionic gradients after every action potential and synaptic event via the Na⁺/K⁺-ATPase, plus baseline resting potential maintenance, adds up to an enormous, continuous ATP demand relative to the organ's small mass.

✗ *D is wrong: The brain notably does NOT store significant glycogen (astrocytes hold a small amount, but nowhere near enough for sustained independent function) — this is exactly why the brain depends so critically on continuous glucose/lactate delivery from the blood, and why hypoglycemia so rapidly impairs brain function.*

† original question, not a verified past-exam item

Q79. Which vitamin, besides thiamine, is a critical cofactor for the glutamate-glutamine cycle and broader amino acid metabolism, with...

✓ **Correct: A. Vitamin B6 (pyridoxine), a cofactor for transaminase/decarboxylase reactions including GABA synthesis**

Pyridoxine (B6) is an essential cofactor for many amino acid-handling enzymes, including glutamate decarboxylase (GABA synthesis) and various transaminases — both deficiency and excess of B6 are associated with peripheral neuropathy.

† original question, not a verified past-exam item

Q80. During intense, sustained neuronal activity beyond what the lactate shuttle can immediately supply, which backup fuel source can neurons...

✓ **Correct: A. Ketone bodies (e.g., beta-hydroxybutyrate), which cross the BBB and can be oxidized for ATP production**

During prolonged fasting or ketogenic states, ketone bodies (produced by the liver from fatty acid oxidation) can cross the blood-brain barrier and be used by neurons as an alternative oxidative fuel source when glucose/lactate supply is limited — the physiological basis of ketogenic diets in some epilepsy treatment.

✗ *B is wrong: Free fatty acids themselves cross the BBB very poorly — the liver must first convert them into ketone bodies, which ARE able to cross efficiently; this distinction is exactly why ketone bodies (not fatty acids directly) are the brain's fasting-adapted alternative fuel.*

† original question, not a verified past-exam item

Lecture 5 — Embryonic Development of the Nervous System (18 questions)

Q81. Which embryonic structure gives rise to the substantia nigra?

✓ **Correct: E. Mesencephalon**

The substantia nigra is a midbrain (mesencephalon) structure — it develops in place as part of the mesencephalic vesicle, not by migrating from another region.

✗ *D is wrong: The diencephalon gives rise to the thalamus, hypothalamus, and related structures — it's immediately rostral to the mesencephalon, an easy neighboring-vesicle mix-up.*

PROF'S TRICK: Memorize the 5 secondary vesicles in order, rostral to caudal: Telencephalon, Diencephalon, Mesencephalon, Metencephalon, Myelencephalon — 'Tell Danny My Mom's Mad.' Substantia nigra = mesencephalon, no exceptions.

Q82. The optic cup is an evagination (outgrowth) of which structure?

✓ **Correct: B. Neuroectoderm of the diencephalon (optic vesicle)**

The optic vesicle (itself an outpouching of the diencephalic neuroectoderm) invaginates to form the double-walled optic cup — the retina's neuroectodermal origin.

✗ *A is wrong: Surface ectoderm forms the LENS placode/vesicle (and later the corneal epithelium) — a separate embryonic contribution that fuses with, but is distinct from, the optic cup.*

PROF'S TRICK: Two different germ-layer contributions build the eye: Neuroectoderm (diencephalon) → optic vesicle → optic cup → retina/optic nerve. Surface ectoderm → lens placode → lens, and → corneal epithelium.

Q83. A meningocele is characterized by protrusion of which structure(s) through a vertebral/cranial defect?

✓ **Correct: A. Meninges and CSF only, no neural tissue**

A meningocele is a herniation of meninges (and the CSF they contain) through a bony defect, WITHOUT neural tissue — generally a milder lesion than a myelomeningocele.

✗ *D is wrong: That combination (meninges + spinal cord) describes a myelomeningocele, the more severe form — a very frequently tested distinction.*

Q84. Which embryonic structure gives rise to the pineal gland?

✓ **Correct: A. Roof of the diencephalon (epithalamus)**

The pineal gland develops as an evagination of the diencephalic roof plate (epithalamus) — consistent with it being a genuine CNS/neuroectodermal structure.

Q85. Which functional component is characteristic of neurons in the hypoglossal nucleus, located at the level of the myelencephalon (medulla)?

✓ **Correct: E. GSE (general somatic efferent)**

The hypoglossal nerve (CN XII) is a pure motor nerve to tongue musculature (derived from occipital somites) — classified functionally as GSE, the same category as the somatic motor nuclei of III, IV, VI.

✗ *D is wrong: SVE (special visceral efferent) is the category for branchiomeric muscle motor nuclei (e.g., facial, trigeminal motor, nucleus ambiguus) — tongue muscles are NOT branchial arch derivatives, so hypoglossal is GSE, not SVE.*

PROF'S TRICK: CN XII (hypoglossal) and CN III/IV/VI (eye movers) are the only pure GSE cranial nerve nuclei — because tongue and extraocular muscles both derive from somites/somitomeres, not pharyngeal arches.

Q86. Which structure does the optic cup differentiate into?

✓ **Correct: C. Retina (both neural and pigmented layers) and epithelium of the iris/ciliary body**

The two layers of the optic cup become the neural retina (inner layer) and retinal pigment epithelium (outer layer), with the cup's rim extending forward to form the epithelial layers of the iris and ciliary body.

Q87. Which birth defect of the spinal cord/vertebral column is classically associated with hydrocephalus (via Arnold-Chiari II malformation)?

✓ **Correct: B. Myelomeningocele**

Myelomeningocele is strongly associated with Arnold-Chiari type II malformation (cerebellar tonsil/vermis herniation through the foramen magnum), which obstructs CSF flow and produces hydrocephalus.

Q88. Failure of the neural folds to fuse (neural tube defect) most directly results in which condition, if it occurs at the cranial end?

✓ **Correct: B. Anencephaly**

Failure of the ANTERIOR neuropore (cranial end of the neural tube) to close results in anencephaly — absence of most of the brain and calvaria; failure at the caudal (posterior neuropore) end produces spina bifida instead.

PROF'S TRICK: Anterior neuropore failure = Anencephaly (both start with 'A' — easy pairing). Posterior neuropore failure = Spina bifida (caudal end).

Q89. What is the developmental origin of the pituitary gland (hypophysis)?

✓ **Correct: B. Dual origin: Rathke's pouch (surface ectoderm, oral roof) forms the anterior lobe; a diencephalic downgrowth forms the posterior lobe**

The pituitary has a classic dual embryonic origin: the adenohypophysis (anterior lobe) arises from Rathke's pouch (an upward invagination of oral/surface ectoderm), while the neurohypophysis (posterior lobe) is a direct downgrowth of the diencephalic floor (neuroectoderm) — explaining why the posterior pituitary counts as CNS tissue.

PROF'S TRICK: This dual origin explains hormone physiology too: posterior pituitary just STORES/releases hormones made by hypothalamic neurons (it's neural tissue), while anterior pituitary actually SYNTHESIZES its own hormones (it's glandular/epithelial ectoderm).

Q90. The corneal endothelium develops from which embryonic source?

✓ **Correct: B. Neural crest cell migration**

Corneal endothelium (and much of the corneal stroma) derives from migrating neural crest cells (periocular mesenchyme), not the surface ectoderm (which instead forms the corneal EPITHELIUM).

✗ **A is wrong:** Surface ectoderm forms the outer corneal epithelium — the endothelium (innermost layer) is a neural crest derivative, a subtle but classic distinction.

Q91. Failure of closure of the anterior neuropore results in which condition?

✓ **Correct: A. Anencephaly**

Same fact restated: anterior neuropore failure = anencephaly.

Q92. The lens of the eye develops from which structure?

✓ **Correct: B. Surface ectoderm (via the lens placode/lens vesicle)**

The lens placode (an ectodermal thickening induced by the adjacent optic vesicle) invaginates to form the lens vesicle, which detaches from the surface ectoderm and differentiates into lens fibers.

Q93. Which structure forms the walls of the lateral ventricles, and what is its embryonic layer of origin?

✓ **Correct: A. Ependyma, derived from the neuroepithelium lining the neural tube lumen**

The ependymal lining of the ventricular system (including the walls of the lateral ventricles) is derived directly from the neuroepithelium that lined the lumen of the embryonic neural tube.

† original question, not a verified past-exam item

Q94. Neural crest cells that fail to migrate normally from the developing neural tube can cause which category of clinical disorders?

✓ **Correct: A. Neurocristopathies (e.g., Hirschsprung disease, some forms of neurofibromatosis)**

Neurocristopathies are a broad disease category caused by abnormal neural crest cell migration, proliferation, or differentiation — Hirschsprung disease (failure of enteric ganglion cell migration) is a classic example.

† original question, not a verified past-exam item

Q95. At which week of embryonic development does neural tube closure (both neuropores) normally complete, making this the critical window for...

✓ **Correct: A. By approximately the end of the 4th week**

Both neuropores normally close by around day 25-28 (end of the 4th week) — folic acid supplementation is recommended BEFORE and in early pregnancy precisely because this critical window often occurs before a woman even knows she's pregnant.

† original question, not a verified past-exam item

PROF'S TRICK: This timing fact is exactly why public health guidance recommends folic acid for ALL women of reproductive age, not just those already pregnant — by the time a missed period is noticed, neural tube closure may already be complete or nearly so.

Q96. A newborn has a tuft of hair and a dimple overlying the lower lumbar spine, with an otherwise normal neurological exam and no palpable...

✓ **Correct: B. Spina bifida occulta**

Spina bifida occulta is the mildest form — just a vertebral arch fusion defect, often with cutaneous markers (hair tuft, dimple, lipoma) directly over it, but with intact skin and typically no neurological deficit.

† original question, not a verified past-exam item

Q97. Which best describes the relationship between the neural tube lumen and the adult ventricular system?

✓ **Correct: B. The neural tube lumen persists and enlarges/reshapes into the ventricular system and central canal**

The central cavity of the embryonic neural tube is continuous with, and gives rise to, the adult ventricular system (lateral, third, fourth ventricles) and the central canal of the spinal cord — this is why the ventricles are lined by ependyma (neuroepithelial derivative), and why CSF fills a truly continuous cavity from the neural tube's original lumen.

† original question, not a verified past-exam item

Q98. Which structure connects the third ventricle to the fourth ventricle, and what would obstruction here (e.g., from a tectal plate tumor)...

✓ **Correct: B. Cerebral aqueduct (of Sylvius); obstruction causes non-communicating (obstructive) hydrocephalus affecting the lateral and third ventricles**

The cerebral aqueduct is a narrow channel especially vulnerable to obstruction (e.g., by a tumor compressing the tectal plate/quadrigeminal region); blocking it causes CSF to back up into the lateral and third ventricles, producing obstructive (non-communicating) hydrocephalus while the fourth ventricle remains normal-sized.

† original question, not a verified past-exam item

Lecture 6 — Development of the Eye (18 questions)

Q99. Sustained high intracranial pressure classically produces which triad of findings (Cushing's triad)?

✓ **Correct: A. Hypertension, bradycardia, irregular respirations**

Cushing's triad (rising ICP) = hypertension (widened pulse pressure), reflex bradycardia (baroreceptor response), and irregular respirations — a late, ominous sign of impending herniation.

PROF'S TRICK: Cushing's triad direction is easy to get backwards under exam pressure: BP goes UP, HR goes DOWN. Remember it's the body trying to maintain cerebral perfusion pressure against the rising ICP, which paradoxically triggers a baroreceptor-mediated bradycardia.

Q100. What does the ventral (inferior) part of the optic vesicle, at the point where it connects to the forebrain, give rise to?

✓ **Correct: B. The retinal fissure (choroid fissure), through which the hyaloid vessels pass into the developing eye**

The ventral surface of the optic vesicle/stalk/cup develops a groove — the retinal (choroid) fissure — allowing the hyaloid artery/vein to enter and supply the developing inner eye structures before it normally closes.

Q101. Failure of the choroid (retinal) fissure to fuse normally results in which congenital anomaly?

✓ **Correct: A. Coloboma (a gap/cleft in the iris, retina, or optic disc)**

A coloboma is a structural defect (notch/gap, classically in the inferior iris/retina/choroid/optic nerve) caused by incomplete fusion of the edges of the choroid fissure during the 5th-7th weeks.

PROF'S TRICK: Colobomas are classically INFERIOR because the choroid fissure runs along the ventral/inferior aspect of the developing optic cup/stalk — if you see 'notch in the iris/retina,' think coloboma, and remember its typical location.

Q102. The spinal trigeminal nucleus (carrying facial pain/temperature) is derived from which embryonic structure?

✓ **Correct: B. Alar plate of the developing brainstem (sensory column)**

Sensory brainstem nuclei (including the spinal trigeminal nucleus) derive from the alar plate — the dorsal, sensory-associated portion of the developing neural tube; motor nuclei derive from the ventral basal plate.

✗ **C is wrong:** The basal plate gives rise to MOTOR nuclei — the sulcus limitans is the developmental landmark separating dorsal alar (sensory) from ventral basal (motor), and this pattern holds throughout the brainstem and spinal cord.

PROF'S TRICK: Alar = Afferent/sensory (dorsal). Basal = Basically motor (ventral). This alar/basal split, established early in neural tube development, explains the entire dorsal-sensory/ventral-motor organization you see throughout the spinal cord and brainstem.

Q103. What is the embryonic origin of the vitreous body of the eye?

✓ **Correct: A. Mesenchyme (partly neural crest, partly mesoderm) within the optic cup, plus contributions from the hyaloid vascular system**

The vitreous develops from mesenchymal cells within the cavity of the optic cup (with an early, transient contribution from the hyaloid vessel system before those vessels regress), not purely from any single ectodermal derivative.

† original question, not a verified past-exam item

Q104. The extraocular muscles that move the eyeball are derived embryologically from which tissue?

✓ **Correct: B. Prechordal and paraxial head mesoderm (specialized somitomeres)**

Extraocular muscles arise from cranial paraxial mesoderm/somitomes (not neural crest, which instead forms the surrounding connective tissue/tendons) — this is why they're innervated by the somatic-efferent (GSE) cranial nerves III, IV, and VI, matching their somite-like origin.

✗ A is wrong: Neural crest contributes the **CONNECTIVE TISSUE** and tendons around the extraocular muscles, and much of the periocular mesenchyme, but the contractile muscle fibers themselves come from paraxial mesoderm.

† original question, not a verified past-exam item

Q105. Persistence of the hyaloid artery after birth (a common incidental finding) reflects failure of which normal developmental process?

✓ **Correct: A. Failure of the hyaloid vessels to regress after supplying the developing lens/vitreous**

The hyaloid vessels normally regress once the lens and vitreous no longer need this transient blood supply; a persistent hyaloid artery (sometimes seen as a Mittendorf dot on the lens or Bergmeister's papilla on the disc) is simply incomplete regression of this embryonic vasculature.

† original question, not a verified past-exam item

Q106. A newborn is found to have microphthalmia (an abnormally small eye) with an associated inferior coloboma. What is the most likely...

✓ **Correct: B. A disruption during optic vesicle/cup formation and/or choroid fissure closure occurring early in the 5th-7th weeks of development**

Both microphthalmia and coloboma often co-occur because they can result from a single early disruption during the critical period of optic vesicle/cup morphogenesis and choroid fissure closure — a shared embryological timing window links these seemingly separate anomalies.

† original question, not a verified past-exam item

PROF'S TRICK: When two anomalies show up together on an exam vignette, always ask: could they share the **SAME** embryological window/process? This kind of linkage question is a favorite way to test whether you understand timing, not just isolated facts.

Q107. Which layer of the optic cup becomes the retinal pigment epithelium (RPE), and which becomes the neurosensory retina?

✓ **Correct: A. Outer layer → RPE; inner layer → neurosensory retina**

The optic cup is double-walled: its thin OUTER layer becomes the single-cell-thick retinal pigment epithelium, while the thicker INNER layer differentiates into the multilayered neurosensory retina (photoreceptors, bipolar cells, ganglion cells, etc.).

† original question, not a verified past-exam item

Q108. Congenital cataract in a newborn (lens opacity present at birth) most directly implies a disruption of which developmental process?

✓ **Correct: A. Lens placode induction/lens vesicle formation and differentiation of lens fibers**

Congenital cataracts reflect abnormal lens development — whether from genetic defects in lens fiber protein (crystallin) synthesis or intrauterine insults (e.g., congenital rubella) disrupting normal lens placode/vesicle differentiation.

† original question, not a verified past-exam item

PROF'S TRICK: Congenital rubella syndrome classically causes cataracts (plus sensorineural deafness and cardiac defects) — a good example of how a single teratogen exposure during the critical embryonic window can hit multiple organ systems developing at that same time.

Q109. Which statement correctly relates the timing of eye development to overall CNS/neural tube development?

✓ **Correct: B. The eye develops as an outgrowth of the diencephalon, so major eye anomalies often co-occur with, or hint at, other forebrain developmental problems**

Because the optic vesicle/cup is literally an evagination of the diencephalon, severe early disruptions to forebrain development (e.g., holoprosencephaly) frequently also produce major eye anomalies (like cyclopia or severe microphthalmia) — the eye is functionally 'part of the brain' embryologically.

† original question, not a verified past-exam item

PROF'S TRICK: Holoprosencephaly (failure of the forebrain to properly divide into two hemispheres) classically presents with midline facial/ocular anomalies (up to cyclopia in severe cases) — a dramatic illustration of just how tightly eye and forebrain development are linked.

Q110. Which of the following eye structures is NOT derived from neuroectoderm?

✓ **Correct: D. Lens**

The lens is a surface ectoderm derivative (via the lens placode/vesicle) — everything else listed (retina, RPE, optic nerve, iris/ciliary epithelium) comes from the neuroectodermal optic cup/stalk.

† original question, not a verified past-exam item

Q111. A patient with anophthalmia (complete absence of one eye) most likely had a disruption at which developmental stage?

✓ **Correct: A. Failure of optic vesicle formation/outgrowth from the diencephalon very early in development**

True anophthalmia (as opposed to severe microphthalmia, which can be hard to distinguish clinically) reflects a very early failure of the optic vesicle to form or evaginate from the diencephalon in the first place — the earliest possible point of failure in the entire eye developmental sequence.

† original question, not a verified past-exam item

Q112. Why are the extraocular muscles innervated by cranial nerves III, IV, and VI classified as GSE (general somatic efferent) rather than...

✓ **Correct: A. Because they derive from paraxial head mesoderm/somitomeres (like body wall/limb somatic muscle), not from pharyngeal arch (branchiomeric) mesoderm**

Functional component classification tracks embryonic muscle origin, not location or function — since extraocular muscles come from somite-like paraxial mesoderm (the same lineage as GSE-innervated body wall muscle), their innervation is classified GSE, unlike branchiomeric (pharyngeal arch-derived) muscles which get SVE innervation (e.g., muscles of mastication, facial expression).

† original question, not a verified past-exam item

PROF'S TRICK: This is a great 'why' question to test true understanding versus rote memorization — the functional component system is entirely embryology-based, not based on where a structure sits or what it superficially does.

Q113. Which statement about the development of the cornea is correct?

✓ **Correct: A. The corneal epithelium is from surface ectoderm; the stroma and endothelium are largely from neural crest-derived mesenchyme**

The cornea is a layered structure with mixed embryonic origins: outer epithelium from surface ectoderm, while the underlying stroma and inner endothelium derive from migrating neural crest-derived mesenchyme — this layered dual origin is a favorite integrative fact.

† original question, not a verified past-exam item

Q114. Which structure gives rise to the ciliary body and its ciliary muscle, important for lens accommodation?

✓ **Correct: B. The rim of the optic cup (neuroectoderm) for the epithelium, with associated mesenchyme for the ciliary muscle**

Like the iris, the ciliary body epithelium derives from the rim of the optic cup (neuroectoderm), with the ciliary muscle itself forming from adjacent neural crest-derived mesenchyme — continuing the pattern of the optic cup's rim giving rise to multiple anterior eye structures.

† original question, not a verified past-exam item

Q115. Aniridia (congenital absence of the iris) is most directly linked to mutations affecting which category of developmental genes?

✓ **Correct: A. PAX6, a master regulatory transcription factor for eye development**

PAX6 is a master transcription factor essential for eye morphogenesis at multiple stages; PAX6 mutations are classically linked to aniridia (and other ocular anomalies) — a good example of how a single 'master switch' gene defect can derail an entire developmental cascade.

† original question, not a verified past-exam item

Q116. Congenital rubella syndrome can cause cataracts, sensorineural deafness, and cardiac defects together. What does this triad best...

✓ **Correct: A. A single teratogen exposure during a critical shared developmental window can simultaneously disrupt multiple organ systems that are undergoing critical morphogenesis at that same time**

The lens, inner ear, and heart are all undergoing critical, vulnerable morphogenesis during a similar early gestational window; a single teratogen (rubella virus) crossing the placenta during that window can disrupt all three simultaneously — illustrating the general teratology principle that WHEN an insult occurs matters as much as WHAT the insult is.

† original question, not a verified past-exam item

Lecture 7 — Spinal Cord: Blood Supply & Clinical Correlation (18 questions)

Q117. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which finding is expected on the SAME (right)...

✓ **Correct: B. Loss of conscious proprioception and vibration, plus spastic paralysis**

Hemisection knocks out the IPSILATERAL dorsal columns (proprioception/vibration) and the ipsilateral lateral corticospinal tract (spastic weakness) below the lesion, because these tracts haven't crossed yet at that spinal level; pain/temperature loss is CONTRALATERAL because the spinothalamic tract has already crossed.

✗ A is wrong: Pain and temperature loss is CONTRALATERAL (opposite side), not ipsilateral — the spinothalamic tract crosses near its level of entry, so damage is felt on the opposite side of the body below the lesion.

PROF'S TRICK: Brown-Séquard's core logic: dorsal columns + corticospinal tract = ipsilateral loss (they cross in the MEDULLA, high up, so below a cord lesion they're still uncrossed on that side). Spinothalamic tract = contralateral loss (it crosses within 1-2 SEGMENTS of entering the cord, so it's already crossed by the time it reaches the lesion level).

Q118. A 65-year-old man has lost vibration, position sense, and joint discrimination in the legs, with preserved pain and temperature...

✓ **Correct: B. Dorsal column-medial lemniscus pathway**

Vibration, proprioception, and fine touch/discrimination are dorsal column-medial lemniscus functions; sparing of pain/temperature (spinothalamic) points to a selective dorsal column lesion — classically seen in conditions like tabes dorsalis (tertiary syphilis) or vitamin B12 deficiency (subacute combined degeneration).

PROF'S TRICK: This 'dissociated sensory loss' pattern (one modality out, another spared) is a classic clinical reasoning trick — always map which specific tract carries which specific modality, because these questions test whether you can localize a lesion from the symptom pattern alone.

Q119. Nuchal rigidity (stiff neck) in a patient is a compensatory sign classically seen with which condition?

✓ **Correct: B. Meningeal irritation (e.g., meningitis or subarachnoid hemorrhage)**

Neck stiffness limiting flexion is a protective reflex against stretching inflamed/irritated meninges, classically seen in meningitis or subarachnoid hemorrhage — tested clinically with Kernig's and Brudzinski's signs.

Q120. An injury damages the LATERAL HORN of the spinal cord gray matter (present at T1-L2/L3). Which function is most directly lost?

✓ **Correct: B. Preganglionic sympathetic output**

The lateral horn (intermediolateral cell column), present only at thoracolumbar levels, contains preganglionic sympathetic neuron cell bodies — its destruction disrupts sympathetic outflow from that segment.

✗ *A is wrong: Somatic motor neurons are in the VENTRAL (anterior) horn, a completely different gray matter region present at ALL spinal levels, not just T1-L2/L3.*

PROF'S TRICK: Lateral horn = ONLY present T1-L2/L3 (sympathetic preganglionic neurons) plus S2-S4 has an analogous parasympathetic cell column. If a question mentions 'lateral horn,' it's testing autonomic (usually sympathetic), not somatic motor, function.

Q121. All of the following statements concerning the anterior spinal artery are correct EXCEPT:

✓ **Correct: E. It is reinforced only at the cervical level, with no other reinforcing radicular arteries lower down**

This is FALSE (hence the correct EXCEPT answer): the anterior spinal artery is reinforced along its length by multiple segmental radicular arteries, most famously the large artery of Adamkiewicz in the lower thoracic/upper lumbar region — reinforcement is not limited to the cervical level.

PROF'S TRICK: Artery of Adamkiewicz (major radicular artery, usually T9-T12) is a classic exam name-drop — its damage (e.g., during aortic surgery) causes anterior spinal artery syndrome affecting the lower cord.

Q122. A 53-year-old man is hospitalized for acute spinal cord infarction (anterior spinal artery territory). Which sensory modality is...

✓ **Correct: C. Vibration and proprioception (dorsal column)**

Anterior spinal artery syndrome spares the dorsal columns (supplied instead by the paired posterior spinal arteries), producing a classic dissociated pattern: loss of motor function, pain, and temperature, with PRESERVED vibration/proprioception.

PROF'S TRICK: Anterior spinal artery syndrome = motor + pain/temp LOST, proprioception/vibration SPARED (dorsal columns have their own separate posterior spinal artery supply). This is the mirror image of the posterior spinal artery syndrome pattern.

Q123. Which of the following is NOT part of the central nervous system?

✓ **Correct: C. Dorsal root ganglion**

The dorsal root ganglion sits just outside the spinal cord (in or near the intervertebral foramen) and is a PNS structure housing primary sensory neuron cell bodies.

Q124. If the anterior spinal artery is occluded, which structure/function is spared, consistent with its blood supply territory?

✓ **Correct: C. Dorsal column (vibration/proprioception) function, supplied by the posterior spinal arteries instead**

Same key fact restated from the syndrome above: the dorsal columns are spared because they're supplied by the posterior spinal arteries, a separate vascular territory from the anterior spinal artery.

Q125. Which finding is a primary MOTOR symptom of a complete spinal cord lesion (transection) below the level of injury?

✓ **Correct: B. Spastic paralysis (initially flaccid in spinal shock, then spastic) below the lesion**

Complete cord transection interrupts descending corticospinal (and other motor) tracts, producing an acute phase of flaccid paralysis/areflexia ('spinal shock') that evolves over days-weeks into chronic spastic paralysis with hyperreflexia below the lesion, once spinal reflex circuits recover from the initial shock.

PROF'S TRICK: Don't forget the *TIME COURSE*: immediately after transection = *FLACCID* (spinal shock), then *LATER* = *SPASTIC*. A question describing the acute setting expects flaccid; one describing weeks later expects spastic — a classic timing trap.

Q126. Which structure extends from the pia mater to the dura mater along the lateral aspect of the spinal cord, anchoring it and separating...

✓ **Correct: A. Denticulate ligament**

The denticulate ligaments are lateral, tooth-like pial extensions that pierce the arachnoid to attach to the dura, stabilizing the spinal cord within the subarachnoid space and providing a useful surgical landmark between dorsal and ventral roots.

✗ **B is wrong:** The *filum terminale* anchors the *CAUDAL* tip of the spinal cord/conus medullaris to the coccyx — a longitudinal, midline structure, not the lateral tooth-like structures separating the roots.

Q127. A patient with posterior spinal artery territory infarction would be expected to lose which function, with relative sparing of motor and...

✓ **Correct: A. Vibration and proprioception (dorsal columns)**

Posterior spinal artery syndrome is the mirror image of anterior spinal artery syndrome — it selectively knocks out the dorsal columns (vibration/proprioception), sparing the anterolaterally-located corticospinal and spinothalamic tracts.

Q128. A patient with a complete spinal cord lesion at C4 would be expected to have which respiratory consequence?

✓ **Correct: B. Potential diaphragmatic paralysis / respiratory failure, since the phrenic nerve (C3-C5) supplying the diaphragm would be disconnected from higher control**

The phrenic nerve arises from C3-C5 ('C3, 4, 5 keep the diaphragm alive'); a lesion at or above C4 disconnects the phrenic motor neurons from higher respiratory drive centers, risking life-threatening diaphragmatic paralysis and ventilator dependence — this is exactly why high cervical cord injuries are so much more dangerous than lower ones.

† original question, not a verified past-exam item

PROF'S TRICK: 'C3, 4, 5 keep the diaphragm alive' — one of the highest-yield mnemonics in all of neuroanatomy for predicting which cord injuries are immediately life-threatening versus 'merely' disabling.

Q129. Which spinal cord region is most vulnerable to ischemia because it lies in a watershed zone between the anterior spinal artery and the...

✓ **Correct: B. The mid-thoracic cord (roughly T4-T8), which has relatively sparse radicular artery reinforcement**

The mid-thoracic cord is a classic vascular watershed zone — it has the fewest reinforcing radicular arteries feeding the anterior spinal artery along its length, making it especially vulnerable to hypotension-related ischemia (a well-known cause of postoperative or perioperative spinal cord injury, e.g., during aortic surgery).

† original question, not a verified past-exam item

Q130. Which combination of findings would distinguish a central cord syndrome from a complete cord transection?

✓ **Correct: A. Central cord syndrome causes disproportionately greater weakness in the arms than the legs, with variable sensory loss, sparing some function; complete transection causes total loss of everything below the level**

Central cord syndrome (classically from hyperextension injury in an already-stenotic cervical canal) preferentially damages the centrally-located cervical/upper limb corticospinal fibers (somatotopically arranged with arm fibers more medial/central than leg fibers), producing arm-greater-than-leg weakness — quite different from the total, symmetric loss below a complete transection.

† original question, not a verified past-exam item

PROF'S TRICK: Somatotopic organization within the corticospinal tract (cervical/arm fibers more medial, sacral/leg fibers more lateral) is exactly why 'incomplete' cord syndromes have such distinctive, localizable patterns — always ask 'what's centrally versus peripherally located in this tract' when a partial-lesion vignette comes up.

Q131. A 25-year-old man sustains a traumatic complete transection of the spinal cord at T6. In the IMMEDIATE hours after injury, which finding...

✓ **Correct: B. Flaccid paralysis with absent reflexes (spinal shock)**

Immediately after complete cord transection, the cord below the lesion enters 'spinal shock' — a temporary state of flaccid paralysis and areflexia — before evolving into the chronic spastic, hyperreflexic pattern over subsequent days to weeks.

✗ *A is wrong: Spasticity/hyperreflexia is the CHRONIC pattern that develops later, not the immediate finding — a classic timing trap already tested elsewhere in this set, worth repeating because it's so frequently missed.*

† original question, not a verified past-exam item

Q132. Which spinal cord segment level is most classically associated with the anal wink reflex (afferent and efferent both via the pudendal...

✓ **Correct: B. S2-S4**

The anal wink (bulbocavernosus-family) reflex tests the sacral (S2-S4) reflex arc via the pudendal nerve — a key bedside test for assessing intact sacral cord function, especially after suspected cauda equina or conus injury.

† original question, not a verified past-exam item

Q133. Which vascular territory watershed makes the conus medullaris and cauda equina especially vulnerable during aortic surgery involving...

✓ **Correct: A. Anterior spinal artery, especially where reinforcement from the artery of Adamkiewicz is compromised**

The lower cord/conus depends heavily on anterior spinal artery flow reinforced by the artery of Adamkiewicz; aortic cross-clamping (or aneurysm repair) can compromise this segmental reinforcement, precipitating spinal cord ischemia in exactly this watershed distribution.

† original question, not a verified past-exam item

Q134. Which spinal level marks the typical inferior extent of the adult spinal cord (conus medullaris), below which lumbar puncture is safely...

✓ **Correct: A. L1-L2 in adults**

Due to differential growth rates between the vertebral column and spinal cord during development, the adult conus medullaris typically terminates around L1-L2 — lumbar puncture is therefore performed at L3-L4 or L4-L5 to safely avoid the cord itself.

† original question, not a verified past-exam item

Lecture 8 — PNS: Cranial & Spinal Nerve Organization (18 questions)

Q135. Which functional component is found in the hypoglossal nucleus?

✓ **Correct: E. GSE**

Repeated core fact: hypoglossal nerve (CN XII) = pure motor to tongue muscles = GSE.

Q136. According to the organization of the PNS, which statement below is INCORRECT?

✓ **Correct: C. There is only one functional component in a peripheral nerve**

This is the FALSE statement (hence correct as the 'INCORRECT' answer): peripheral nerves are typically MIXED, containing multiple functional components simultaneously (e.g., GSA + GVE + SVE in a single cranial nerve like the facial nerve).

Q137. Collections of nerve cell bodies located inside the peripheral nervous system are called:

✓ **Correct: C. Ganglia**

Ganglia = PNS cell body clusters. The CNS equivalent term is 'nucleus'; a bundle of axons is a 'tract' in the CNS or a 'nerve' in the PNS.

✗ *E is wrong: 'Nuclei' is the correct term, but specifically for CNS cell body collections — a PNS collection is called a ganglion, not a nucleus.*

PROF'S TRICK: Cell bodies: CNS = nucleus, PNS = ganglion. Axon bundles: CNS = tract, PNS = nerve. Four terms, two locations, memorize the grid.

Q138. Which functional component is NOT found in the facial nerve (CN VII)?

✓ **Correct: E. GVA (general visceral afferent from thoracic/abdominal viscera)**

The facial nerve is a genuinely mixed nerve with SVE, GVE, SVA, and a small GSA component (external ear) — but it does NOT carry GVA fibers from thoracic/abdominal viscera; that's characteristic of the vagus nerve instead.

PROF'S TRICK: CN VII is famous for carrying FOUR different functional components — a favorite 'which one does it NOT have' trick question, since students often assume a nerve either has one component or 'all of them.'

Q139. Which spinal nerves are NOT included in a somatic nerve plexus?

✓ **Correct: B. T2-T12 (thoracic nerves, which mostly run as individual intercostal nerves)**

Most thoracic spinal nerves (roughly T2-T12) do NOT form a plexus — they run individually as intercostal (and subcostal) nerves, unlike the cervical, brachial, and lumbosacral levels which do interweave into plexuses.

Q140. A 41-year-old woman has recurring episodes of sharp, electric-shock-like facial pain triggered by chewing and touching her cheek. Which...

✓ **Correct: B. Trigeminal nerve (trigeminal neuralgia)**

This is the classic presentation of trigeminal neuralgia — brief, severe, electric-shock-like facial pain in a trigeminal (V2/V3 most commonly) distribution, often triggered by light touch, chewing, or talking.

Q141. Which functional component of nerve fibers is NOT found in a typical spinal nerve?

✓ **Correct: E. SVE**

SVE (special visceral efferent, i.e., branchiomeric muscle motor) is exclusive to certain CRANIAL nerves (V, VII, IX, X, XI) whose target muscles derive from pharyngeal arches — spinal nerves never carry SVE, since body wall/limb muscles are GSE, not SVE.

PROF'S TRICK: Spinal nerves carry only 4 functional components: GSA, GVA, GSE, GVE — never SVA or SVE (those are cranial-nerve-only categories tied to special senses and branchiomeric muscle).

Q142. Which nucleus contains general visceral efferent (GVE) preganglionic parasympathetic neurons supplying the salivary and lacrimal glands?

✓ **Correct: B. Superior and inferior salivatory nuclei**

The superior salivatory nucleus (via CN VII) supplies lacrimal, submandibular, and sublingual glands; the inferior salivatory nucleus (via CN IX) supplies the parotid gland — both are GVE parasympathetic nuclei.

✗ **D is wrong:** Edinger-Westphal is also a GVE parasympathetic nucleus, but it controls the pupillary sphincter/ciliary muscle (via CN III), not salivary/lacrimal glands — a commonly confused parasympathetic nucleus pairing.

PROF'S TRICK: Four cranial parasympathetic (GVE) nuclei to keep straight: Edinger-Westphal (CN III → pupil/lens), Superior salivatory (CN VII → lacrimal/submandibular/sublingual), Inferior salivatory (CN IX → parotid), Dorsal motor nucleus of vagus (CN X → thoracic/abdominal viscera).

Q143. Which statement about a transverse section of the caudal pons is correct?

✓ **Correct: A. It contains the abducens (VI) and facial (VII) nerve nuclei/fascicles, plus the facial colliculus on the 4th ventricle floor**

The caudal pons houses CN VI and VII nuclei; the facial nerve fibers loop around the abducens nucleus internally, creating the visible bump of the facial colliculus on the floor of the 4th ventricle — a classic internal-brainstem landmark.

✗ **B is wrong:** The hypoglossal nucleus is in the MEDULLA, not the pons — one level too caudal for this question.

Q144. A patient cannot detect light with the optic disc (physiologic blind spot). What is the anatomical reason?

✓ **Correct: A. The optic disc is where retinal blood vessels and the optic nerve exit, and it contains no photoreceptors**

The optic disc is simply the exit point for retinal ganglion cell axons (forming the optic nerve) and the entry/exit of central retinal vessels — there's no room for photoreceptors there, creating the normal physiological blind spot.

✗ **D is wrong:** The fovea (highest photoreceptor/cone density, sharpest vision) is a DIFFERENT location on the retina, lateral to the optic disc — mixing these two up is a very common error.

PROF'S TRICK: Optic disc = blind spot (no photoreceptors, where the nerve exits). Fovea = sharpest vision (photoreceptor-dense, especially cones). They're anatomically separate points on the retina — don't confuse 'exit point' with 'sharpest vision point.'

Q145. Which dermatome landmark is listed INCORRECTLY?

✓ **Correct: E. S1 - forehead**

S1 corresponds to the lateral foot/heel/little toe area, NOT the forehead — the forehead is trigeminal (CN V) territory, not a spinal dermatome at all, making 'S1 = forehead' the clearly false pairing.

PROF'S TRICK: High-yield dermatome landmarks: C6=thumb, C7=middle finger, C8=little finger, T4=nipple, T10=umbilicus, L1=groin, L4=medial malleolus/knee, S1=lateral foot/little toe. Memorize this ladder cold — it's tested constantly.

Q146. Which nerve fiber type is responsible for increased salivation and constriction of the urinary sphincter... wait, which fiber type...

✓ **Correct: B. Increased salivation = parasympathetic GVE (cranial outflow); urinary sphincter contraction = sympathetic GVE (thoracolumbar outflow)**

Salivation is a classic parasympathetic (craniosacral outflow, GVE) effect; contraction of the internal urethral/urinary sphincter is a sympathetic (thoracolumbar outflow, GVE) alpha-1-mediated effect — both are GVE fibers, but from opposite autonomic divisions.

✗ **E is wrong:** This reverses both assignments — salivation is parasympathetic, and sphincter contraction is sympathetic, not the other way around.

Q147. A lesion of the vagus nerve (CN X) on one side would be expected to cause which finding on indirect laryngoscopy/palate exam?

✓ **Correct: B. Uvula deviates AWAY from the side of the lesion (toward the intact side), because the weak side cannot elevate the palate**

On phonation ('say ah'), the palate on the healthy side elevates normally while the weak (denervated) side sags — the uvula gets pulled toward the stronger, intact side, i.e., AWAY from the lesion.

✗ **A is wrong:** This is the exact opposite — the uvula moves away from (not toward) the side of a vagus/palate weakness, a frequently reversed clinical sign.

† original question, not a verified past-exam item

PROF'S TRICK: 'The curtain rises unevenly' — uvula points away from the weak side. This is the mirror-image logic of tongue deviation with CN XII lesions, where the tongue deviates TOWARD the weak (lesioned) side instead — don't mix up these two opposite deviation rules!

Q148. Which structure, if compressed by a mass at the cerebellopontine angle, would classically cause combined sensorineural hearing loss AND...

✓ **Correct: B. A mass affecting both CN VII and CN VIII together as they run through the internal acoustic meatus (e.g., vestibular schwannoma / acoustic neuroma)**

CN VII and VIII run together through the internal acoustic meatus at the cerebellopontine angle; a vestibular schwannoma (arising from CN VIII's vestibular division) growing large enough can compress the adjacent facial nerve too, producing the classic combination of hearing loss/tinnitus/vertigo PLUS ipsilateral facial weakness.

† original question, not a verified past-exam item

PROF'S TRICK: Cerebellopontine angle tumors are a favorite 'which nerves travel together' integration question — CN V, VII, and VIII are all anatomically close there, so a large enough mass can pick off any combination as it grows.

Q149. A patient shows loss of the corneal reflex and facial sensory loss, but normal facial MOVEMENT (motor function intact). Which nerve is...

✓ **Correct: B. Trigeminal nerve (CN V), sensory (ophthalmic) division**

Since motor function (facial movement) is intact but sensation and the corneal reflex's AFFERENT limb are lost, the lesion is isolated to CN V (sensory), not CN VII (motor) — a clean dissociation between the two nerves that together mediate the full corneal reflex arc.

† original question, not a verified past-exam item

Q150. Which of the following is the correct composition of a typical mixed spinal nerve regarding its roots?

✓ **Correct: A. Dorsal root (sensory, GSA+GVA) + Ventral root (motor, GSE+GVE) unite to form the mixed spinal nerve**

This is the Bell-Magendie law: dorsal (posterior) root = sensory afferents (GSA, GVA) with the dorsal root ganglion; ventral (anterior) root = motor efferents (GSE, GVE) with no ganglion — they unite just distal to the DRG to form the mixed spinal nerve.

† original question, not a verified past-exam item

Q151. Which cranial nerve provides both taste to the anterior 2/3 of the tongue AND general sensation to a small area of the external ear?

✓ **Correct: C. Facial nerve (CN VII)**

CN VII is famously multi-component: SVE (facial expression), GVE (lacrimal/salivary glands), SVA (anterior 2/3 tongue taste via chorda tympani), and a small GSA contribution to part of the external ear/ear canal — the same nerve tested elsewhere in this set for its four functional components.

† original question, not a verified past-exam item

Q152. Damage to the recurrent laryngeal nerve (a branch of the vagus) would be expected to cause which finding?

✓ **Correct: A. Hoarseness/vocal cord paralysis, due to loss of motor innervation to most intrinsic laryngeal muscles**

The recurrent laryngeal nerve supplies most intrinsic laryngeal muscles (controlling vocal cord movement); its injury (e.g., during thyroid surgery, given its close anatomical relationship to the thyroid gland) classically causes hoarseness or vocal cord paralysis.

† original question, not a verified past-exam item

Lecture 9 — Brainstem I: Internal Structures & Localization (17 questions)

Q153. An MRI shows a tumor compressing the right tuberculum cinereum at the closed (caudal) medulla. Which nerve's nucleus is most directly...

✓ **Correct: A. Trigeminal nerve (spinal trigeminal nucleus/tract)**

The tuberculum cinereum overlies the spinal trigeminal nucleus and tract in the closed medulla — a classic surface landmark for underlying trigeminal sensory structures.

PROF'S TRICK: 'Tuberculum cinereum' = trigeminal spinal nucleus landmark. If you see this phrase anywhere in a stem, your brain should immediately go to CN V (trigeminal), specifically its spinal sensory nucleus for pain/temperature of the face.

Q154. Which reflex test best represents intact function of the medulla oblongata?

✓ **Correct: A. Gag reflex**

The gag reflex depends on CN IX (afferent) and CN X (efferent), both with nuclei in the medulla — a classic 'brainstem is intact' bedside test, especially relevant in assessing brain death/coma.

X E is wrong: The corneal reflex tests CN V (afferent) and CN VII (efferent), both PONTINE nuclei — one level rostral to the medulla, a common mix-up with the gag reflex.

PROF'S TRICK: Brainstem reflex ladder (rostral to caudal): Pupillary light reflex (midbrain, CN II/III) → Corneal reflex (pons, CN V/VII) → Gag reflex (medulla, CN IX/X). Testing this ladder in order is exactly how brain death exams localize the level of brainstem dysfunction.

Q155. A 20-year-old man has loss of muscle power on the LEFT side of his body plus strabismus and ptosis of the RIGHT eye. Where is the lesion?

✓ **Correct: C. Rostral midbrain, right side (crossed hemiplegia — Weber syndrome)**

This is a classic Weber syndrome (medial midbrain syndrome): a right-sided lesion damages the exiting right CN III fascicles (ipsilateral oculomotor palsy — ptosis, strabismus) AND the right cerebral peduncle/corticospinal fibers before they've crossed (causing LEFT, contralateral, hemiplegia).

PROF'S TRICK: 'Crossed' brainstem syndromes (cranial nerve deficit on ONE side, body weakness on the OTHER) are the single highest-yield brainstem localization pattern — the cranial nerve tells you WHICH side and WHICH level; the body deficit is always contralateral because corticospinal fibers haven't crossed yet at that point.

Q156. Which neuronal pathway performs voluntary control of body and limb movements?

✓ **Correct: E. Lateral corticospinal (pyramidal) tract**

The lateral corticospinal tract is THE primary pathway for voluntary, skilled limb movement, descending from motor cortex, crossing at the pyramidal decussation, and synapsing on lower motor neurons.

✗ **A is wrong:** The reticulospinal tract is involved in more automatic postural/gross movement control (an extrapyramidal pathway) — it is NOT the pathway for fine voluntary control, a common trap distractor.

Q157. Which CNS structure lies at approximately the same horizontal (axial) level as the tentorial notch/incisura, useful for localizing...

✓ **Correct: B. Midbrain**

The midbrain passes through the tentorial incisura (the opening in the tentorium cerebelli) — this is exactly why UNCAL (transtentorial) herniation compresses the midbrain first, producing the classic ipsilateral CN III palsy (blown pupil) of a rapidly expanding supratentorial mass.

Q158. A 32-year-old man has loss of muscle power and a positive Babinski sign on the LEFT side, with right-sided tongue deviation on...

✓ **Correct: B. Right medial medulla (medial medullary syndrome — affecting CN XII and the pyramid/medial lemniscus)**

Medial medullary syndrome (occlusion of the anterior spinal artery/vertebral artery branches) damages the ipsilateral (right) hypoglossal nerve fascicles — causing tongue deviation TOWARD the lesion (right) on protrusion — plus the pyramid (contralateral, left, corticospinal weakness/Babinski) and medial lemniscus (contralateral dorsal column loss).

PROF'S TRICK: Tongue deviates TOWARD the side of a CN XII (hypoglossal) lesion because the genioglossus muscle on the weak side can't push the tongue away — combine that with contralateral body weakness and you've localized a crossed medial medullary syndrome instantly.

Q159. A patient has a left pupil larger than the right, plus ptosis and an eye deviated 'down and out.' Which structure is most likely compressed?

✓ **Correct: B. Left oculomotor nerve (CN III), e.g., from uncal herniation**

A dilated ('blown'), unreactive pupil plus ptosis plus a down-and-out eye is the classic complete CN III palsy triad — the parasympathetic fibers controlling pupil constriction run on the OUTSIDE of the nerve, making them especially vulnerable to compression (as opposed to ischemic CN III palsies, which tend to spare the pupil).

PROF'S TRICK: 'Surgical' (compressive) CN III palsy = pupil involved (blown, dilated) because the parasympathetic fibers sit peripherally and get compressed first. 'Medical' (ischemic, e.g., diabetic) CN III palsy = pupil SPARED, because the vasa nervorum ischemia hits the central fibers first, sparing the peripheral parasympathetic fibers. This pupil-sparing distinction is one of the most classic tested contrasts in all of neurology.

Q160. A patient has loss of pain and temperature sensation on the CONTRALATERAL body plus an IPSILATERAL facial pain/temperature deficit, plus...

✓ **Correct: B. Lateral medullary (Wallenberg) syndrome — PICA territory**

Wallenberg syndrome (lateral medullary syndrome, classically from PICA or vertebral artery occlusion) hits the spinal trigeminal nucleus/tract (ipsilateral facial pain/temp loss), the lateral spinothalamic tract (contralateral body pain/temp loss), the inferior cerebellar peduncle (ipsilateral ataxia), descending sympathetics (ipsilateral Horner's), and nucleus ambiguus (hoarseness/dysphagia) — a huge, high-yield symptom cluster.

PROF'S TRICK: Wallenberg = 'everything BUT motor' (the pyramids/corticospinal tract are anteromedial, spared in this LATERAL medullary infarct) — that absence of weakness is itself a diagnostic clue, since most other brainstem strokes DO cause some limb weakness.

Q161. Which tract's second-order neurons cross the midline at the level of the medulla oblongata (rather than immediately upon entering the...

✓ **Correct: B. Dorsal column-medial lemniscus pathway (crosses as internal arcuate fibers in the caudal medulla)**

Dorsal column axons ascend ipsilaterally all the way up to the caudal medulla, synapse in the gracile/cuneate nuclei, and THEN the second-order neurons cross as internal arcuate fibers to form the contralateral medial lemniscus — a distinctly later crossing point than the spinothalamic tract's early spinal cord crossing.

✗ **A is wrong:** Spinothalamic tract fibers cross much earlier and lower — within about 1-2 spinal segments of entering the cord via the anterior white commissure — not at the medulla.

PROF'S TRICK: This crossing-LEVEL difference (spinothalamic crosses low, in the cord; DCML crosses high, in the medulla) is EXACTLY why Brown-Séquard syndrome produces opposite-side patterns for these two pathways below a cord lesion, but why a medullary lesion can selectively hit DCML crossing fibers alone.

Q162. Damage to which major component of the Ascending Reticular Activating System (ARAS) most directly impairs consciousness?

✓ **Correct: D. Pons and midbrain (rostral brainstem reticular formation)**

The ARAS core is the pontomesencephalic (pons + midbrain) reticular formation, which projects to the thalamus and cortex to maintain wakefulness — bilateral damage here (or to its thalamic/cortical targets) is a classic cause of coma.

Q163. What is the major anatomical site whose bilateral damage produces coma via ARAS disruption?

✓ **Correct: B. Pons and midbrain reticular formation**

Same core concept: bilateral pontomesencephalic reticular formation damage (or bilateral thalamic/diffuse cortical damage) is required to produce coma — UNILATERAL cortical damage alone typically does not cause coma given the bilateral/diffuse nature of ARAS projections.

PROF'S TRICK: Coma requires either (1) bilateral brainstem ARAS damage, or (2) bilateral (diffuse) cortical/thalamic damage — a UNILATERAL hemispheric lesion alone, however large, usually does NOT cause coma unless it secondarily compresses the brainstem (e.g., via herniation).

Q164. A patient presents with altered consciousness, irregular breathing patterns, and decorticate/decerebrate posturing. This clinical...

✓ **Correct: B. Brainstem (reticular formation) dysfunction, possibly from herniation or direct injury**

Altered consciousness plus abnormal respiratory patterns plus abnormal posturing together localize to brainstem/reticular formation dysfunction (often from rising ICP causing herniation) — respiratory pattern and posturing type can even help localize the LEVEL of brainstem compromise (e.g., decorticate = above red nucleus, decerebrate = below it, at/below midbrain).

PROF'S TRICK: Decorticate (arms flexed, legs extended) = lesion ABOVE the red nucleus (better prognosis). Decerebrate (arms AND legs extended) = lesion AT OR BELOW the red nucleus, more caudal, worse prognosis. As the lesion progresses caudally, posturing gets worse — a useful serial-exam trend to recognize.

Q165. Which cranial nerve nuclei are found at the level of the pons (not the medulla or midbrain)?

✓ **Correct: B. CN V, VI, VII, VIII**

CN V (trigeminal motor/main sensory), VI (abducens), VII (facial), and VIII (vestibulocochlear) nuclei are all located in the pons — a useful grouping for localizing pontine lesions by which nerves are affected.

† original question, not a verified past-exam item

PROF'S TRICK: Cranial nerve nuclei by level: Midbrain = III, IV. Pons = V, VI, VII, VIII. Medulla = IX, X, XI, XII (plus V's spinal nucleus extends down here too). Memorizing this ladder lets you localize ANY brainstem lesion just from which cranial nerves are affected.

Q166. Which structure, if damaged bilaterally in the pons, classically produces 'locked-in syndrome' (preserved consciousness and vertical eye...

✓ **Correct: A. Bilateral ventral pons (corticospinal and corticobulbar tracts), sparing the tegmentum/reticular formation**

Locked-in syndrome results from a lesion (classically basilar artery occlusion) confined to the VENTRAL pons, destroying descending motor pathways (corticospinal/corticobulbar) while SPARING the dorsal tegmentum where the ARAS and vertical gaze centers live — hence consciousness and vertical eye movement are preserved even though the patient cannot move or speak.

† original question, not a verified past-exam item

PROF'S TRICK: This is a devastating and famous localization lesson: ventral pons = 'the highway out' (motor tracts passing through). Dorsal pons/tegmentum = 'the control room' (reticular formation, cranial nerve nuclei). Damage the highway but spare the control room, and you get a fully conscious, aware patient trapped in a paralyzed body — locked-in syndrome.

Q167. Why does a lesion of the medial longitudinal fasciculus (MLF) classically cause internuclear ophthalmoplegia (INO)?

✓ **Correct: A. The MLF connects the abducens nucleus on one side to the contralateral oculomotor (medial rectus) subnucleus, coordinating conjugate horizontal gaze**

The MLF is the internuclear connection allowing the abducens nucleus (lateral gaze) to simultaneously signal the contralateral oculomotor nucleus's medial rectus subnucleus, so both eyes move together during horizontal conjugate gaze; damaging it (classically in multiple sclerosis) causes impaired adduction of the ipsilateral eye on attempted lateral gaze, with nystagmus in the abducting eye.

† original question, not a verified past-exam item

PROF'S TRICK: INO is a favorite MS localization question: the affected side is named for which eye fails to ADDUCT (not the side of the primary MLF lesion's laterality convention some students misremember) — always work out the pathway from scratch rather than memorizing a shortcut, since the naming convention trips people up.

Q168. Which best explains why a lesion of one medial longitudinal fasciculus impairs ADDUCTION of the ipsilateral eye specifically, rather...

✓ **Correct: A. Because the MLF specifically carries the signal from the contralateral abducens nucleus to the ipsilateral oculomotor nucleus's medial rectus subnucleus, which is what drives eye adduction during conjugate horizontal gaze**

The MLF is specifically the relay from the contralateral abducens nucleus (lateral gaze signal) to the ipsilateral oculomotor medial rectus subnucleus — since medial rectus is the adductor muscle, damaging this specific relay selectively impairs adduction on attempted conjugate gaze, sparing other eye movements that don't depend on this particular circuit.

† original question, not a verified past-exam item

Q169. Which best explains why lesions affecting the pyramids (corticospinal tract) at the level of the medulla produce contralateral limb...

✓ **Correct: A. Because the pyramidal decussation is where corticospinal fibers cross from one side to the other; a lesion at ANY point along this pathway (above or at the decussation, but before it has crossed) will still ultimately produce weakness on the side OPPOSITE the lesion, once you account for where the crossing occurs relative to the lesion**

Understanding exactly WHERE a tract crosses (the decussation) relative to a lesion is what lets you correctly predict whether a deficit will be ipsilateral or contralateral to that lesion — since corticospinal fibers cross at the cervicomedullary junction (pyramidal decussation), any lesion ABOVE this point (cortex, internal capsule, midbrain, pons, upper medulla) produces CONTRALATERAL weakness, while a lesion BELOW this point (in the spinal cord, after crossing) produces IPSILATERAL weakness.

† original question, not a verified past-exam item

PROF'S TRICK: This single organizing question — 'has this tract crossed yet at the level of the lesion?' — is THE master key for localizing ANY motor or sensory deficit anywhere in the CNS. Work it out from first principles (where does this tract cross?) rather than memorizing lesion-by-lesion lists, and you can solve localization questions you've never even seen before.

Lecture 10 — Reticular Formation & ARAS (17 questions)

Q170. Which of the following correctly pairs a reticular formation column with its function?

✓ **Correct: A. Median column (raphe nuclei) — ascending projections for arousal and attention**

The median column (raphe nuclei) sends ascending serotonergic projections toward the cortex/thalamus, important for arousal, attention, and mood regulation.

PROF'S TRICK: Three reticular formation columns, three jobs: Median (raphe) = ascending, arousal/attention/mood. Medial (central) = intrinsic brainstem reflexes (eye movement, swallowing). Lateral = descending, muscle tone/respiration/blood pressure control.

Q171. Fibers from which reticular formation structure synapse with the motor nuclei of CN V and CN VII to help mediate brainstem reflexes like...

✓ **Correct: B. Medial (central) column, via short intrinsic interneuronal connections**

The medial (central) reticular column provides the local, intrinsic interneuronal connections linking sensory input to motor cranial nerve nuclei for brainstem reflexes (eye movements, swallowing, and other coordinated reflex arcs).

Q172. Which region does the Ascending Reticular Activating System (ARAS) NOT directly project to?

✓ **Correct: E. The cerebellar deep nuclei as its primary/defining target**

The ARAS's defining projections are to the thalamus (intralaminar nuclei), basal forebrain, and hypothalamus, which then relay diffusely to the cortex to maintain arousal — the cerebellum is not its primary/defining target pathway.

Q173. What is the general term for the diffuse network of interconnected neurons running through the core of the brainstem, from the medulla...

✓ **Correct: A. Reticular formation**

The reticular formation is exactly this: a diffuse, interconnected neuronal network extending through the core of the brainstem from the medulla up through the midbrain, involved in arousal, autonomic regulation, and motor/reflex modulation.

† original question, not a verified past-exam item

Q174. The reticular formation contributes to autonomic regulation of which vital functions via its lateral (descending) column?

✓ **Correct: A. Respiration and blood pressure (cardiovascular) control**

The lateral reticular column sends descending projections to the spinal cord that help regulate muscle tone, respiratory rhythm (via connections to respiratory centers), and blood pressure (via connections to vasomotor centers) — vital, life-sustaining autonomic functions.

† original question, not a verified past-exam item

PROF'S TRICK: This is exactly why brainstem strokes/injuries at this level are so dangerous — the same reticular formation regulating consciousness is intermingled with the circuits controlling breathing and blood pressure, so brainstem injury can simultaneously threaten wakefulness AND vital signs.

Q175. Damage to serotonergic raphe nuclei neurons has been implicated in which category of clinical conditions?

✓ **Correct: A. Mood disorders (e.g., depression) and sleep-wake regulation abnormalities**

The raphe nuclei are the brain's principal source of serotonergic projections; dysfunction here is mechanistically linked to mood disorders and to sleep-wake/circadian regulation abnormalities, which is also why many antidepressants target serotonin reuptake.

† original question, not a verified past-exam item

Q176. Which reticular formation nucleus group is the principal source of noradrenergic (norepinephrine) projections implicated in arousal and...

✓ **Correct: B. Locus coeruleus**

The locus coeruleus (in the pontine tegmentum, part of the reticular formation's noradrenergic system) is the principal source of norepinephrine to the CNS, driving arousal and stress-related attentional shifts.

✗ **C is wrong:** Substantia nigra is DOPAMINERGIC (relevant to Parkinson's disease and motor control), a different neurotransmitter system entirely from the noradrenergic locus coeruleus — easy to confuse since both are small, pigmented, brainstem-associated nuclei.

† original question, not a verified past-exam item

PROF'S TRICK: Three key brainstem neurotransmitter 'factories': Locus coeruleus = norepinephrine. Raphe nuclei = serotonin. Substantia nigra (+ VTA) = dopamine. All feed diffusely into forebrain circuits regulating arousal, mood, and reward — a favorite triad to test.

Q177. Why is the reticular formation considered anatomically and functionally distinct from discrete, well-bounded nuclei like the facial...

✓ **Correct: A. It consists of a diffuse, interconnected network of neurons with widespread dendritic/axonal arborization rather than a tight, well-circumscribed nuclear boundary**

The 'reticular' (net-like) name itself reflects this diffuse, interwoven organization — unlike a sharply-bounded motor nucleus with a specific, singular function, the reticular formation is a loosely-organized network integrating many different inputs and outputs across multiple modalities (arousal, autonomic, motor, sensory modulation).

† original question, not a verified past-exam item

Q178. Which best describes the circadian rhythm-related function attributed in part to reticular formation and related brainstem/hypothalamic...

✓ **Correct: A. Regulating the sleep-wake cycle in coordination with the suprachiasmatic nucleus**

Brainstem arousal circuits (ARAS) work together with the hypothalamic suprachiasmatic nucleus (the master circadian clock) to coordinate the sleep-wake cycle — ARAS provides the arousal 'switch,' while the SCN provides circadian timing input.

† original question, not a verified past-exam item

Q179. A patient with a lesion selectively damaging the lateral reticular formation column (sparing the median and medial columns) would most...

✓ **Correct: B. Disrupted descending control of muscle tone, respiration, and blood pressure, with relatively preserved arousal and intrinsic brainstem reflexes**

Since the lateral column specifically carries DESCENDING projections to the spinal cord for tone/respiration/blood pressure control, a selective lesion there (theoretically) would primarily disrupt those descending autonomic/motor functions, while the median (arousal) and medial (intrinsic reflex) columns' functions would be relatively preserved — a good test of whether you understand that these three columns are functionally, not just anatomically, distinct.

† original question, not a verified past-exam item

PROF'S TRICK: Whenever an exam question describes a SELECTIVE lesion of one subsystem, always ask what SHOULD be spared, not just what's lost — testing both halves of the prediction (what's affected AND what's preserved) is exactly how a rigorous prof checks true understanding versus memorized association.

Q180. Which of the following would be the MOST specific bedside test to assess pontomesencephalic (upper brainstem) ARAS integrity in a...

✓ **Correct: B. Pupillary light reflex and oculoccephalic (dolls eyes) / oculovestibular (caloric) reflexes**

Pupillary and oculoccephalic/oculovestibular reflexes directly test midbrain and pontine circuits central to the ARAS pathway — their presence or absence is one of the most specific bedside tools for localizing the level of brainstem dysfunction in coma, more so than spinal reflexes or blood pressure alone.

† original question, not a verified past-exam item

PROF'S TRICK: Preserved brainstem reflexes (pupils reactive, dolls-eyes present) with NO purposeful response = suggests a metabolic/diffuse cortical cause of coma with an intact brainstem, a very different (and often more reversible) situation than absent brainstem reflexes, which point to primary brainstem injury/herniation — a crucial clinical distinction.

Q181. Which neurotransmitter system, originating largely from the reticular formation, is most associated with the transition FROM NREM sleep...

✓ **Correct: A. Increased norepinephrine (locus coeruleus) and serotonin (raphe), decreased acetylcholine**

The transition to wakefulness is driven by increased noradrenergic (locus coeruleus) and serotonergic (raphe) tone, along with cholinergic and histaminergic (tuberomammillary nucleus) arousal systems — a reversal of the pattern seen during REM sleep, when NE/serotonin drop and ACh rises.

✗ **B is wrong:** This pattern (↓NE/serotonin, ↑ACh) actually characterizes the shift INTO REM sleep from NREM, not the transition to wakefulness — an easy directionality mix-up.

† original question, not a verified past-exam item

PROF'S TRICK: Sleep-wake neurotransmitter direction is a favorite reversed-logic trap: Wake = high NE/serotonin/ACh/histamine (all arousal systems firing). REM = low NE/serotonin but high ACh (a very particular, 'paradoxical' combination). NREM = intermediate, generally low arousal-system activity across the board.

Q182. The reticular formation's role in modulating pain (e.g., via descending projections to the dorsal horn) is mediated in part through...

✓ **Correct: A. Descending raphespinal (serotonergic) and reticulospinal (noradrenergic) pathways that can inhibit dorsal horn pain transmission**

Descending projections from the raphe nuclei (serotonergic) and other reticular formation regions (noradrenergic, e.g., via the locus coeruleus) synapse in the dorsal horn and can inhibit ascending pain transmission — part of the endogenous descending pain modulation system, which also interacts with the gate-control/enkephalin mechanisms.

† original question, not a verified past-exam item

PROF'S TRICK: This descending pain modulation system is exactly why certain antidepressants (SNRIs, which boost both serotonin AND norepinephrine) are also effective for chronic pain conditions — the same descending raphespinal/reticulospinal pathways are being pharmacologically enhanced.

Q183. Damage to the ARAS from a large bilateral thalamic stroke would most likely produce which clinical picture?

✓ **Correct: B. Coma or severely depressed consciousness, since the thalamus is a critical relay for ARAS projections to the cortex**

Bilateral thalamic damage (particularly involving the intralaminar nuclei, the key ARAS relay to the cortex) can produce coma or severely depressed consciousness even without direct brainstem damage — because the thalamus is the obligatory relay station between the brainstem ARAS and the diffuse cortical arousal it's trying to produce.

† original question, not a verified past-exam item

Q184. Why is a UNILATERAL cortical stroke, however large, generally insufficient to cause coma on its own (unless secondary herniation occurs)?

✓ **Correct: A. Because the ARAS provides bilateral, diffuse projections to both hemispheres, so the intact hemisphere and brainstem can still maintain arousal**

Since ARAS projections diffusely and bilaterally innervate the cortex/thalamus, an intact contralateral hemisphere plus intact brainstem can still sustain a baseline level of arousal even after a large unilateral cortical injury — true coma generally requires bilateral hemispheric/thalamic dysfunction OR direct bilateral brainstem injury, OR secondary mass effect/herniation compressing the brainstem from a unilateral lesion.

† original question, not a verified past-exam item

PROF'S TRICK: This is why any unilateral stroke patient who becomes comatose should immediately raise concern for a *SECONDARY* process — rising intracranial pressure, hemorrhagic transformation, or herniation compressing the brainstem — rather than assuming the original unilateral lesion itself is now 'big enough' to directly cause coma.

Q185. Which best distinguishes the concepts of 'arousal' (wakefulness) from 'awareness' (content of consciousness), both relevant to reticular...

✓ **Correct: B. Arousal (eyes-open wakefulness, ARAS-dependent) can be present without full awareness (as in a vegetative state), showing these are dissociable aspects of consciousness**

Vegetative state patients demonstrate this dissociation clinically — they can have preserved sleep-wake cycles and eyes-open arousal (ARAS-dependent, brainstem intact) while lacking any evidence of awareness or purposeful behavior (which requires more diffuse, intact cortical function) — proving arousal and awareness are separable components of consciousness, not a single unified process.

† original question, not a verified past-exam item

PROF'S TRICK: This arousal-vs-awareness distinction is exactly why the vegetative state (eyes open, no awareness) is clinically and prognostically completely different from coma (eyes closed, no arousal OR awareness) — a subtle but high-yield conceptual distinction a rigorous exam will test.

Q186. Which best explains why damage to the reticular formation can simultaneously threaten both consciousness AND vital signs (breathing,...

✓ **Correct: A. Because the reticular formation's different functional columns (arousal, autonomic, motor-reflex) are anatomically intermingled within the same compact brainstem core, so injury severe enough to affect one system often affects the others too**

Unlike the cortex (where different functions are more spatially segregated across different lobes/gyri), the reticular formation's arousal (median), reflex/intrinsic (medial), and autonomic (lateral) columns are packed closely together within the compact brainstem — this anatomical proximity is exactly why brainstem strokes/injuries so often produce the dangerous combination of depressed consciousness PLUS respiratory/cardiovascular instability simultaneously, unlike a cortical stroke of similar size.

† original question, not a verified past-exam item

Lecture 11 — Special Sense Organs: Histology & Development (17 questions)

Q187. The inner ear (membranous labyrinth) develops embryologically from which structure?

✓ **Correct: B. Otic placode**

The otic placode invaginates to form the otic vesicle (otocyst), which gives rise to the entire membranous labyrinth.

† original question, not a verified past-exam item

PROF'S TRICK: Ear develops from *THREE* different embryonic pouches/clefts/placodes: external ear (1st cleft), middle ear (1st pouch), inner ear (otic placode, neuroectoderm) — three separate structures fusing into one organ, a favorite integrative exam theme.

Q188. The external auditory meatus develops from which embryonic structure?

✓ **Correct: B. First pharyngeal cleft**

The first pharyngeal cleft forms the external auditory meatus, while the corresponding first pharyngeal pouch forms the tympanic cavity and auditory tube.

✗ *A is wrong: Pouches (endoderm-lined) form the MIDDLE ear cavity; clefts (ectoderm-lined) form the EXTERNAL ear canal — an easy pouch/cleft swap.*

† original question, not a verified past-exam item

Q189. The malleus and incus (middle ear ossicles) are derived embryologically from which cartilage?

✓ **Correct: B. Meckel's cartilage (1st arch)**

Meckel's cartilage (1st pharyngeal arch) gives rise to the malleus and incus; Reichert's cartilage (2nd arch) gives rise to the stapes.

† original question, not a verified past-exam item

PROF'S TRICK: 1st arch (Meckel's) → malleus + incus. 2nd arch (Reichert's) → stapes. This mismatch (2 ossicles from arch 1, only 1 from arch 2) is a favorite 'which arch gives which ossicle' trick.

Q190. Which cell type in the olfactory epithelium serves as the stem cell population for continuous regeneration of olfactory receptor neurons?

✓ **Correct: C. Basal cells**

Basal cells are the progenitor/stem cell population of the olfactory epithelium, continuously regenerating olfactory receptor neurons throughout life.

† original question, not a verified past-exam item

PROF'S TRICK: Basal cells are the universal stem cell answer for BOTH olfactory epithelium AND taste buds — an easy fact to double up on two different exam questions.

Q191. The nasal placodes are induced by signals from which adjacent embryonic structure?

✓ **Correct: A. The developing forebrain (telencephalon)**

Nasal placodes form on the frontonasal prominence under inductive signals from the underlying forebrain — another example (alongside the eye) of the CNS directly inducing an adjacent special sense organ.

† original question, not a verified past-exam item

Q192. Which taste bud cell type is directly responsible for chemoreceptive taste transduction?

✓ **Correct: C. Gustatory (taste receptor) cells**

Taste receptor cells (Type II/III) within the taste bud are the chemosensory cells that transduce taste stimuli via microvilli projecting into the taste pore.

† original question, not a verified past-exam item

Q193. Which special sense epithelium is uniquely capable of continuous neurogenesis (producing new sensory NEURONS, not just supporting cells)...

✓ **Correct: C. Olfactory epithelium**

Olfactory receptor neurons are continuously regenerated from basal stem cells — a distinctive feature not shared by cochlear hair cells or retinal photoreceptors, which do not regenerate in adult humans.

✗ *E is wrong: Taste bud cells DO turn over regularly, but the taste receptor cells are specialized epithelial cells, not true neurons — the olfactory epithelium is unique in regenerating actual bipolar sensory NEURONS with axons that must find their way back into the CNS (olfactory bulb).*

† original question, not a verified past-exam item

Q194. The cochlear duct (scala media) contains which fluid, and how does its ionic composition differ from the adjacent scalae?

✓ **Correct: B. Endolymph, which is unusually high in K⁺ (more like intracellular fluid)**

The cochlear duct (scala media) is filled with endolymph, notable for its high K⁺ / low Na⁺ composition (unlike typical extracellular fluid) — this unusual ionic gradient, maintained by the stria vascularis, is what actually drives the receptor (transduction) current when hair cell channels open.

† original question, not a verified past-exam item

PROF'S TRICK: Endolymph (scala media) = high K⁺, resembles intracellular fluid. Perilymph (scala vestibuli & scala tympani) = low K⁺/high Na⁺, resembles typical extracellular fluid/CSF. This K⁺-driven transduction current (not Na⁺, unusually) is a favorite physiology trick question.

Q195. Cochlear hair cells are innervated by bipolar neurons whose cell bodies reside in which structure?

✓ **Correct: B. Spiral ganglion (within the modiolus of the cochlea)**

The spiral ganglion, coiled within the bony modiolus at the center of the cochlea, contains the bipolar first-order sensory neurons whose peripheral processes contact hair cells and whose central processes form the cochlear nerve.

✗ **C is wrong:** The vestibular ganglion (Scarpa's ganglion) serves the vestibular (balance) hair cells of the semicircular canals/utricle/sacculle, a separate ganglion from the cochlear spiral ganglion — even though both eventually join CN VIII.

† original question, not a verified past-exam item

Q196. Inner hair cells make up what proportion of the afferent fibers of the cochlear nerve, despite being far fewer in number than outer hair...

✓ **Correct: C. About 95%**

Despite inner hair cells being a minority of total hair cells, they receive about 95% of the afferent (sensory) innervation from the cochlear nerve — they are the true sensory transducers sending sound information to the brain; outer hair cells are mainly EFFERENT-controlled amplifiers (cochlear mechanical tuning), not primary sensory transmitters.

✗ **A is wrong:** 5% is actually the OUTER hair cell share of afferent innervation — the numbers are commonly swapped in student memory since outer hair cells are more numerous overall but carry far less of the actual afferent signal.

† original question, not a verified past-exam item

PROF'S TRICK: Outer hair cells: more numerous, mostly EFFERENT-controlled amplifiers (cochlear amplifier function), only ~5% of afferent fibers. Inner hair cells: fewer in number, but ~95% of afferent fibers — they're the ones actually 'talking' to the brain about sound.

Q197. Stereocilia of vestibular/cochlear hair cells are anchored where, and how do they relate to the kinocilium in mechanotransduction?

✓ **Correct: A. They are embedded in an overlying membrane (e.g., tectorial membrane in the cochlea) and their deflection relative to the kinocilium determines depolarization vs. hyperpolarization**

Stereocilia (graded in height, embedded in the tectorial membrane in the cochlea, or the otolithic/cupular membrane in the vestibular system) bend when the overlying membrane shifts; bending TOWARD the tallest stereocilia/kinocilium opens mechanically-gated K⁺ channels causing depolarization, while bending AWAY closes them, causing hyperpolarization.

† original question, not a verified past-exam item

PROF'S TRICK: Direction relative to kinocilium is everything: TOWARD kinocilium = depolarization = more neurotransmitter release = more afferent firing. AWAY from kinocilium = hyperpolarization = less firing. This single directional rule underlies hearing, balance (linear and angular acceleration detection), and is tested constantly.

Q198. The three semicircular canals detect which type of head movement?

✓ **Correct: B. Angular (rotational) acceleration, each canal oriented in a different plane**

The three semicircular canals (anterior, posterior, lateral/horizontal), each roughly orthogonal to the others, detect ANGULAR (rotational) acceleration in their respective planes via endolymph movement deflecting the cupula.

✗ **A is wrong:** Linear acceleration (including gravity/head tilt) is detected by the OTOLITH organs (utricle and sacculle), not the semicircular canals — a fundamental and frequently tested vestibular distinction.

† original question, not a verified past-exam item

PROF'S TRICK: Semicircular canals = ANGULAR/rotational acceleration (spinning). Utricle + sacculle (otolith organs) = LINEAR acceleration + static head position relative to gravity (tilting, elevator start/stop). Two different physical quantities, two different organ types.

Q199. Which histological feature distinguishes the utricle and saccule (otolith organs) from the semicircular canal ampullae?

✓ **Correct: A. The maculae of the utricle/saccule have an otolithic membrane embedded with calcium carbonate crystals (otoconia), adding inertial mass; the ampullary crista instead has a gelatinous cupula with no crystals**

Otoconia (calcium carbonate crystals) embedded in the otolithic membrane give the maculae extra inertial mass, allowing them to lag behind during linear acceleration/gravity shifts and deflect the underlying hair cell stereocilia — the ampullary cupula has no such crystals, relying instead on endolymph flow during rotational movement.

† original question, not a verified past-exam item

Q200. Which structure supports the organ of Corti and is set into vibration by traveling sound waves, with different regions tuned to...

✓ **Correct: B. Basilar membrane**

The basilar membrane's mechanical properties vary along its length (narrow/stiff at the base = high frequencies; wide/flexible at the apex = low frequencies), creating tonotopic organization — the physical basis of frequency discrimination.

† original question, not a verified past-exam item

PROF'S TRICK: Base of cochlea = narrow, stiff basilar membrane = HIGH frequency. Apex = wide, floppy basilar membrane = LOW frequency. This base-high/apex-low tonotopic map is the anatomical basis for frequency coding and gets tested via both anatomy and clinical (presbycusis pattern) questions.

Q201. Presbycusis (age-related hearing loss) classically affects which frequency range first, and which cochlear structure is most vulnerable?

✓ **Correct: B. High frequencies first, affecting outer hair cells at the cochlear base**

Presbycusis classically begins with HIGH-frequency hearing loss, reflecting cumulative damage to outer (and eventually inner) hair cells at the cochlear BASE — the base being tuned to high frequencies and cumulatively exposed to the most mechanical stress over a lifetime.

† original question, not a verified past-exam item

Q202. Which structure transmits sound vibrations from the tympanic membrane to the oval window, providing mechanical impedance matching...

✓ **Correct: A. The three ossicles (malleus, incus, stapes) acting as a lever system**

The ossicular chain (malleus-incus-stapes) provides mechanical impedance matching (via lever action and the area ratio between the larger tympanic membrane and smaller oval window) — without this amplification, most sound energy would simply reflect off the fluid-air interface rather than being transmitted into the cochlea.

† original question, not a verified past-exam item

Q203. The Eustachian (auditory) tube connects the middle ear to which structure, and what is its main physiological function?

✓ **Correct: A. The nasopharynx; equalizing middle ear pressure with atmospheric pressure**

The Eustachian tube connects the middle ear cavity to the nasopharynx, allowing pressure equalization (e.g., the 'ear popping' sensation during altitude changes) — dysfunction here (e.g., from congestion) is a common cause of middle ear pressure symptoms and conductive hearing changes.

† original question, not a verified past-exam item

Lecture 12 — Somatosensory Pathways (17 questions)

Q204. An MRI shows right-sided hemisection of the spinal cord (Brown-Séquard syndrome) at L2. Which neuronal tract/fasciculus passes through...

✓ **Correct: C. Lateral spinothalamic tract, as its second-order fibers cross to the opposite side**

Spinothalamic second-order neurons cross the midline via the anterior (ventral) white commissure, just anterior to the central canal, within 1-2 segments of entering the cord — this is the anatomical crossing point tested repeatedly in Brown-Séquard-type questions.

Q205. A patient has loss of proprioception on the LEFT side and loss of pain/temperature on the RIGHT side of the body below a spinal lesion....

✓ **Correct: B. Left-sided Brown-Séquard hemisection (ipsilateral dorsal column loss, contralateral spinothalamic loss)**

This exact split (proprioception lost on the SAME side as the lesion, pain/temp lost on the OPPOSITE side) is the signature of Brown-Séquard hemisection — here specifically a LEFT-sided lesion (ipsilateral dorsal column = left proprioception loss; contralateral spinothalamic = right pain/temp loss).

Q206. Which tract lesion classically gives a positive Romberg test (patient sways/falls with eyes closed but is stable with eyes open)?

✓ **Correct: B. Dorsal column-medial lemniscus pathway (proprioception loss compensated visually)**

A positive Romberg sign specifically indicates a SENSORY (proprioceptive, i.e., dorsal column) ataxia — vision compensates for the lost proprioceptive input, so the patient is stable with eyes open but becomes unsteady once that visual compensation is removed (eyes closed).

X D is wrong: Cerebellar (truly ataxic) patients are UNSTEADY REGARDLESS of whether their eyes are open or closed — that's a Romberg-negative pattern, and distinguishing a positive from negative Romberg is exactly how you differentiate sensory ataxia (dorsal column) from cerebellar ataxia.

PROF'S TRICK: Positive Romberg = worse with eyes closed = sensory (dorsal column) problem, vision was compensating. Unsteady with eyes OPEN too = NOT a true positive Romberg, points toward cerebellar disease instead. This distinction is tested constantly and often worded in tricky, negatively-phrased ways.

Q207. Which of the following conditions primarily affects the PNS rather than the CNS?

✓ **Correct: B. Guillain-Barré syndrome**

Guillain-Barré syndrome is an autoimmune, ascending demyelinating polyneuropathy that targets PERIPHERAL nerve myelin (Schwann cells) — the other options are all primarily CNS diseases.

Q208. A patient has right-sided pain and temperature loss below T12, and left-sided pain and temperature loss below T10 (staggered levels)....

✓ **Correct: B. An intramedullary lesion (e.g., syrinx) expanding asymmetrically, or two separate/expanding lesions affecting decussating spinothalamic fibers at slightly different levels on each side**

Because spinothalamic fibers cross close to their level of entry, an asymmetric or expanding central lesion (like a syrinx) can catch decussating fibers at DIFFERENT levels on each side as it grows, producing this classic 'staggered' sensory level pattern rather than one clean, single transverse level.

PROF'S TRICK: A 'staggered' or 'dissociated, asymmetric level' sensory pattern is a strong clue for syringomyelia — the crossing spinothalamic fibers right next to the expanding central cavity get hit unevenly on each side as the syrinx grows, whereas a discrete traumatic hemisection tends to give one clean, sharp level.

Q209. Which structure is responsible for relaying UNCONSCIOUS proprioceptive information from the lower limb (via the dorsal spinocerebellar)...

✓ **Correct: B. First-order neuron → Clarke's nucleus (second-order neuron) → dorsal spinocerebellar tract → cerebellum**

Unconscious proprioceptive input from the lower limb synapses first onto second-order neurons in Clarke's nucleus (dorsal nucleus, present roughly C8-L2/3), which then form the DORSAL spinocerebellar tract, an uncrossed pathway carrying fine proprioceptive/movement information to the ipsilateral cerebellum for real-time motor coordination.

PROF'S TRICK: Clarke's nucleus is present only C8-L2/3 — below L2/3, unconscious lower-limb proprioception instead uses an alternate route (spinal border cells → ventral spinocerebellar tract) since there's no Clarke's nucleus at those lower levels to synapse on directly. This nucleus-only-at-certain-levels fact is a favorite trick.

Q210. What structure is responsible for unconscious proprioceptive input from the SACRAL spinal cord levels (S2-S4), given that Clarke's...

✓ **Correct: B. Sacral proprioceptive afferents ascend and effectively synapse in the region of Clarke's nucleus equivalent higher up, or use the ventral spinocerebellar pathway**

Since Clarke's nucleus proper doesn't extend into the sacral cord, unconscious proprioceptive signals from S2-S4 instead ascend and effectively relay via the ventral spinocerebellar tract pathway (using spinal border cells as the functional equivalent further down), ultimately still reaching the cerebellum for lower-limb coordination.

Q211. Which pathway carries CONSCIOUS proprioception, fine touch, and vibration sense, as opposed to the unconscious proprioceptive...

✓ **Correct: B. Dorsal column-medial lemniscus (DCML) pathway**

The DCML pathway is the one that reaches conscious awareness via the thalamus (VPL) and primary somatosensory cortex — the spinocerebellar tracts, by contrast, carry proprioceptive information that never reaches consciousness, feeding directly into cerebellar motor coordination circuits instead.

† original question, not a verified past-exam item

PROF'S TRICK: Key distinction: DCML = CONSCIOUS proprioception (you can tell where your limb is). Spinocerebellar tracts = UNCONSCIOUS proprioception (used only by the cerebellum for automatic movement coordination, never reaches your awareness). Same general 'sense,' two totally different destinations and purposes.

Q212. A lesion of the dorsal columns bilaterally (e.g., from vitamin B12 deficiency, subacute combined degeneration) would be expected to...

✓ **Correct: B. Sensory ataxia with a positive Romberg sign, loss of vibration and proprioception, normal pain/temperature**

Subacute combined degeneration classically affects the dorsal columns (and lateral corticospinal tracts) bilaterally, sparing the spinothalamic tracts — producing the classic dissociated pattern of proprioceptive/vibratory loss with sensory ataxia (positive Romberg), while pain and temperature sensation remain intact.

† original question, not a verified past-exam item

Q213. Which spinal cord tract carries fine touch, vibration, and conscious proprioception from the LOWER limb specifically, ascending in the...

✓ **Correct: B. Fasciculus gracilis (lower limb, medial dorsal column)**

The dorsal columns are somatotopically organized: fasciculus gracilis (medial) carries LOWER limb/lower trunk input, while fasciculus cuneatus (lateral, added above T6) carries UPPER limb/upper trunk input — a favorite localization detail.

✗ **A is wrong:** Fasciculus cuneatus is the UPPER limb pathway, and it's positioned LATERAL (not medial) to the gracile fasciculus — the exact opposite pairing (nerve/position) from what this lower-limb question describes.

† original question, not a verified past-exam item

PROF'S TRICK: Dorsal column somatotopy: Gracile (medial, 'G for legs, Ground-level, lower/medial') = lower body. Cuneatus (lateral, added above T6) = upper body. This medial-leg/lateral-arm organization is preserved all the way up into the medial lemniscus too.

Q214. A patient has a central spinal cord lesion (e.g., early syringomyelia) that selectively damages decussating spinothalamic fibers at the...

✓ **Correct: A. Bilateral, symmetric cape-like loss of pain and temperature sensation over the shoulders/arms, with preserved proprioception/vibration and motor strength**

Because the expanding central cavity of a syrinx classically first damages the crossing spinothalamic fibers in the anterior white commissure (bilaterally, at the levels involved), it produces the classic 'cape' or 'shawl' distribution of bilateral pain/temperature loss over the shoulders and arms, while sparing the more peripherally-located dorsal columns and corticospinal tracts (at least early in the disease).

† original question, not a verified past-exam item

PROF'S TRICK: This 'cape distribution' dissociated sensory loss (pain/temp gone, touch/proprioception preserved, bilateral, at a cervical level) is THE classic syringomyelia pattern — recognizing this pattern instantly should trigger 'syrinx' in your differential, one of neurology's most satisfying pattern-recognition diagnoses.

Q215. Which fiber type conducts the FASTEST among peripheral sensory afferents, corresponding to proprioceptive information from muscle...

✓ **Correct: D. A-alpha fibers (largest diameter, most heavily myelinated, fastest conducting)**

A-alpha fibers (which include Ia muscle spindle afferents and alpha motor neuron efferents) are the largest-diameter, most heavily myelinated, and fastest-conducting peripheral nerve fibers — appropriate for the split-second timing demands of reflexes like the stretch reflex.

† original question, not a verified past-exam item

PROF'S TRICK: Fiber conduction speed ladder (fastest to slowest): A-alpha (proprioception, motor) > A-beta (touch/pressure) > A-delta (fast/sharp pain, cold) > B (autonomic preganglionic) > C (slow/dull pain, warmth, unmyelinated). Bigger + more myelin = faster, every time.

Q216. Which best explains why a lesion of the dorsal columns produces a POSITIVE Romberg sign, while a cerebellar lesion produces a NEGATIVE...

✓ **Correct: A. Dorsal column loss removes proprioceptive input that vision can otherwise substitute for; cerebellar ataxia is a movement-coordination problem that visual input cannot compensate for at all**

Dorsal column loss removes one specific INPUT (proprioception) that the CNS can partially substitute for using vision — hence stability with eyes open, instability with eyes closed (removing that substitute). Cerebellar disease is a problem with the COORDINATION/PROCESSING of movement itself, not a missing sensory input, so no amount of visual compensation can fix an intrinsically uncoordinated motor output — hence instability regardless of eye status.

† original question, not a verified past-exam item

PROF'S TRICK: This is a beautiful 'mechanism explains the clinical sign' question — understanding WHY the Romberg test works the way it does (a sensory-substitution logic, not just 'move your eyes and see what happens') is exactly the kind of reasoning that separates rote memorizers from students who can apply the physiology to a novel scenario.

Q217. Two-point discrimination (the ability to distinguish two separate touch points as distinct) is best in which body region, and why?

✓ **Correct: B. The fingertips/lips, because of high mechanoreceptor density and correspondingly larger cortical representation (small receptive fields)**

Two-point discrimination correlates with mechanoreceptor density (especially Meissner's/Merkel) and the size of cortical representation (via the sensory homunculus) — fingertips and lips have small receptive fields and disproportionately large cortical maps, giving the finest spatial resolution anywhere on the body.

† original question, not a verified past-exam item

PROF'S TRICK: Small receptive field = fine discrimination (fingertips, lips). Large receptive field = coarse discrimination (back, thigh). This inverse relationship between receptive field SIZE and discrimination ABILITY, combined with the disproportionate cortical 'real estate' given to high-discrimination areas (see: sensory homunculus), is a foundational somatosensory physiology concept.

Q218. Which best describes the concept of a 'dermatome' and why overlapping dermatomes make it hard to detect a single-root sensory loss on exam?

✓ **Correct: A. A dermatome is the skin area innervated by a single spinal nerve root; because adjacent dermatomes overlap significantly, a single nerve root lesion may produce only SUBTLE or no detectable sensory loss on exam**

Because neighboring dermatomes overlap substantially (each skin region typically receives contributions from more than one adjacent root), an isolated single-root lesion (e.g., one herniated disc affecting one nerve root) often produces only mild or patchy sensory changes rather than a sharply demarcated, complete sensory loss — which is exactly why radiculopathy diagnosis often relies more on motor findings, reflexes, and pain distribution than on a 'clean' sensory exam alone.

† original question, not a verified past-exam item

PROF'S TRICK: This overlap concept is DIFFERENT from peripheral nerve territories (which have much less overlap and DO tend to produce sharper, more complete sensory deficits when damaged) — a classic distinction between root-level (dermatomal, overlapping, subtle) versus peripheral-nerve-level (non-overlapping, sharp) sensory loss patterns on exam.

Q219. Which best describes the difference between 'fast pain' and 'slow pain' in terms of BOTH fiber type AND the resulting...

✓ **Correct: A. Fast pain (A-delta) is sharp, well-localized, and prompts immediate withdrawal; slow pain (C fiber) is dull, poorly-localized, and drives a more prolonged, emotionally unpleasant, protective response**

This is the integrative summary of the fast/slow pain distinction tested throughout this exam: A-delta-mediated fast pain triggers an immediate reflexive withdrawal (protecting from further injury in the moment), while C-fiber-mediated slow pain (with its broader, more diffuse projections including to limbic structures) drives the longer-lasting unpleasant, motivating quality of pain that promotes rest/protection of the injured area during healing.

† original question, not a verified past-exam item

PROF'S TRICK: Notice how fast pain's ANATOMY (direct spinothalamic projection to VPL/cortex) explains its FUNCTION (precise localization for reflexive withdrawal), while slow pain's ANATOMY (diffuse paleospinothalamic projections to reticular formation/limbic system) explains ITS function (the emotional, motivating 'stay still and protect this' quality) — structure and function are never arbitrary in physiology; they always explain each other.

Q220. Which best summarizes why understanding the ANATOMICAL LEVEL at which a sensory tract crosses the midline is the single most useful tool...

✓ **Correct: A. Because it lets you predict, for any given tract, whether a lesion at a specific spinal level will produce a deficit on the SAME side or OPPOSITE side of the body relative to the lesion, and matching that predicted pattern against the patients actual exam findings is how you pinpoint both the level AND the tract(s) involved**

This is the unifying principle behind essentially every lesion-localization question in this entire exam: dorsal columns cross late (medulla) so lesions below that point cause ipsilateral loss; spinothalamic tracts cross early (within 1-2 cord segments) so a cord lesion causes contralateral loss; corticospinal tracts cross at the pyramidal decussation, so lesions above it cause contralateral weakness and lesions below it cause ipsilateral weakness. Master this one framework, and you can work out the expected findings for a lesion ANYWHERE in the neuraxis, for ANY tract, without memorizing each scenario as an isolated fact.

† original question, not a verified past-exam item

PROF'S TRICK: If you remember only ONE big-picture lesson from this entire 300-question set, make it this one: always ask 'has this specific tract crossed the midline yet at this specific level?' That single question, applied systematically, is more powerful than memorizing hundreds of individual lesion-symptom pairings — and it's exactly the kind of first-principles thinking that separates a good doctor from a memorization machine. You've got this.

Lecture 13 — Taste, Sensory Receptors & Neurotransmitters (17 questions)

Q221. The ganglia housing first-order taste neurons are located in which structures?

✓ **Correct: A. Geniculate, Petrosal, Nodose ganglia (for CN VII, IX, X respectively)**

Taste afferents from the anterior 2/3 tongue (CN VII, geniculate ganglion), posterior 1/3 tongue (CN IX, petrosal/inferior ganglion), and epiglottis/pharynx (CN X, nodose/inferior ganglion) all carry their first-order cell bodies in these three respective ganglia.

PROF'S TRICK: Taste = 3 cranial nerves, 3 ganglia, 3 tongue regions: VII/geniculate = anterior 2/3, IX/petrosal = posterior 1/3, X/nodose = epiglottis/pharynx. Memorize this trio as a set — questions love to ask about any one piece in isolation.

Q222. Which stem cell type regenerates sensory receptor cells in BOTH the tongue (taste buds) and the olfactory epithelium?

✓ **Correct: C. Basal cells**

Basal cells are the shared regenerative stem cell population for both taste bud cells and olfactory receptor neurons — a nice unifying fact across two different special senses.

Q223. The gustatory (taste) pathway includes all of the following EXCEPT:

✓ **Correct: E. Trapezoid body**

The trapezoid body is part of the AUDITORY pathway (a brainstem relay in the ascending auditory pathway), not the gustatory pathway — a structure-swap trap between two unrelated special sense pathways.

PROF'S TRICK: Full gustatory pathway ladder: Taste bud → CN VII/IX/X → Geniculate/petrosal/nodose ganglion → Solitary nucleus (rostral part) → VPM thalamus → Insula/frontal operculum (gustatory cortex). Five steps, memorize the whole chain, not just pieces.

Q224. An unpleasant or distorted taste sensation is medically termed:

✓ **Correct: C. Dysgeusia**

Dysgeusia = distorted/unpleasant taste. The other 'dys-' terms all refer to unrelated deficits: dysphagia (swallowing), dysphonia (voice), dysarthria (articulation), dysmetria (cerebellar movement accuracy).

PROF'S TRICK: 'Dys-' word bank, don't mix these up: -phagia=swallowing, -phonia=voice/phonation, -geusia=taste, -arthria=speech articulation, -metria=cerebellar movement accuracy, -osmia=smell (anosmia=no smell). Five to seven similar-sounding terms is a classic examiner trap — build your own glossary.

Q225. When stereocilia of a taste/hair cell bend TOWARD the kinocilium, what is the physiological result?

✓ **Correct: C. Depolarization occurs**

Bending toward the kinocilium opens mechanically-gated cation channels, causing depolarization and increased neurotransmitter release/afferent firing — the opposite of bending away, which hyperpolarizes the cell.

Q226. In the Gate-Control theory of pain, activated inhibitory spinal interneurons release enkephalins that act by:

✓ **Correct: C. Inhibiting presynaptic Ca²⁺ influx and opening postsynaptic K⁺ channels, reducing substance P release and dampening pain transmission**

Enkephalins act at opioid receptors to REDUCE presynaptic Ca²⁺ influx (less neurotransmitter/substance P release from the primary nociceptive afferent) AND increase postsynaptic K⁺ conductance (hyperpolarizing the second-order pain neuron) — a dual pre- and postsynaptic inhibitory mechanism dampening the ascending pain signal.

PROF'S TRICK: Gate-control full circuit: large-diameter mechanoreceptor (A-β) input → activates inhibitory spinal interneurons → enkephalin release → presynaptic Ca²⁺↓ AND postsynaptic K⁺↑ → less substance P, less pain transmission. This is literally why rubbing an injury ('gating' pain with touch input) actually works physiologically.

Q227. Which of the following correctly matches a neurotransmitter to its vesicular/reuptake transporter pair?

✓ **Correct: A. Glutamate = VGLUT (vesicular) / EAAT (reuptake)**

Correct pairings: Glutamate uses VGLUT (loading into vesicles) and EAAT (reuptake); GABA uses VGAT (vesicular) and GAT (reuptake); Monoamines (serotonin, dopamine, norepinephrine) use VMAT (vesicular) and their specific transporters (e.g., SERT for serotonin) for reuptake.

PROF'S TRICK: Three transporter families, three neurotransmitter classes: Glutamate → VGLUT/EAAT. GABA → VGAT/GAT. Monoamines → VMAT/(SERT, DAT, NET). Build this as a three-row table and drill it — these letter-soup transporters are exactly what examiners love to scramble.

Q228. During long-term potentiation (LTP), after Ca²⁺ enters through NMDA receptors, which retrograde messenger diffuses back to the...

✓ **Correct: A. Nitric oxide (NO)**

Postsynaptic Ca²⁺ influx (via NMDA receptors) activates nitric oxide synthase, producing NO, a gaseous retrograde messenger that diffuses backward across the synapse to the presynaptic terminal, enhancing further glutamate release — reinforcing the potentiated synapse from both sides.

Q229. What is the mechanism of action of diazepam (a benzodiazepine) at the GABA-A receptor?

✓ **Correct: B. It increases the FREQUENCY of Cl⁻ channel opening, but only in the presence of GABA (allosteric potentiation)**

Benzodiazepines are positive allosteric modulators — they bind a distinct site on the GABA-A receptor and increase the FREQUENCY of GABA-gated Cl⁻ channel opening, but they require GABA to be present/bound to have any effect (unlike barbiturates, which can directly open the channel at high doses even without GABA).

PROF'S TRICK: Benzodiazepines ↑ *FREQUENCY* of Cl⁻ channel opening (GABA-dependent). Barbiturates ↑ *DURATION* of channel opening, and at high enough doses can open the channel directly even without GABA — which is exactly why barbiturate overdose is so much more dangerous/lethal than benzodiazepine overdose (much higher risk of fatal respiratory depression).

Q230. Which sensory receptor type is a rapidly-adapting mechanoreceptor responsible for detecting vibration and deep pressure, with a...

✓ **Correct: B. Pacinian corpuscle**

The Pacinian corpuscle's distinctive lamellated ('onion-like') capsule is both its histological signature and the reason for its fast adaptation — the layers mechanically filter out sustained pressure, allowing it to respond only to rapid, transient deformation (vibration).

† original question, not a verified past-exam item

Q231. Which type of thermoreceptor channel (TRP family) is activated by noxious heat and capsaicin (the 'hot' component of chili peppers)?

✓ **Correct: B. TRPV1**

TRPV1 is activated by noxious heat (>43°C) and by capsaicin, which is exactly why capsaicin produces a burning sensation — it's directly activating the same channel that heat would.

✗ *A is wrong: TRPM8 is the COLD/menthol receptor (activated by cool temperatures and menthol) — the opposite end of the thermal spectrum from TRPV1/capsaicin, and a very commonly swapped pair.*

† original question, not a verified past-exam item

PROF'S TRICK: TRPV1 = heat + capsaicin ('V for very hot'). TRPM8 = cold + menthol ('M for minty/cold'). This hot/cold, capsaicin/menthol pairing is one of the cleanest mnemonic opportunities in all of sensory physiology — use it.

Q232. Lateral inhibition in a sensory system serves what functional purpose?

✓ **Correct: A. It sharpens spatial resolution/contrast by suppressing the response of less-stimulated neighboring afferents, enhancing the perceived edge/peak of a stimulus**

Lateral inhibition (via inhibitory interneurons connecting neighboring sensory afferents) suppresses the surrounding, less-intensely-stimulated neurons, sharpening the contrast between the most-stimulated point and its surroundings — improving two-point discrimination and edge detection across touch, vision, and other modalities.

† original question, not a verified past-exam item

Q233. A 71-year-old man reports his face 'feels funny' on one side. Examination shows loss of light touch, temperature, and taste sensation in...

✓ **Correct: B. CN V, VII, and VIII (given their close anatomical relationship at the cerebellopontine angle)**

CN V (facial sensation), VII (taste/facial motor), and VIII (hearing/balance) are all anatomically close at the pons/cerebellopontine angle, so a mass or lesion in this region can plausibly affect any combination of sensory and taste modalities carried by these nerves together.

Q234. Which taste modality is most closely associated with glutamate detection (via a specific metabotropic receptor), important in the...

✓ **Correct: B. Umami (via T1R1/T1R3 receptor heterodimer, and related metabotropic glutamate receptors)**

Umami taste (savory, glutamate-rich foods like meat, cheese, MSG) is transduced via specific metabotropic receptors (T1R1/T1R3 heterodimer, among others) tuned to detect glutamate and related amino acids.

† original question, not a verified past-exam item

PROF'S TRICK: Five basic tastes, five transduction logics: Sweet/umami/bitter = metabotropic GPCR-based (T1R and T2R families). Sour = direct H⁺ ion detection. Salty = direct Na⁺ channel detection. Two 'simple ion' tastes, three 'GPCR' tastes — a clean way to organize this topic.

Q235. Which taste modality is transduced by direct H⁺ ion entry/detection, rather than a G-protein-coupled receptor mechanism?

✓ **Correct: C. Sour**

Sour taste is detected via direct H⁺ (proton) interaction with ion channels on taste receptor cells (rather than a metabotropic GPCR cascade like sweet/umami/bitter) — reflecting the fact that 'sourness' is fundamentally just a measure of acidity/H⁺ concentration.

† original question, not a verified past-exam item

Q236. A patient reports that food 'tastes like metal' and has reduced overall taste sensitivity after starting a new medication. Which...

✓ **Correct: A. The medication is chelating/depleting zinc, impairing normal taste receptor cell regeneration/turnover (basal cell-dependent)**

Zinc is a known cofactor important for normal taste bud cell turnover/regeneration (a basal-cell-dependent process); certain medications (e.g., some chelating agents, ACE inhibitors in some patients) can cause zinc-related dysgeusia by disrupting this turnover, producing metallic taste distortion and reduced sensitivity — connecting basic taste bud histology (basal cell regeneration) to a real clinical pharmacology phenomenon.

† original question, not a verified past-exam item

PROF'S TRICK: This question deliberately links two facts you already know from this lecture set — basal cells regenerate taste receptor cells, AND zinc is a cofactor for that process — into a single integrative clinical vignette. This is exactly the kind of 'connect two facts you learned separately' question that rewards deep understanding over rote memorization.

Q237. Why can a strong, unpleasant taste (e.g., bitter) trigger a protective gag or spitting reflex even before conscious taste identification...

✓ **Correct: A. Because gustatory afferents also connect to brainstem reflex circuitry (via the solitary nucleus and its connections to motor nuclei) that can trigger protective reflexes rapidly, in parallel with the slower pathway to conscious cortical taste perception**

Just as pain pathways branch off to trigger fast spinal reflexes (withdrawal) in parallel with the slower pathway to conscious perception, gustatory afferents synapsing in the solitary nucleus can also trigger brainstem-level protective reflexes (gagging, salivation changes) via local connections to motor nuclei — a general principle across sensory systems where 'fast, local reflex' and 'slower, conscious perception' pathways run in parallel from the same initial sensory input.

† original question, not a verified past-exam item

PROF'S TRICK: This parallel-pathway principle — a sensory input branching into both a fast subconscious reflex arc AND a slower conscious perceptual pathway — shows up again and again across the nervous system (pain/withdrawal, taste/gag, light/pupillary reflex vs. conscious vision). Recognizing this as a recurring design pattern, rather than a series of unrelated facts, is exactly the kind of big-picture thinking a great professor is trying to cultivate.

Lecture 14 — Sensory Receptors, Neurotransmitters & Pharmacology (22 questions)

Q238. A woman presents with depressive symptoms and suicidal thoughts. Which neurotransmitter is classically implicated as abnormal in this case?

✓ **Correct: C. Serotonin**

Serotonergic dysfunction is the classic neurotransmitter abnormality implicated in major depression, which is why SSRIs are first-line pharmacotherapy.

✗ **E is wrong:** Dopamine is more classically linked to reward/motivation/psychosis and Parkinson's disease — a tempting but less specific answer for a straightforward depression vignette.

Q239. What is the mechanism of action of botulinum toxin?

✓ **Correct: C. Blocks presynaptic fusion of neurotransmitter vesicles (cleaves SNARE proteins)**

Botulinum toxin is a protease that cleaves SNARE proteins (e.g., SNAP-25, syntaxin, VAMP) required for vesicle docking/fusion, directly preventing ACh release regardless of whether Ca^{2+} has entered the terminal.

✗ *D is wrong: Blocking Ca^{2+} channels would indirectly reduce release (this is closer to Lambert-Eaton's autoimmune mechanism), but botulinum's actual target is the SNARE fusion machinery ITSELF, downstream of Ca^{2+} entry — a subtle but important mechanistic distinction.*

Q240. An 80-year-old man has progressive short-term memory loss and word-finding difficulty, diagnosed with Alzheimer disease. Which statement...

✓ **Correct: E. Blocking of dopamine reuptake transporters**

Dopamine reuptake blockade has nothing to do with Alzheimer's pathogenesis — that's a completely different neurotransmitter system, more relevant to Parkinson's disease or stimulant pharmacology, making it the clearly false statement.

PROF'S TRICK: Alzheimer's core pathology triad: amyloid-beta plaques (extracellular) + neurofibrillary tangles (intracellular, hyperphosphorylated tau) + cortical atrophy (especially hippocampus/temporal-parietal). Beta-secretase (BACE1) is the enzyme that cleaves APP toward the amyloidogenic pathway — increased activity worsens plaque formation.

Q241. A patient arrives with continuous generalized tonic-clonic seizures for 15 minutes (status epilepticus). What is the most appropriate...

✓ **Correct: A. Diazepam (benzodiazepine)**

Benzodiazepines (IV lorazepam or diazepam) are first-line for actively seizing status epilepticus because of their rapid onset via GABA-A potentiation; second-line agents (phenytoin, valproate, levetiracetam) are used if seizures persist.

✗ *D is wrong: Fluoxetine (an SSRI antidepressant) has no role in acute seizure termination — an obviously wrong-category distractor testing whether you're just pattern-matching to any 'drug name' rather than the actual clinical indication.*

PROF'S TRICK: Status epilepticus drug ladder: Benzodiazepine FIRST (fast GABA-A potentiation) → then phenytoin/fosphenytoin or valproate/levetiracetam SECOND-line if seizures continue → then general anesthesia (e.g., propofol) as a last resort for refractory status.

Q242. The summation of EPSPs arriving from numerous different presynaptic fibers converging onto the same postsynaptic neuron at roughly the...

✓ **Correct: A. Spatial summation**

Multiple different synapses (different locations = spatial) firing together add their depolarizations — spatial summation.

✗ *B is wrong: Temporal summation is the SAME synapse firing repeatedly in rapid succession, a time-based (not location-based) mechanism — the question specifically describes 'numerous fibers,' which is a spatial, not temporal, scenario.*

Q243. A 20-year-old college student has depressive symptoms including sleep disturbance and anorexia for 1 month, with no prior mood disorder...

✓ **Correct: C. Inhibition of the serotonin transporter (SERT), increasing synaptic serotonin availability**

SSRIs (selective serotonin reuptake inhibitors) work by blocking SERT, the presynaptic serotonin reuptake transporter, leaving more serotonin available in the synaptic cleft over time.

Q244. A woman presents with progressive muscle weakness, dysphagia, and ptosis that worsens with repeated blinking. Which finding is most...

✓ **Correct: C. Anti-nicotinic acetylcholine receptor antibody (myasthenia gravis)**

Fatigable weakness worsening with repeated activity (repeated blinking causing ptosis to worsen) is classic myasthenia gravis, caused by autoantibodies against postsynaptic nicotinic ACh receptors.

✗ *B is wrong: Lambert-Eaton syndrome antibodies target presynaptic Ca^{2+} channels and classically cause weakness that IMPROVES with repeated activity — the opposite pattern from this vignette's worsening-with-use presentation.*

Q245. A 45-year-old man has a deep gardening wound, then develops jaw stiffness (lockjaw) and muscle spasms. Which mechanism best explains the...

✓ **Correct: D. Blocks release of inhibitory neurotransmitters (glycine/GABA) from spinal inhibitory interneurons, causing disinhibited, excessive motor neuron firing (tetanus)**

This is tetanus (*Clostridium tetani*), not botulism — tetanospasmin travels retrogradely to the CNS and specifically cleaves SNARE proteins in INHIBITORY (glycinergic/GABAergic) interneurons, removing normal inhibition of motor neurons and causing sustained muscle rigidity/spasm — the opposite clinical picture from botulinum toxin's flaccid paralysis, despite a similar SNARE-cleaving mechanism.

PROF'S TRICK: Botulinum and tetanus toxins both cleave SNARE proteins by a nearly identical molecular mechanism, but they travel to and act on completely different neurons — botulinum stays at the NMJ (blocking excitatory ACh release → flaccid paralysis), while tetanus toxin travels retrogradely up the axon into the CNS to block INHIBITORY interneurons (→ spastic rigidity/spasm). Same weapon, opposite target, opposite clinical picture — one of the best 'contrast pair' questions in all of pharmacology.

Q246. A chemotherapy patient has nausea and vomiting. Which antiemetic receptor target is most appropriate?

✓ **Correct: A. 5-HT₃ receptor (e.g., ondansetron)**

Chemotherapy-induced nausea/vomiting is strongly mediated via 5-HT₃ receptors in the gut/area postrema; 5-HT₃ antagonists like ondansetron are first-line antiemetics for this specific cause.

Q247. Which of the following agents does NOT stimulate nociceptors (i.e., is not itself an inflammatory/pain-sensitizing mediator)?

✓ **Correct: E. Vibration**

Vibration is a MECHANICAL stimulus detected by mechanoreceptors (like Pacinian corpuscles), not a chemical inflammatory mediator that sensitizes/activates nociceptors — the other four are all classic chemical mediators of the 'inflammatory soup' that sensitizes pain fibers.

PROF'S TRICK: The 'inflammatory soup' sensitizing nociceptors: prostaglandins, bradykinin, K⁺, H⁺, histamine, substance P, ATP — memorize this list as a set, since 'which of these does NOT belong' questions are extremely common.

Q248. What is the term for the time delay between an action potential arriving at a presynaptic terminal and its effect on the postsynaptic...

✓ **Correct: A. Synaptic delay**

Synaptic delay (~0.5-1ms) reflects the time needed for Ca²⁺ entry, vesicle fusion, neurotransmitter diffusion, and receptor binding — the real, measurable time cost of chemical (as opposed to electrical) synaptic transmission.

✗ **B is wrong:** Synaptic fatigue refers to a DECREASE in synaptic efficacy with sustained/repeated use (neurotransmitter depletion), a completely different phenomenon from the fixed time delay of a single transmission event.

Q249. Which of the following receptors is ionotropic (directly gates an ion channel upon ligand binding)?

✓ **Correct: A. AMPA receptor**

AMPA receptors are ionotropic glutamate receptors (ligand-gated Na⁺/K⁺ channels), providing fast excitatory transmission — the other three listed (mGluR6, GABA-B, 5-HT₁) are all metabotropic (G-protein-coupled), acting through slower second-messenger cascades.

PROF'S TRICK: Ionotropic = fast, direct ion channel (AMPA, NMDA, GABA-A, glycine, nicotinic ACh, 5-HT₃). Metabotropic = slow, G-protein-coupled (mGluRs, GABA-B, muscarinic ACh, most 5-HT subtypes, dopamine, adrenergic). '3' is the magic ionotropic serotonin number to remember: 5-HT₃ is the ONE ionotropic serotonin receptor; all other 5-HT subtypes are metabotropic.

Q250. What is the mechanism of fluoxetine?

✓ **Correct: A. Selective serotonin reuptake inhibitor (blocks SERT)**

Fluoxetine is a classic SSRI, blocking SERT to increase synaptic serotonin.

Q251. What is the mechanism of the antiemetic drug ondansetron?

✓ **Correct: C. Blocks peripheral and central 5-HT₃ receptors**

Ondansetron is a 5-HT₃ receptor antagonist, blocking serotonin's action at this specific ionotropic receptor subtype involved in the vomiting reflex.

Q252. What is the mechanism of haloperidol?

✓ **Correct: B. Blocks D₂ dopamine receptor**

Haloperidol is a typical (first-generation) antipsychotic, blocking D₂ dopamine receptors.

***PROF'S TRICK:** Three drugs, three completely different single-target mechanisms tested back to back (fluoxetine=SERT, ondansetron=5-HT₃, haloperidol=D₂) — a favorite examiner trick is to present these consecutively and swap the mechanisms between them.*

Q253. A patient is splashed with hot cooking oil. Which channel is the primary receptor for NOXIOUS heat in this scenario?

✓ **Correct: A. TRPV2 (very high heat threshold, extreme noxious heat)**

For truly extreme/noxious heat (like hot oil, well above the ~43°C threshold of TRPV1), TRPV2 is the higher-threshold heat nociceptor channel recruited — distinct from TRPV1's more moderate noxious-heat range.

***PROF'S TRICK:** Thermal TRP channel ladder by temperature: TRPM8 (cool, ~menthol) < TRPV4/TRPV3 (warm) < TRPV1 (hot/noxious, ~43°C+, capsaicin) < TRPV2 (extreme heat, >52°C). The hotter the stimulus, the higher up this ladder you go.*

Q254. A woman presents with fatigue, social withdrawal, and depressed mood. Which neurotransmitter system is most likely implicated?

✓ **Correct: C. Serotonin**

Same core depression-serotonin link tested again — a very frequently repeated exam fact worth over-learning.

Q255. Anosmia and ageusia (loss of smell and taste) can result from downregulation of which receptor, also notable for its role as the entry...

✓ **Correct: C. ACE2 (angiotensin-converting enzyme 2)**

ACE2 is expressed on olfactory support cells (sustentacular cells) and is both the receptor SARS-CoV-2 uses for cellular entry and a plausible mechanistic link to COVID-19-associated anosmia/ageusia — a great example of a basic science fact suddenly becoming extremely clinically relevant.

Q256. Which neurotransmitter-transporter pairing is correct?

✓ **Correct: A. VGAT - GABA (correct vesicular pairing)**

VGAT loads GABA into synaptic vesicles — a correctly matched pair. The other four options mismatch a transporter/enzyme with the wrong neurotransmitter (SERT is for serotonin not dopamine; DAT is for dopamine not ACh; EAATs are for glutamate not norepinephrine; glutaminase is an enzyme, not a transporter, and it converts glutamine TO glutamate, it doesn't transport glutamine per se).

Q257. A patient overdoses on a drug that directly opens GABA-A chloride channels even in the complete absence of GABA, causing profound,...

✓ **Correct: B. Barbiturates, because at high doses they can directly gate the Cl⁻ channel independent of GABA, producing unopposed, potentially fatal CNS/respiratory depression**

Barbiturates, unlike benzodiazepines, can directly open the GABA-A chloride channel even without GABA bound, at sufficiently high concentrations — this GABA-independent action removes any physiological 'ceiling' on inhibition, making barbiturate overdose far more likely to cause fatal respiratory arrest than benzodiazepine overdose (where the effect is capped by needing GABA to be present).

† original question, not a verified past-exam item

***PROF'S TRICK:** This 'why is X more dangerous than Y' framing is a hallmark of a well-designed critical-thinking question — always be ready to explain not just the mechanism, but its clinical CONSEQUENCE (here: no ceiling effect = more dangerous in overdose).*

Q258. Why does chronic SSRI use often take 2-4 weeks to produce clinical antidepressant benefit, even though synaptic serotonin levels rise...

✓ **Correct: A. The delay reflects downstream neuroadaptive changes — including desensitization of presynaptic serotonin autoreceptors (allowing sustained increased release) and slower changes in postsynaptic receptor density/signaling and neurotrophic factors (e.g., BDNF) — not simply the acute rise in synaptic serotonin itself**

The acute pharmacologic effect (SERT blockade, more synaptic serotonin) happens fast, but the clinical antidepressant response depends on slower, downstream adaptive changes in receptor sensitivity and neuroplasticity (e.g., autoreceptor desensitization, altered gene expression, increased BDNF/neurotrophic signaling) — this mismatch between immediate pharmacology and delayed clinical effect is a genuinely important, frequently tested concept in psychiatric pharmacology.

† original question, not a verified past-exam item

PROF'S TRICK: This kind of 'the obvious mechanism doesn't explain the observed timeline' question is exactly the sort of thing a good prof uses to separate students who memorized 'SSRI = blocks SERT' from students who understand psychopharmacology at a deeper, more clinically useful level.

Q259. Which best explains the clinical observation that opioid analgesics are highly effective for nociceptive (tissue-injury) pain but often...

✓ **Correct: A. Opioids primarily act on mu-opioid receptors in the pain-modulating circuits (periaqueductal gray, dorsal horn) engaged by normal nociceptive signaling, whereas neuropathic pain often involves aberrant, opioid-less-responsive mechanisms (ectopic firing, central sensitization, altered ion channel expression) in damaged nerves themselves**

Nociceptive pain (from ongoing tissue injury/inflammation activating normal nociceptor pathways) responds well to opioids acting at mu-receptors along the normal pain-modulating circuitry; neuropathic pain instead often arises from damaged nerve fibers generating aberrant signals through mechanisms (ectopic ion channel expression, central sensitization) that aren't primarily opioid-receptor-dependent — explaining why neuropathic pain often instead responds better to anticonvulsants (e.g., gabapentin, targeting Ca²⁺ channels) or antidepressants (e.g., SNRIs, boosting descending inhibition) than to opioids alone.

† original question, not a verified past-exam item

PROF'S TRICK: This mechanistic 'why does drug X work for pain type A but not pain type B' reasoning is exactly the sort of integrative pharmacology-meets-physiology question that separates surface-level memorization from the kind of understanding you'll actually need as a practicing physician choosing pain medications.

Lecture 15 — Visual System (19 questions)

Q260. A patient exhibits a near-vision reflex defect. Which of the following is NOT a normal characteristic of the near vision reflex?

✓ **Correct: B. Levator palpebrae superioris contraction (eyelid elevation)**

The near reflex TRIAD is convergence + pupillary constriction + accommodation (increased lens convexity) — eyelid elevation (levator palpebrae) is not part of this triad at all, making it the correct 'NOT characteristic' answer.

PROF'S TRICK: Near reflex triad = 3 C's roughly: Convergence, Constriction (pupil), (increased) Convexity of lens. If an option mentions eyelids, that's an automatic red flag — eyelid function is not part of the near triad.

Q261. A patient has a tumor compressing the optic chiasm. Which visual field defect results?

✓ **Correct: C. Bitemporal heteronymous hemianopia**

The optic chiasm is where NASAL retinal fibers (carrying TEMPORAL visual field information) cross; compressing the chiasm classically damages these crossing fibers bilaterally, causing loss of both temporal visual fields — bitemporal hemianopia (classic pituitary tumor presentation).

PROF'S TRICK: Optic chiasm compression (classically from a pituitary adenoma pressing upward) = bitemporal hemianopia, the single most tested visual field defect in all of neuroanatomy. 'Tunnel vision' from losing both peripheral fields is the classic clinical description.

Q262. Which part of the retina has NO photoreceptor cells at all?

✓ **Correct: D. Optic disc (the physiologic blind spot)**

The optic disc is purely the exit point for ganglion cell axons/central vessels, with zero photoreceptors — the anatomical basis of the physiologic blind spot.

Q263. What structure in the eye is primarily responsible for absorbing stray light, reducing internal reflection/glare?

✓ **Correct: A. Choroid (its pigmented layer)**

The heavily pigmented choroid absorbs stray light that passes through the retina, preventing internal scatter/reflection that would otherwise degrade image quality — functioning like the black interior of a camera.

Q264. Which condition is indicated by an intraocular pressure (IOP) greater than 25 mmHg?

✓ **Correct: D. Glaucoma**

Elevated IOP (normal roughly 10-21 mmHg) is the hallmark of glaucoma, which can progressively damage the optic nerve (via mechanical/ischemic injury to ganglion cell axons) if untreated.

Q265. Which correctly arranges retinal layers from the rod/cone (photoreceptor) layer toward the optic nerve fiber layer?

✓ **Correct: C. Rods/cones → Outer nuclear → Inner nuclear → Ganglion cell → Optic nerve fiber**

Correct order from photoreceptor toward the vitreous/optic nerve: photoreceptor segments → outer nuclear layer (photoreceptor cell bodies) → outer plexiform → inner nuclear layer (bipolar/horizontal/amacrine cell bodies) → inner plexiform → ganglion cell layer → nerve fiber layer.

***PROF'S TRICK:** Remember: light must pass THROUGH most retinal layers before hitting the photoreceptors (the retina is 'inside-out' relative to a camera) — but the SIGNAL then flows the opposite direction, from photoreceptor outward toward the ganglion cells and optic nerve, which is the direction this question tests.*

Q266. A patient has a left pupil that does not constrict and is larger than the right, with associated ptosis and 'down and out' eye...

✓ **Correct: D. CN III (oculomotor nerve)**

Dilated, fixed pupil + ptosis + down-and-out eye = complete CN III palsy — the parasympathetic pupillary fibers on the outside of the nerve are compressed, explaining the pupil involvement.

Q267. Surface ectoderm gives rise to which eye structure?

✓ **Correct: A. Lens**

The lens is the classic surface ectoderm derivative (via lens placode/vesicle) — the other options are neuroectoderm (iris epithelium, retina) or neural crest/mesoderm (corneal endothelium, sclera) derived.

Q268. Which specialization of the fovea centralis increases visual acuity there?

✓ **Correct: C. Lateral displacement of overlying ganglion/inner retinal cell layers, minimizing light scatter before reaching photoreceptors**

At the fovea, the inner retinal layers (ganglion cells, bipolar cells, etc.) are pushed laterally out of the way, letting light hit the densely-packed cone photoreceptors with minimal obstruction/scatter — a key structural adaptation for maximal visual acuity.

✗ D is wrong: The fovea is actually CONE-dominated (essentially no rods at its very center) — increased ROD density is the opposite of what makes the fovea special; rods dominate the peripheral retina instead, favoring sensitivity over acuity.

***PROF'S TRICK:** Fovea = cones only, best acuity, worst in dim light (no rods = can't see faint stars if you look straight at them — that's why astronomers use averted vision, shifting the image to the rod-rich periphery).*

Q269. What structure is damaged to produce a RIGHT homonymous hemianopia?

✓ **Correct: A. Left optic tract (or fibers further downstream on the left visual pathway)**

A right homonymous hemianopia (loss of the right visual field in BOTH eyes) results from a lesion of the LEFT visual pathway AFTER the chiasm (left optic tract, radiation, or occipital cortex) — post-chiasmal lesions always cause a homonymous (same-side) field cut, on the side OPPOSITE the lesion.

PROF'S TRICK: Golden rule: PRE-chiasmal lesions (optic nerve) = monocular defects. AT the chiasm = bitemporal (heteronymous). POST-chiasmal (tract, radiation, cortex) = homonymous, on the side of the visual field OPPOSITE to the lesion side. Always work out which 'zone' the lesion is in first.

Q270. Which abnormality of retinal cells causes the majority of color blindness cases?

✓ **Correct: C. Abnormal cone cells (cone photopigment gene defects)**

Color vision depends on the three cone photopigment types (red/green/blue-sensitive); most color blindness (e.g., red-green color blindness, an X-linked trait) results from defects in the genes encoding these cone opsins.

Q271. What is the effect of sympathetic fiber activation on the eye?

✓ **Correct: C. Pupillary dilation (mydriasis)**

Sympathetic activation (via the dilator pupillae muscle, alpha-1 receptors) causes pupillary dilation (mydriasis) — the classic 'fight or flight' pupil response, opposite to the parasympathetic-mediated constriction of the near/light reflexes.

Q272. What is the physiological effect of retinal detachment?

✓ **Correct: A. Loss of blood supply to the photoreceptor layer (from the choroid), causing photoreceptor ischemia/degeneration**

The outer retina (photoreceptors) depends on diffusion from the underlying choroidal blood supply; when the neurosensory retina detaches from the retinal pigment epithelium/choroid, photoreceptors are cut off from this nutrient/oxygen source and begin to degenerate — the urgency behind emergency retinal detachment repair.

Q273. A lesion of the RIGHT Meyer's loop (temporal lobe optic radiation) causes which visual field defect?

✓ **Correct: A. Left homonymous SUPERIOR quadrantanopia**

Meyer's loop carries fibers for the CONTRALATERAL SUPERIOR visual field (from the inferior retina); a RIGHT-sided lesion therefore causes a LEFT, SUPERIOR quadrant defect — 'pie in the sky' on the opposite side.

PROF'S TRICK: Meyer's loop = temporal lobe = 'pie in the sky' = contralateral SUPERIOR quadrantanopia. The parietal lobe's optic radiation fibers instead carry the INFERIOR field, so parietal lesions give contralateral INFERIOR quadrantanopia ('pie on the floor'). Temporal=sky, Parietal=floor — a classic paired mnemonic.

Q274. A patient with a right temporal lobe lesion (affecting Meyer's loop) most likely shows which visual field defect?

✓ **Correct: C. Left homonymous superior quadrantanopia**

Same Meyer's loop logic restated: right temporal lobe = left, superior quadrant defect.

Q275. Touching the RIGHT cornea produces no blink in either eye; touching the LEFT cornea produces bilateral blinking. Where is the lesion?

✓ **Correct: C. Right trigeminal nerve (afferent limb of the corneal reflex)**

The corneal reflex's AFFERENT limb is CN V (trigeminal); if stimulating the right cornea produces NO response at all (bilaterally absent), the afferent (sensory) side on the right must be damaged — since the left side still triggers a normal bilateral (consensual) blink, the EFFERENT (CN VII, facial) pathways on both sides must be intact.

✗ A is wrong: If the LEFT facial nerve (efferent) were damaged, you'd expect the LEFT eye specifically to fail to blink regardless of which cornea is touched — but here BOTH eyes blink normally when the left cornea is stimulated, ruling out any efferent (facial nerve) problem entirely.

PROF'S TRICK: Corneal reflex logic puzzle: figure out afferent (CN V) vs efferent (CN VII) by asking 'does touching ONE side ever produce ANY blink?' If touching side X produces NO blink in either eye, but touching the other side produces blinking in BOTH eyes, the problem must be AFFERENT (sensory, CN V) on side X — the efferent (motor, CN VII) pathways on both sides are clearly working fine since they respond correctly when properly triggered from the other side.

Q276. A patient has anisocoria noted incidentally — pupils differ in size — but reacts normally to light and accommodation, worse in dim...

✓ **Correct: B. Physiologic (simple) anisocoria, a common benign variant present in a meaningful percentage of the healthy population**

Physiologic anisocoria (a small, stable pupil size difference, more noticeable in dim light, with normal light/accommodation reflexes and no other neurological findings) is a common, entirely benign variant seen in a meaningful fraction of healthy people — critical to distinguish from pathological causes (CN III palsy, Horner's, pharmacologic mydriasis) that instead show abnormal reflexes or other red-flag findings.

† original question, not a verified past-exam item

PROF'S TRICK: Not every asymmetry is pathological! A good clinician (and a good exam question) always asks: are the REFLEXES normal? Are there OTHER signs? Physiologic anisocoria is the 'nothing to see here' answer specifically because everything else checks out normal — this is a favorite way to test whether you can recognize when NOT to chase a zebra.

Q277. Which layer of the eye is most directly responsible for providing nutrients to the outer retina via diffusion (choriocapillaris), making...

✓ **Correct: B. Choroid**

The choroid's dense capillary bed (choriocapillaris) diffuses oxygen/nutrients to the avascular outer retina (photoreceptors, RPE) — dysfunction here (e.g., drusen deposition, neovascularization) underlies both dry and wet forms of age-related macular degeneration.

† original question, not a verified past-exam item

Q278. Why does damage to the optic nerve (before the chiasm) cause a MONOCULAR visual deficit, while damage to the optic tract (after the...

✓ **Correct: A. Because the chiasm is the point where fibers from both eyes' nasal retinas cross and combine with the intact temporal retina fibers of the other eye, so everything AFTER the chiasm carries information from BOTH eyes; everything BEFORE the chiasm carries information from only ONE eye**

This is the master organizing principle of the entire visual pathway: pre-chiasmal structures (optic nerve) are eye-specific (one nerve = one eye's full visual field), while the chiasm reorganizes fibers by VISUAL FIELD rather than by eye — so everything post-chiasmal (tract, radiation, cortex) carries a mixed, binocular representation of one half of visual space, explaining why post-chiasmal lesions always produce homonymous (matching, same-side) field cuts in both eyes rather than a single blind eye.

† original question, not a verified past-exam item

PROF'S TRICK: This single principle — 'before the chiasm = organized by EYE, after the chiasm = organized by VISUAL FIELD' — is the master key that lets you derive EVERY visual field defect pattern from first principles instead of memorizing them individually. Once you truly understand this, questions about optic nerve vs. chiasm vs. tract vs. radiation lesions become logic problems, not memorization problems.

Lecture 16 — Auditory & Vestibular Systems (22 questions)

Q279. Which structure is found at the level of the rostral pons?

✓ **Correct: D. Motor and principal (chief) sensory nuclei of CN V**

The trigeminal motor and principal sensory nuclei sit at the level of the rostral (mid-upper) pons — a classic internal brainstem landmark question.

✗ **B is wrong:** The abducens nucleus is at the level of the CAUDAL pons (with the facial colliculus), one level more caudal than this question is asking about.

Q280. A patient has facial nerve damage resulting in hyperacusis (abnormally loud perception of sound). Which structure's function is most...

✓ **Correct: D. Stapedius reflex (stapedius muscle, innervated by CN VII)**

The stapedius muscle (dampening ossicle movement in response to loud sound) is innervated by the facial nerve; facial nerve damage disables this protective reflex, causing hyperacusis (sounds seem abnormally loud).

✗ *C is wrong: The tensor tympani is innervated by CN V (trigeminal), not CN VII — a facial nerve lesion would not directly affect this muscle's reflex.*

PROF'S TRICK: Two ossicle-dampening muscles, two different cranial nerves: Tensor tympani = CN V. Stapedius = CN VII. A facial nerve lesion knocks out only the stapedius reflex, producing hyperacusis specifically — a great example of how remembering WHICH nerve innervates WHICH muscle lets you predict the exact clinical finding.

Q281. Which vestibular nucleus sends axons directly to the oculomotor nucleus (contributing to the vestibulo-ocular reflex)?

✓ **Correct: C. Superior vestibular nucleus**

The superior vestibular nucleus is the primary source of ascending vestibulo-ocular reflex connections (via the MLF) to the oculomotor and other extraocular motor nuclei, coordinating eye movements with head movements.

✗ *B is wrong: The lateral vestibular nucleus (Deiters' nucleus) instead gives rise to the lateral vestibulospinal tract, a DESCENDING pathway to the spinal cord for postural/antigravity muscle tone — a different job entirely from the ascending oculomotor connections.*

Q282. Which structure releases (dissipates) the pressure wave of sound energy out of the inner ear, allowing continued fluid movement within...

✓ **Correct: C. Round window**

The round window (a flexible membrane at the base of the scala tympani) bulges outward to release the pressure wave transmitted through the cochlear fluids, allowing continuous fluid movement despite the fluid's incompressibility — without it, the oval window's inward pushes would have nowhere to 'go.'

✗ *B is wrong: The oval window is the ENTRY point (where the stapes footplate pushes IN), the exact opposite role from the round window's pressure-RELEASE function — these two windows are constantly swapped in student answers.*

PROF'S TRICK: Oval window = IN (stapes pushes fluid in). Round window = OUT (releases pressure, bulges outward). Sound energy flow: air → tympanic membrane → ossicles → oval window → cochlear fluid wave → round window releases the pressure.

Q283. Which statement regarding sound stimulation and cochlear transduction is TRUE?

✓ **Correct: C. Inner hair cells are stimulated by relative movement against the tectorial membrane and generate action potentials carried via the peripheral processes of the SPIRAL ganglion**

Inner hair cell stereocilia deflection (relative to the tectorial membrane, as the basilar membrane moves) generates the receptor potential that ultimately triggers action potentials in the peripheral processes of spiral ganglion neurons — the actual primary afferent pathway for hearing.

✗ *D is wrong: High frequencies are encoded at the BASE of the cochlea (stiff, narrow basilar membrane), not the apex — the apex is tuned to LOW frequencies, the exact opposite of what this option claims.*

Q284. What structure detects vertical linear acceleration, as when an elevator starts moving up or down?

✓ **Correct: D. Sacculle (oriented to detect vertical linear acceleration)**

The sacculle's maculae are oriented in a roughly VERTICAL plane, making it the primary detector of vertical linear acceleration (elevator start/stop, jumping) — while the utricle's horizontally-oriented macula detects horizontal linear acceleration instead.

✗ *E is wrong: The utricle detects HORIZONTAL linear acceleration (like a car starting to move forward), the perpendicular plane from what this vertical-elevator scenario is testing.*

PROF'S TRICK: Sacculle = VERTICAL linear acceleration (elevator, jumping) — think 'Sacculle = Straight up and down.' Utricle = HORIZONTAL linear acceleration (car accelerating forward) — think 'Utricle = yoU're driving forward.'

Q285. What structure primarily detects tilting the head UP (vertical linear head position change)?

✓ **Correct: B. Saccul**

Same saccul/vertical-plane logic as the elevator question — tilting the head to change its vertical position relative to gravity is detected by the vertically-oriented saccular macula.

Q286. What structure is primarily activated by nodding the head forward (in the sagittal/vertical rotational plane)?

✓ **Correct: D. Superior (anterior) semicircular canal**

Nodding ('yes' motion) is a rotational movement in the sagittal plane, detected by the superior (anterior) semicircular canal, which is oriented to sense exactly this plane of angular acceleration.

PROF'S TRICK: Semicircular canal orientation-to-movement mapping: Horizontal canal = shaking head 'no' (yaw). Anterior/superior canal = nodding 'yes' (pitch). Posterior canal = tilting head toward shoulder (roll). Each canal is tuned to detect rotation specifically in its own anatomical plane.

Q287. When you extend your head backward (posterior tilt), which vestibular receptor is primarily stimulated?

✓ **Correct: A. Superior semicircular canal**

Extending the head backward is a rotational (pitch-plane) movement, sensed by the anterior/superior semicircular canal — the same canal responsible for the forward-nodding motion, since it detects rotation in that plane in EITHER direction.

Q288. Where is the vomiting center located?

✓ **Correct: C. Area postrema**

Area postrema (a circumventricular organ with a leaky BBB) functions as the chemoreceptor trigger zone that detects blood-borne emetogenic substances and directly triggers vomiting.

PROF'S TRICK: Note the connection to vestibular input here too: motion sickness works because vestibular nuclei ALSO project to the vomiting center/area postrema — that's the whole physiological basis of why spinning/motion makes you nauseated.

Q289. What structure reduces sensitivity to background/surrounding sound (by inhibiting outer hair cells) to help focus on speech in a noisy...

✓ **Correct: A. Stapedius muscle (and the broader olivocochlear efferent system inhibiting outer hair cells)**

The medial olivocochlear efferent system (with the stapedius reflex contributing too) can inhibit outer hair cell activity, effectively 'turning down' background noise amplification and improving signal-to-noise discrimination for speech — a genuine efferent (top-down) control loop within the auditory system.

Q290. Trouble hearing LOW-frequency sounds suggests a possible lesion/occlusion where along the basilar membrane?

✓ **Correct: B. Apex of the basilar membrane (tuned to low frequencies)**

Low frequencies are encoded at the cochlear APEX (wide, flexible basilar membrane) — trouble with low-frequency hearing points to pathology there, the opposite end from presbycusis's classic high-frequency, base-first pattern.

Q291. Which of the following correctly describes the central auditory pathway?

✓ **Correct: A. Cochlear nerve → Cochlear nucleus → Superior olivary complex → Medial geniculate body (of the thalamus)**

The auditory pathway runs Cochlear nerve (1st order) → Cochlear nucleus (2nd order, brainstem) → Superior olivary complex (first major bilateral relay) → (lateral lemniscus →) Inferior colliculus → Medial geniculate body (thalamus) → primary auditory cortex.

✗ **C is wrong:** The auditory pathway relays through the MEDIAL geniculate body, not the lateral geniculate body — the lateral geniculate body is the THALAMIC relay for the VISUAL pathway instead, a classic cross-modality mix-up.

PROF'S TRICK: Thalamic relay nuclei by sense: Vision = Lateral Geniculate. Hearing = Medial Geniculate. ('Lateral for Looking, Medial for Music' — or simply remember alphabetical: Auditory/Medial both later in the alphabet than Visual/Lateral, whatever memory trick sticks for you.)

Q292. An 80-year-old patient presents with presbycusis (age-related hearing loss, 50-8000 Hz range affected). Which structure is most likely...

✓ **Correct: B. Outer hair cells (cumulative damage, especially at the cochlear base)**

Presbycusis classically results from cumulative outer hair cell damage/loss, especially at the high-frequency-encoding cochlear base, progressing over a lifetime of noise/metabolic stress exposure.

Q293. A man slips and is about to fall; he reflexively extends his upper and lower limbs to catch himself. Which pathway mediates this reflex?

✓ **Correct: B. Lateral vestibulospinal tract (extensor/antigravity tone control)**

The lateral vestibulospinal tract (from the lateral vestibular/Deiters' nucleus) facilitates extensor (antigravity) muscle tone throughout the limbs — exactly the reflex needed to extend the limbs and catch a fall.

PROF'S TRICK: Lateral vestibulospinal tract = limb extensor tone, uncrossed, whole-body postural reflexes (catching a fall). Medial vestibulospinal tract = mostly neck/head/upper axial muscle control (via MLF), for stabilizing gaze/head position during movement — different target muscle groups, different jobs.

Q294. What is the underlying pathological cause of presbycusis?

✓ **Correct: C. Progressive damage to cochlear hair cells (especially outer hair cells at the base)**

Presbycusis is fundamentally a SENSORINEURAL process — cumulative cochlear hair cell damage — not a conductive problem (ossicle stiffening would instead be otosclerosis, a different, conductive hearing loss mechanism).

✗ **B is wrong:** Ossicle stiffening describes OTOSCLEROSIS, a conductive hearing loss mechanism — a commonly confused alternative cause of hearing loss in an aging population, but mechanistically and audiometrically distinct from presbycusis.

PROF'S TRICK: Always sort hearing loss into CONDUCTIVE (outer/middle ear — wax, otosclerosis, perforation) versus SENSORINEURAL (inner ear/cochlear nerve — presbycusis, noise damage, ototoxic drugs) — this single sorting question underlies almost every hearing-loss vignette you'll ever see.

Q295. What is INCORRECT about stimulation of the crista ampullaris (within the semicircular canal ampulla)?

✓ **Correct: D. It detects LINEAR acceleration of the head (this is the false statement — that is the otolith organs' job)**

The crista ampullaris (semicircular canal sensory organ) detects ANGULAR (rotational), not linear, acceleration — linear acceleration is the job of the utricle/sacculae (otolith organs with their otoconia-laden maculae), making this the clearly false statement.

Q296. Which tract can inhibit/modulate the vestibulospinal reflex's effect on limb muscles, allowing voluntary override of purely reflexive...

✓ **Correct: B. Corticospinal tract (descending voluntary motor control that can override/modulate reflexive vestibulospinal output)**

The corticospinal tract provides descending VOLUNTARY motor control that can override or modulate the more automatic/reflexive vestibulospinal postural adjustments — a nice example of higher cortical control superimposed on brainstem reflex circuitry.

Q297. A presbycusis test is performed on patients aged 45, 60, and 90. At which decibel (dB) level does hearing loss typically begin to be...

✓ **Correct: B. 25-50 dB**

Presbycusis progresses gradually, with noticeable threshold shifts typically beginning in the mild-to-moderate 25-50 dB range before progressing further with age — consistent with a slow, cumulative sensorineural process rather than a sudden, severe loss.

Q298. Why does damage to the outer hair cells (rather than inner hair cells) cause a disproportionately large loss of hearing SENSITIVITY,...

✓ **Correct: A. Outer hair cells function as the cochlear amplifier — using electromotility to actively amplify basilar membrane vibration at specific frequencies BEFORE inner hair cells transduce it, so losing them means inner hair cells receive a much weaker mechanical signal to begin with**

Outer hair cells actively amplify sound-induced basilar membrane motion via electromotility (the 'cochlear amplifier'), sharpening frequency tuning and boosting sensitivity to quiet sounds — even though they send relatively little afferent information to the brain themselves, their PRE-processing amplification is what inner hair cells depend on for a strong enough signal; lose the amplifier, and even a fully intact inner hair cell population receives a much weaker mechanical stimulus, explaining the outsized sensitivity loss.

† original question, not a verified past-exam item

PROF'S TRICK: This is a genuinely elegant 'why' question: it's not a contradiction that a cell type carrying only 5% of afferent fibers can cause such large sensitivity loss when damaged — because its job isn't to CARRY the signal, it's to AMPLIFY it before the signal-carrying cells (inner hair cells) even see it. Understanding 'amplifier vs. carrier' roles is exactly the kind of systems-level thinking a great professor rewards.

Q299. Why does conductive hearing loss (e.g., from otosclerosis or a perforated eardrum) typically preserve the ability to hear via bone...

✓ **Correct: A. Conductive loss is a problem in the outer/middle ear mechanical transmission pathway (which bone conduction bypasses entirely by vibrating the skull/cochlea directly); sensorineural loss is a problem in the cochlea/nerve itself, which bone conduction cannot bypass**

Bone conduction testing (e.g., Rinne/Weber tests) works by vibrating the skull to directly stimulate the cochlea, bypassing the outer/middle ear entirely — so it remains NORMAL in purely conductive hearing loss (the problem is upstream, in sound transmission), but is IMPAIRED in sensorineural loss (the problem is in the cochlea/nerve itself, which bone conduction cannot route around).

† original question, not a verified past-exam item

PROF'S TRICK: This is exactly the physiological logic behind the Rinne and Weber tests, two of the most classic bedside hearing tests in all of clinical medicine — understanding WHY bone conduction bypasses the outer/middle ear (rather than just memorizing 'AC>BC is normal, BC>AC suggests conductive loss') lets you actually reason through any tuning-fork exam scenario instead of pattern-matching.

Q300. Why does spinning in place and then suddenly stopping cause a strong sensation of continued rotation (post-rotatory vertigo/nystagmus)?

✓ **Correct: A. Endolymph in the semicircular canals has inertia and continues moving briefly after the head stops, still deflecting the cupula and signaling rotation even though the head is now still**

Endolymph fluid inertia means it keeps moving briefly even after the skull (and semicircular canal walls) have actually stopped rotating — this continued relative endolymph flow still deflects the cupula, generating a false 'still rotating' signal, producing the classic post-rotatory vertigo and nystagmus everyone has experienced (e.g., after spinning as a child).

† original question, not a verified past-exam item

PROF'S TRICK: This is a great everyday-experience physiology question — nearly everyone has felt post-spinning dizziness, and being able to explain WHY it happens (fluid inertia lagging behind the now-stationary canal walls) using the same cupula-deflection mechanism tested throughout this vestibular section shows genuine understanding, not just memorization.